

SOLACE ARCHITECT REPORT

Payment Solutions V1

Payment solutions platform. New build. Cards + UPI. Multi-region India/US. 5K TPS. PCI-DSS. No pattern match — custom architecture.

SKILLS

ARTIFACTS

EXECUTION

USER WAIT

17

47

48m 54s

34m 30s

Summary

Payment solutions platform. New build. Cards + UPI. Multi-region India/US. 5K TPS. PCI-DSS. No pattern match — custom architecture.

COMPLETED SKILLS	TOTAL ARTIFACTS	SYSTEMS	DECISIONS	REVIEW FINDINGS	OPEN ITEMS
17 of 17	47 files	12	19	0	14

Scope & Inputs

Key inputs from discovery that drive downstream architecture decisions.

⚙️ CONNECTED SYSTEMS

12 SYSTEMS

Core Banking Platf...	producer/consumer	Credit Card Proces...	producer/consumer	UPI Gateway	producer/consumer
Fraud Detection En...	consumer/producer	Risk Management	consumer/producer	2FA Service	consumer/producer
Customer Notification Serv...	consumer	Merchant Portal	consumer	Customer-Facing A...	producer/consumer
Internal Operations Dashbo...	consumer	Settlement Engine	consumer/producer	Analytics Platform	consumer

☰ CURRENT LANDSCAPE

EXISTING MESSAGING

API-based (REST). No existing message broker or event streaming platform.

PROTOCOLS

REST, MQTT, WebSocket

REFERENCE ARCHITECTURE

None — custom architecture. Shares banking-sector traits with Pattern 1 (PCI-DSS, audit trail) but is a payment processing system, not an AI assistant.

📏 EVENT TYPES

9 TYPES

Payment initiated ~2,000/sec

Payment authorized ~1,800/sec

Payment captured/settled ~1,500/sec

Fraud check request/response ~2,000/sec

2FA challenge/response ~1,000/sec

Settlement batch events ~100/sec bursts

Customer notifications ~3,000/sec

Merchant settlement reports ~50/sec

Analytics events ~5,000/sec

☑️ REQUIREMENTS

🚩 GOALS & CONSTRAINTS

DELIVERY GUARANTEE

Mixed — Guaranteed messaging for payment transactions, fraud alerts, 2FA, settlement. Direct messaging for notifications, analytics, dashboard updates.

ORDERING

Per-key partitioned queues with transaction ID as partition key

LATENCY TARGET

Sub-second for the hot path (payment auth + fraud check + 2FA)

SCALE

5,000 TPS peak, 10% year-on-year growth (~8,050 TPS in 5 years)

TOPOLOGY

Multi-region from day one — AWS Mumbai (India) + AWS US-East (US)

PROCESSING GUARANTEE

At-least-once with idempotent consumers (Solace provides at-least-once via Guaranteed messaging; exactly-once requires application-level idempotency)

DATA RESIDENCY

India payment data stays in India (RBI regulations). US payment data stays in US. Cross-region flows limited to shared reference data and aggregated analytics.

PROJECT TYPE

New build — greenfield event-driven payment processing system

DRIVER

Launching a new payment product supporting credit/debit cards and UPI

TIMELINE

4 months to production

BUDGET

Cost-optimized — preference for cloud-managed to reduce ops burden

TEAM

10 people, dedicated platform team, moderate Solace/EDA experience

Recommended Architecture

Platform: Event broker service (Cloud), Enterprise class — two instances, one per region. Multi-region from day one — AWS Mumbai (India) + AWS US-East (US).

Engagement Summary

SKILL	STATUS	EXECUTION	ARTIFACTS
<u>Discovery</u>	Complete	5m 21s	<u>1 files</u>
<u>Topic Design</u>	Complete	3m 45s	<u>3 files</u>
<u>Broker Selection</u>	Complete	1m 42s	<u>1 files</u>
<u>Protocol Selection</u>	Complete	1m 40s	<u>1 files</u>
<u>Mesh Design</u>	Complete	1m 21s	<u>2 files</u>
<u>HA/DR</u>	Complete	2m 1s	<u>2 files</u>
<u>Integration</u>	Complete	2m 5s	<u>1 files</u>
<u>Architect Review</u>	Complete	2m 33s	<u>1 files</u>
<u>Ops Review</u>	Complete	2m 33s	<u>1 files</u>
<u>Security Review</u>	Complete	2m 33s	<u>1 files</u>
<u>Dev Review</u>	Complete	2m 33s	<u>1 files</u>
<u>Validation</u>	Complete	41s	<u>1 files</u>
<u>Technical Blueprint</u>	Complete	6m 35s	<u>15 files</u>
<u>Diagrams</u>	Complete	6m 42s	<u>23 files</u>
<u>Business Case</u>	Complete	4m 50s	<u>3 files</u>

SKILL	STATUS	EXECUTION	ARTIFACTS
solace-intake	Complete	1m 59s	3 files

Decisions

DECISION	SKILL	VALUE	RATIONALE
project-type	Discovery	New build	Greenfield payment platform, no existing Solace deployment
delivery-mode	Discovery	Mixed — Guaranteed for payments/fraud/settlement/2FA, Direct for notifications/analytics	PCI-DSS audit trail requires lossless delivery for financial events. Notifications and analytics are ephemeral.
ordering-strategy	Discovery	Per-key partitioned queues with transaction ID as partition key	Per-transaction ordering without global bottleneck across 5,000 TPS
latency-tier	Discovery	Sub-second for hot path (payment auth + fraud check + 2FA)	UPI mandates sub-second authorization. Card networks expect similar. User waiting on payment completion.
topology	Discovery	Multi-region from day one — AWS Mumbai (India) + AWS US-East (US)	India and US data residency is a hard requirement. NPCI must be in India. Card networks serve both markets.
topic-taxonomy	Topic Design	Domain/Noun/Verb/v1/{region}/{property} — 17 Guaranteed, 3 Direct	9 domain namespaces, 20 topics. Region as first property after version for data residency enforcement. Transaction ID in message properties, not topic string.
subscription-strategy	Topic Design	Region-scoped subscriptions. No cross-region payment data over DMR external links.	Every subscription scoped to local region. Selective property subscriptions (riskLevel, severity) instead of broad wildcards. 36 total subscriptions across 12 consumers.
analytics-delivery-mode	Topic Design	Direct messaging for analytics/transaction/mirrored	Analytics mirror at 5K/sec is fan-out data. Per the antipattern guidance, fan-out data uses Direct messaging. Analytics Platform tolerates gaps.
broker-type	Broker Selection	Event broker service (Cloud), Enterprise class — two instances, one per region	10-person team, moderate experience, 4-month timeline, cost-optimized OpEx preference. Sub-second latency (not sub-microsecond) is well within Cloud capability. AWS Mumbai and US-East regions available.
broker-sizing	Broker Selection	Enterprise class per region. Upgrade to Enterprise+ at ~year 3 as volume grows.	5K TPS peak, ~2KB avg payload, 12 systems with 24-120 connections, ~36GB spool for 1hr replay. Enterprise class accommodates. Enterprise+ upgrade path around year 3.
protocol-strategy	Protocol Selection	SMF (8 server-side systems), WebSocket (3 browser apps), MQTT 5.0 (mobile), REST/RDP (notifications)	Hot path servers use SMF for lowest latency and partitioned queue access. Browser apps use WebSocket. Mobile uses MQTT 5.0. Notification service uses REST delivery point. Broker handles all protocol mediation.
encryption-in-transit	Protocol Selection	TLS mandatory on all connections. Non-TLS ports disabled.	PCI-DSS requires encryption for payment data in transit. All protocols support TLS.

DECISION	SKILL	VALUE	RATIONALE
mesh-topology	Mesh Design	Two-site federation via compressed DMR external link. Single broker per region. < 25 msg/sec cross-link.	Two sites, clear data sovereignty boundaries, <5K TPS per region fits single broker. No intra-site cluster needed. External link for reference data only.
cross-region-data-policy	Mesh Design	No payment/fraud/auth/settlement/risk topics cross DMR link. Reference and analytics summaries only.	India (RBI) and US data residency. Payment data never crosses regional boundary. Cross-region traffic limited to reference data and aggregated analytics summaries.
ha-strategy	HA/DR	Cloud Enterprise built-in HA per region. No manual HA configuration.	Cloud Enterprise HA is built-in: three-node redundancy, multi-AZ, auto failover. Team configures client reconnection and idempotent consumers.
dr-strategy	HA/DR	Phase 1: Cloud HA + application-level recovery. Phase 2: replication groups within same data residency boundary.	Data residency prevents cross-region DR. Event broker is not system of record. Core Banking ledger and card networks are authoritative. Full-region failure is rare. 4-month timeline favors Phase 1 simplicity.
replay-log	HA/DR	Replay log enabled, 1hr minimum retention per region.	Enterprise class supports replay. Needed for recovery and reprocessing after failover.
integration-strategy	Integration	Zero Micro-Integrations. All connections via native Solace APIs (SMF, MQTT, WebSocket) + 1 REST delivery point for notifications.	Greenfield platform. All 12 systems are custom applications connecting directly via Solace APIs. No third-party backends needing Micro-Integrations.
analytics-integration	Integration	Direct consumer via Solace API. Kafka bridge (Broker Integrated) as conditional alternative.	Advisory decision defaulted to direct consumer. Kafka bridge available if analytics stack is Kafka-based.

Open Items

14 of 14 items remain open or in progress.

2 BLOCKING

2 HIGH

5 MEDIUM

5 ADVISORY

ID	SEVERITY	DESCRIPTION	SOURCE	RESOLUTION PATH	STATUS
OI-001	BLOCKING	Card network connectivity model — direct ISO 8583 socket connections to Visa/Mastercard/RuPay, or through a payment processor/switch?	Discovery	Confirm with network operations team. Current design assumes direct connections. If processor-mediated, update protocol selection and security model. source	OPEN
OI-002	BLOCKING	NPCI integration model — UPI 2.0 PSP (Payment Service Provider) or TPP (Third Party Provider)?	Discovery	Confirm with payments team. PSP assumed in current design. If TPP, update UPI topic flows (collect vs pay vs mandate), 2FA triggers, and settlement logic. source	OPEN
OI-003	HIGH	Encryption at rest — verify Solace Cloud message spool encryption meets PCI-DSS Requirement 3 for stored cardholder data.	Security Review	Verify with Solace sales/support that Cloud Enterprise spool encryption meets PCI-DSS. If insufficient, implement application-level field encryption for cardholder data before publishing. source	OPEN

ID	SEVERITY	DESCRIPTION	SOURCE	RESOLUTION PATH	STATUS
OI-004	HIGH	ACL profiles — define and provision per-system publish/subscribe access control profiles before production.	Security Review	Provision the ACL profiles documented in the Security Model section of the blueprint. Each of the 12 systems needs explicit publish/subscribe permissions. Verify with integration tests. source	OPEN
OI-005	MEDIUM	DMQ on all production queues — specify dead message queue configuration for every Guaranteed messaging queue.	Validation	Configure DMQ for all Guaranteed messaging queues (not just the RDP queue). Set DMQ message TTL. Add alerting on DMQ depth > 0. source	OPEN
OI-006	MEDIUM	Alert thresholds — finalize and configure monitoring thresholds in Datadog.	Ops Review	Configure the thresholds from the Operational Model section (queue depth, DMQ depth, message rate deviation, connection drops, DMR link status, spool usage, hot path latency) in Datadog. source	OPEN
OI-007	MEDIUM	Operational runbooks — write the 4 runbooks identified in HA/DR design.	Ops Review	Author runbooks for: (1) broker failover procedures, (2) DMQ review and poison message handling, (3) regional failure response, (4) spool exhaustion and capacity management. source	OPEN
OI-008	MEDIUM	Slow consumer mitigation — set per-queue max spool quota and define horizontal scaling policy for hot path consumers.	Ops Review	Set max spool quota on each queue. Use non-exclusive queues for Fraud Detection and 2FA Service to enable horizontal scaling under sustained load spikes. source	OPEN
OI-009	MEDIUM	Notification loss path — Direct messaging notifications have no recovery mechanism if the consumer is down.	Architect Review	Implement application-level retry in source systems (Core Banking, Settlement, Fraud Detection) for lost Direct messaging notifications. Consider upgrading to Guaranteed messaging if notification SLA tightens. source	OPEN
OI-010	ADVISORY	Merchant onboarding — currently ops-managed, self-service deferred to Phase 2.	Discovery	Plan self-service merchant onboarding for Phase 2. No impact on current architecture. source	OPEN
OI-011	ADVISORY	Dispute and chargeback handling — currently batch, event-driven deferred to Phase 2.	Discovery	Plan event-driven dispute/chargeback flow for Phase 2. Will require new topics in the payment domain. source	OPEN
OI-012	ADVISORY	Analytics platform technology — Kafka bridge available if analytics stack changes to Kafka-based.	Integration	If analytics moves to Kafka, use Apache Kafka (Broker Integrated) Micro-Integration. Unidirectional Solace-to-Kafka only. source	OPEN
OI-013	ADVISORY	Replication groups for broker-level DR — deferred to Phase 2.	HA/DR	Evaluate when PCI-DSS auditor requires broker-level DR or business needs < 30-second RTO. Deploy standby broker within same data residency boundary. source	OPEN
OI-014	ADVISORY	Event Portal for schema management and topic governance — not yet adopted.	Dev Review	Evaluate Event Portal adoption for schema registry, topic governance, and event catalog after initial production deployment stabilizes. source	OPEN

Connected Systems (12)

SYSTEM	ROLE	DESCRIPTION
Core Banking Platform	producer / consumer	account data, balance management, ledger updates
Credit Card Processing	producer / consumer	Visa, Mastercard, RuPay direct network integration
UPI Gateway	producer / consumer	NPCI direct integration for UPI payments (pay, collect, mandate flows)
Fraud Detection Engine	consumer / producer	real-time transaction scoring, fraud alerts
Risk Management	consumer / producer	risk scoring, threshold enforcement, regulatory reporting
2FA Service	consumer / producer	OTP and app-based two-factor authentication for payment authorization
Customer Notification Service	consumer	SMS, push notification, email delivery
Merchant Portal	consumer	merchant dashboards, settlement reports, transaction history
Customer-Facing App	producer / consumer	mobile app and web for end customers initiating payments
Internal Operations Dashboard	consumer	ops team monitoring, dispute management, reconciliation
Settlement Engine	consumer / producer	end-of-day and real-time settlement processing
Analytics Platform	consumer	transaction analytics, business intelligence, reporting

DISCOVERY

Discovery Brief: Payment Solutions V1

System landscape

- **Systems:**
 - Core Banking Platform (producer/consumer) — account data, balance management, ledger updates
 - Credit Card Processing (producer/consumer) — Visa, Mastercard, RuPay direct network integration
 - UPI Gateway (producer/consumer) — NPCI direct integration for UPI payments (pay, collect, mandate flows)
 - Fraud Detection Engine (consumer/producer) — real-time transaction scoring, fraud alerts
 - Risk Management (consumer/producer) — risk scoring, threshold enforcement, regulatory reporting
 - 2FA Service (consumer/producer) — OTP and app-based two-factor authentication for payment authorization
 - Customer Notification Service (consumer) — SMS, push notification, email delivery
 - Merchant Portal (consumer) — merchant dashboards, settlement reports, transaction history
 - Customer-Facing App (producer/consumer) — mobile app and web for end customers initiating payments
 - Internal Operations Dashboard (consumer) — ops team monitoring, dispute management, reconciliation
 - Settlement Engine (consumer/producer) — end-of-day and real-time settlement processing
 - Analytics Platform (consumer) — transaction analytics, business intelligence, reporting
- **Existing messaging:** API-based (REST). No existing message broker or event streaming platform.
- **Protocols in play:** REST, MQTT, WebSocket
- **Event types (estimated rates at 5,000 TPS peak):**
 - Payment initiated (~2,000/sec) — card and UPI combined
 - Payment authorized (~1,800/sec) — after fraud check + 2FA
 - Payment captured/settled (~1,500/sec)
 - Fraud check request/response (~2,000/sec) — inline with payment flow
 - 2FA challenge/response (~1,000/sec)
 - Settlement batch events (~100/sec bursts)
 - Customer notifications (~3,000/sec) — SMS, push, email triggers
 - Merchant settlement reports (~50/sec)

- Analytics events (~5,000/sec) — mirrors all transaction events
- **Payload format:** JSON, industry-standard payment schemas (ISO 8583 for card, UPI XML for NPCI)
- **Matched reference architecture:** None — custom architecture. Shares banking-sector traits with Pattern 1 (PCI-DSS, audit trail) but is a payment processing system, not an AI assistant.
- **Micro-Integration availability:**
 - Core Banking: No cataloged MI. Custom integration via REST or JMS.
 - Credit Card Networks (Visa/Mastercard/RuPay): No cataloged MI. Custom integration via proprietary network protocols.
 - NPCI (UPI): No cataloged MI. Custom integration via NPCI API specs.
 - Fraud Detection Engine: No cataloged MI. Custom application consuming from event broker.
 - Datadog: No cataloged MI. Solace Insights provides broker-level metrics exportable to Datadog.
 - Analytics Platform: If Kafka-based, Apache Kafka (Broker Integrated) MI available. Otherwise custom.

Requirements

- **Delivery guarantee:** Mixed — Guaranteed messaging for payment transactions, fraud alerts, 2FA, settlement. Direct messaging for notifications, analytics, dashboard updates.
- **Ordering:** Per-key partitioned queues with transaction ID as partition key
- **Processing guarantee:** At-least-once with idempotent consumers (Solace provides at-least-once via Guaranteed messaging; exactly-once requires application-level idempotency)
- **Latency target:** Sub-second for the hot path (payment auth + fraud check + 2FA)
- **Scale:** 5,000 TPS peak, 10% year-on-year growth (~8,050 TPS in 5 years)
- **Topology:** Multi-region from day one — AWS Mumbai (India) + AWS US-East (US)
- **Data residency:** India payment data stays in India (RBI regulations). US payment data stays in US. Cross-region flows limited to shared reference data and aggregated analytics.

Goals

- **Project type:** New build — greenfield event-driven payment processing system
- **Driver:** Launching a new payment product supporting credit/debit cards and UPI
- **Timeline:** 4 months to production
- **Budget:** Cost-optimized — preference for cloud-managed to reduce ops burden
- **Team:** 10 people, dedicated platform team, moderate Solace/EDA experience
- **Observability:** Datadog (existing, must align)
- **CI/CD:** Jenkins

Open questions

- **[Blocking]** Card network connectivity model — direct ISO 8583 socket connections to Visa/Mastercard/RuPay, or through a payment processor/switch? Affects: protocol selection, Micro-Integration design, security model.
- **[Blocking]** NPCI integration model — UPI 2.0 PSP (Payment Service Provider) model or TPP (Third Party Provider)? Affects: topic design (collect vs pay vs mandate flows), 2FA triggers, settlement logic.
- **[Advisory]** Merchant onboarding flow — self-service or ops-managed? Default assumption: ops-managed initially, self-service in Phase 2.
- **[Advisory]** Dispute and chargeback handling — real-time event-driven or batch? Default assumption: batch processing initially, event-driven in Phase 2.
- **[Advisory]** Analytics platform technology — Kafka-based (use Broker Integrated MI) or direct consumer? Default assumption: direct consumer via Solace client library.
- **[Advisory]** 2FA provider — in-house OTP generation or third-party (Twilio, MSG91)? Default assumption: third-party via REST webhook.

Recommended next steps

- `/solace-plan` — orchestrate the full engagement. The plan will sequence: topic design, broker selection, protocol selection, mesh design (multi-region), HA/DR, integration, then reviews, validation, and blueprint.

- Individual skills if preferred: `/solace-topic-design` for payment event taxonomy, `/solace-broker-select` for cloud-managed sizing.

Antipattern Validation Report: Topic Design

Result: PASS

All 20 topics and 36 subscriptions checked against the Solace Antipattern Library.

CHECK	STATUS	DETAIL
No environment names in topics	✓ PASS	Environment separation via message VPNs, not topic strings.
No tracing/audit IDs in topics	✓ PASS	Transaction ID carried in message properties as partition key, not in topic hierarchy.
Wildcard subscriptions scoped appropriately	✓ PASS	Every subscription scoped to local region. No broad <code>domain/></code> across DMR external links. Risk Management and Ops Dashboard use selective property values instead of <code>></code> .
Properties after version in taxonomy	✓ PASS	All topics follow <code>Domain/Noun/Verb/v1/Properties</code> . Region and secondary properties follow version.
No selector-based filtering	✓ PASS	All routing via topic hierarchy. No JMS selectors or message property filters for routing.
No special characters in published topics	✓ PASS	Template variables (<code>{region}</code> , <code>{paymentMethod}</code> , etc.) are substituted at publish time. Published strings contain only alphanumeric and <code>/</code> .
No mixed delivery modes on same topic	✓ PASS	17 topics Guaranteed end-to-end, 3 topics Direct end-to-end. No topic has mixed mode subscribers.
Fan-out data uses Direct messaging	✓ PASS	Analytics mirror (~5K/sec) and notification delivery confirmations use Direct. No per-subscriber acks on high-volume fan-out.
Naming conventions correct	✓ PASS	"Guaranteed messaging" and "Direct messaging" used throughout. No QoS references. No connector/adaptor terminology.

Notes

- The Customer Notification Service subscribes to `auth/twoFactor/failed` and `payment/transaction/declined` topics which are Guaranteed. Since Customer Notification is a Direct messaging consumer by design, these subscriptions use topic subscriptions without queue binding. The publisher still publishes Guaranteed, and queue-bound consumers (Core Banking) receive via Guaranteed. This is not the "mixed delivery mode" antipattern because each consumer's subscription type is consistent with its delivery mode requirement. The notification path is explicitly designed to tolerate loss.
- No negative subscriptions (`!` prefix) are needed. Selective subscriptions by property value (e.g., `medium`, `high` for `riskLevel`) achieve the same filtering without the complexity of negation.

Topic Taxonomy: Payment Solutions V1

Topic naming convention

`Domain/Noun/Verb/v1/Properties` where properties are ordered least-specific to most-specific.

- Region** (`IN`, `US`) is always the first property. It is the primary routing dimension for data residency enforcement.
- Secondary properties** vary by domain: `paymentMethod` (`card`, `upi`), `riskLevel` (`low`, `medium`, `high`), `method` (`otp`, `app`), `channel` (`sms`, `push`, `email`), `network` (`visa`, `mastercard`, `rupay`), `flowType` (`pay`, `collect`, `mandate`), `severity` (`low`, `medium`, `high`, `critical`), `merchantId`, `eventType`.
- Hard limits: 250 characters max, 128 levels max. camelCase used throughout.

Topic catalog

#	TOPIC STRING	DELIVERY MODE	PRODUCER	CONSUMER(S)	RATIONALE
1	payment/transaction/initiated/v1/{region}/{paymentMethod}	Guaranteed	Customer-Facing App	Core Banking, Fraud Detection Engine	Hot path entry point. Region + method enables per-broker, per-method filtering.
2	fraud/check/requested/v1/{region}/{paymentMethod}	Guaranteed	Core Banking	Fraud Detection Engine	Inline fraud scoring on every transaction before authorization.
3	fraud/check/completed/v1/{region}/{riskLevel}	Guaranteed	Fraud Detection Engine	Core Banking, Risk Management	riskLevel property lets Risk Management subscribe selectively to medium+high only.
4	auth/twoFactor/challenged/v1/{region}/{method}	Guaranteed	Core Banking	2FA Service	method = otp or app. 2FA Service routes to correct verification channel.
5	auth/twoFactor/verified/v1/{region}/{method}	Guaranteed	2FA Service	Core Banking	Success path. Core Banking resumes payment authorization flow.
6	auth/twoFactor/failed/v1/{region}/{method}	Guaranteed	2FA Service	Core Banking, Customer Notification Service	Failure triggers decline flow and customer notification.
7	payment/transaction/authorized/v1/{region}/{paymentMethod}	Guaranteed	Core Banking	Credit Card Processing / UPI Gateway, Settlement Engine, Merchant Portal, Customer Notification	Fan-out to downstream processors. paymentMethod routes to correct network handler.

#	TOPIC STRING	DELIVERY MODE	PRODUCER	CONSUMER(S)	RATIONALE
8	payment/transaction/declined/v1/{region}/{paymentMethod}	Guaranteed	Core Banking	Customer-Facing App, Customer Notification, Analytics	Decline events carry reason codes in payload for dispute tracking.
9	card/networkResponse/received/v1/{region}/{network}	Guaranteed	Credit Card Processing	Core Banking	network = visa, mastercard, rupay. ISO 8583 response from card network.
10	upi/networkResponse/received/v1/{region}/{flowType}	Guaranteed	UPI Gateway	Core Banking	flowType = pay, collect, mandate per NPCI specification. India region only.
11	payment/transaction/captured/v1/{region}/{paymentMethod}	Guaranteed	Settlement Engine	Core Banking, Merchant Portal, Analytics	Capture confirmation after network-level settlement.
12	payment/transaction/settled/v1/{region}/{paymentMethod}	Guaranteed	Settlement Engine	Core Banking, Merchant Portal	Final settlement. Audit trail anchor for PCI-DSS compliance.
13	settlement/batch/initiated/v1/{region}	Guaranteed	Settlement Engine	Core Banking, Internal Ops Dashboard	End-of-day or scheduled batch processing start.
14	settlement/batch/completed/v1/{region}	Guaranteed	Settlement Engine	Merchant Portal, Internal Ops Dashboard	Batch completion with summary metrics.
15	settlement/report/generated/v1/{region}/{merchantId}	Guaranteed	Settlement Engine	Merchant Portal	Per-merchant settlement report. merchantId is most-specific property.
16	fraud/alert/raised/v1/{region}/{severity}	Guaranteed	Fraud Detection Engine	Risk Management, Internal Ops Dashboard	severity = low, medium, high, critical. Ops subscribes to high+critical.

#	TOPIC STRING	DELIVERY MODE	PRODUCER	CONSUMER(S)	RATIONALE
17	risk/assessment/completed/v1/{region}/{riskLevel}	Guaranteed	Risk Management	Core Banking, Internal Ops Dashboard	Post-fraud-check risk scoring for authorization decisions.
18	notification/customer/requested/v1/{region}/{channel}	Direct	Core Banking, Settlement Engine, Fraud Detection	Customer Notification Service	channel = sms, push, email. Direct: loss triggers retry from source app.
19	notification/customer/delivered/v1/{region}/{channel}	Direct	Customer Notification Service	Analytics	Delivery confirmation for tracking delivery rates.
20	analytics/transaction/mirrored/v1/{region}/{eventType}	Direct	All producing systems	Analytics Platform	eventType mirrors source event name. Direct: analytics tolerates gaps.

Domain breakdown

DOMAIN	TOPICS	DELIVERY MODE	VOLUME (PEAK)
payment	5 topics (#1, 7, 8, 11, 12)	Guaranteed	~7,000/sec combined
fraud	3 topics (#2, 3, 16)	Guaranteed	~4,050/sec combined
auth	3 topics (#4, 5, 6)	Guaranteed	~2,000/sec combined
card	1 topic (#9)	Guaranteed	~1,200/sec
upi	1 topic (#10)	Guaranteed	~800/sec
settlement	3 topics (#13, 14, 15)	Guaranteed	~250/sec
risk	1 topic (#17)	Guaranteed	~500/sec
notification	2 topics (#18, 19)	Direct	~6,000/sec combined
analytics	1 topic (#20)	Direct	~5,000/sec

Delivery mode summary

- Guaranteed messaging (17 topics):** All payment processing, fraud, 2FA, settlement, risk events. PCI-DSS audit trail requires lossless delivery. At-least-once with application-level idempotency using transaction ID.
- Direct messaging (3 topics):** Customer notifications, notification delivery confirmations, analytics mirror. These are ephemeral or recoverable from source.

Data residency enforcement

Region is the first property after version in every topic. This enables:

- Mumbai broker: subscribes only to topics with `IN` region
- US-East broker: subscribes only to topics with `US` region

- No payment transaction data crosses DMR external links
- Cross-region flows limited to shared reference data (not in this topic catalog; handled at the application layer)

Ordering strategy

Per-key partitioned queues with transaction ID as partition key. The partition key is carried in message properties (not in the topic string). All events for a single payment transaction are processed in order by the same consumer instance within a consumer group.

Wildcard Subscription Map: Payment Solutions V1

Subscription design principles

1. Every subscription is scoped to the local region (`IN` or `US`). No consumer subscribes across regions.
2. Where a consumer needs only a subset of a property's values, explicit subscriptions replace `>` wildcards. This keeps bandwidth tight, especially for DMR external links.
3. Guaranteed messaging consumers bind to partitioned queues. Direct messaging consumers use topic subscriptions without queue binding.

Consumer subscription map

Core Banking Platform

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
payment/transaction/initiated/v1/{localRegion}/>	All local payment initiations	Partitioned (txnId)	Hot path entry.
fraud/check/completed/v1/{localRegion}/>	All fraud check results	Partitioned (txnId)	Resumes auth flow.
auth/twoFactor/verified/v1/{localRegion}/>	2FA success	Partitioned (txnId)	Resumes auth flow.
auth/twoFactor/failed/v1/{localRegion}/>	2FA failure	Partitioned (txnId)	Triggers decline.
card/networkResponse/received/v1/{localRegion}/>	Card network responses	Partitioned (txnId)	ISO 8583 results.
upi/networkResponse/received/v1/{localRegion}/>	UPI responses	Partitioned (txnId)	India region only.
payment/transaction/captured/v1/{localRegion}/>	Capture confirmations	Partitioned (txnId)	Ledger update.
payment/transaction/settled/v1/{localRegion}/>	Settlement confirmations	Partitioned (txnId)	Final ledger.
settlement/batch/initiated/v1/{localRegion}	Batch starts	Non-exclusive	Ops awareness.
risk/assessment/completed/v1/{localRegion}/>	Risk scores	Partitioned (txnId)	For authorization decisions.

Total subscriptions: 10 — Core Banking is the central orchestrator of the payment flow.

Fraud Detection Engine

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
payment/transaction/initiated/v1/{localRegion}/>	All local initiations	Partitioned (txnId)	Scores every transaction.
fraud/check/requested/v1/{localRegion}/>	Explicit check requests	Partitioned (txnId)	For re-check scenarios.

Total subscriptions: 2

Risk Management

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
fraud/check/completed/v1/{localRegion}/medium	Medium-risk results	Non-exclusive	Selective. Skips low.
fraud/check/completed/v1/{localRegion}/high	High-risk results	Non-exclusive	Needs human review.

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
fraud/alert/raised/v1/{localRegion}/>	All fraud alerts	Non-exclusive	All severities.

Total subscriptions: 3 — Deliberately excludes low riskLevel to reduce processing noise.

2FA Service

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
auth/twoFactor/challenged/v1/{localRegion}/>	All 2FA challenges	Partitioned (txnId)	Routes by method (otp/app).

Total subscriptions: 1

Credit Card Processing

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
payment/transaction/authorized/v1/{localRegion}/card	Card authorizations only	Partitioned (txnId)	Filtered to card. UPI excluded.

Total subscriptions: 1 — Hard filter on card paymentMethod. No wildcards.

UPI Gateway

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
payment/transaction/authorized/v1/IN/upi	UPI authorizations	Partitioned (txnId)	India-only. Hard filter IN + upi.

Total subscriptions: 1 — Hardcoded to India region. UPI does not exist in the US market.

Settlement Engine

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
payment/transaction/authorized/v1/{localRegion}/>	All local authorizations	Partitioned (txnId)	Feeds settlement pipeline.

Total subscriptions: 1

Customer Notification Service

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
notification/customer/requested/v1/{localRegion}/>	All notification requests	Topic subscription (Direct)	Routes by channel.
auth/twoFactor/failed/v1/{localRegion}/>	2FA failures	Topic subscription (Direct)	Customer alert on failure.
payment/transaction/declined/v1/{localRegion}/>	Payment declines	Topic subscription (Direct)	Customer alert on decline.

Total subscriptions: 3 — Direct messaging for notifications. No queue binding.

Merchant Portal

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
payment/transaction/authorized/v1/{localRegion}/>	Authorizations	Non-exclusive	Dashboard updates.
payment/transaction/captured/v1/{localRegion}/>	Captures	Non-exclusive	
payment/transaction/settled/v1/{localRegion}/>	Settlements	Non-exclusive	
settlement/batch/completed/v1/{localRegion}	Batch completion	Non-exclusive	

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
settlement/report/generated/v1/{localRegion}/>	Per-merchant reports	Non-exclusive	Filter by merchantId at app level.

Total subscriptions: 5

Customer-Facing App

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
payment/transaction/declined/v1/{localRegion}/>	Payment declines	Topic subscription	WebSocket/MQTT push to mobile/web.

Total subscriptions: 1 — Real-time decline notification to end user.

Internal Operations Dashboard

SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
settlement/batch/initiated/v1/{localRegion}	Batch start	Non-exclusive	Batch monitoring.
settlement/batch/completed/v1/{localRegion}	Batch completion	Non-exclusive	
fraud/alert/raised/v1/{localRegion}/high	High fraud alerts	Non-exclusive	Not > . Selective.
fraud/alert/raised/v1/{localRegion}/critical	Critical fraud alerts	Non-exclusive	Pager-worthy.
risk/assessment/completed/v1/{localRegion}/>	Risk scores	Non-exclusive	

Total subscriptions: 5 — Ops Dashboard uses selective severity subscriptions, not broad wildcards.

Analytics Platform

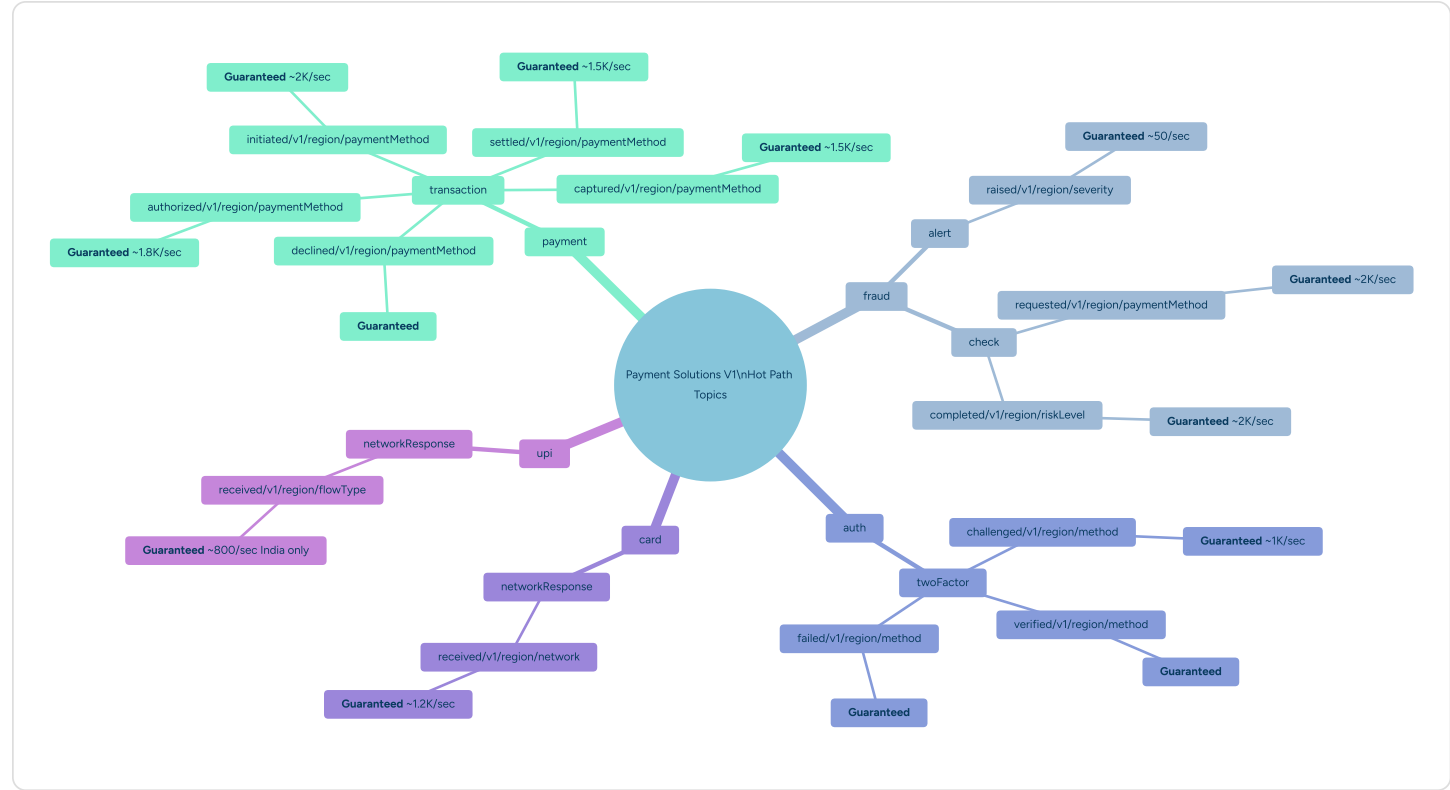
SUBSCRIPTION	EVENTS MATCHED	QUEUE TYPE	NOTES
analytics/transaction/mirrored/v1/{localRegion}/>	All mirrored events	Topic subscription (Direct)	~5K/sec. Direct messaging.
notification/customer/delivered/v1/{localRegion}/>	Delivery confirmations	Topic subscription (Direct)	
payment/transaction/declined/v1/{localRegion}/>	Declines	Topic subscription (Direct)	

Total subscriptions: 3 — All Direct messaging. Analytics tolerates gaps.

DMR external link subscription hygiene

- **No cross-region payment subscriptions.** Each broker subscribes only to its local region.
- **Cross-region flows (future):** If aggregated analytics or reference data needs to cross regions, add explicit narrow subscriptions. Never `analytics/>` across the link.
- **UPI Gateway** is hardcoded to `IN` . It will never appear on the US-East broker.
- **Total unique subscriptions across all consumers: 36** (18 on India broker, 18 on US broker, minus UPI which is India-only).

topic hierarchy



Queue Subscriptions Detail

Companion to `queue-subscriptions.mermaid` and `queue-subscriptions-02-operations.mermaid` . Full subscription patterns, wildcard expressions, and partition keys.

Core Banking Platform (10 subscriptions)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	PARTITION KEY
q-core-banking-initiated	payment/transaction/initiated/v1/{localRegion}/>	Partitioned	txnId
q-core-banking-fraud	fraud/check/completed/v1/{localRegion}/>	Partitioned	txnId
q-core-banking-2fa-ok	auth/twoFactor/verified/v1/{localRegion}/>	Partitioned	txnId
q-core-banking-2fa-fail	auth/twoFactor/failed/v1/{localRegion}/>	Partitioned	txnId
q-core-banking-card-resp	card/networkResponse/received/v1/{localRegion}/>	Partitioned	txnId
q-core-banking-upi-resp	upi/networkResponse/received/v1/{localRegion}/>	Partitioned	txnId
q-core-banking-captured	payment/transaction/captured/v1/{localRegion}/>	Partitioned	txnId
q-core-banking-settled	payment/transaction/settled/v1/{localRegion}/>	Partitioned	txnId
q-core-banking-batch	settlement/batch/initiated/v1/{localRegion}	Non-exclusive	n/a
q-core-banking-risk	risk/assessment/completed/v1/{localRegion}/>	Partitioned	txnId

Fraud Detection Engine (2 subscriptions)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	PARTITION KEY
q-fraud-initiated	payment/transaction/initiated/v1/{localRegion}/>	Partitioned	txnId
q-fraud-requested	fraud/check/requested/v1/{localRegion}/>	Partitioned	txnId

2FA Service (1 subscription)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	PARTITION KEY
q-2fa-challenged	auth/twoFactor/challenged/v1/{localRegion}/>	Partitioned	txnId

Credit Card Processing (1 subscription)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	PARTITION KEY
q-card-authorized	payment/transaction/authorized/v1/{localRegion}/card	Partitioned	txnId

Hard filter: card paymentMethod only. No wildcards.

UPI Gateway (1 subscription)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	PARTITION KEY
q-upi-authorized	payment/transaction/authorized/v1/IN/upi	Partitioned	txnId

Hardcoded to India region and UPI payment method. India-only deployment.

Settlement Engine (1 subscription)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	PARTITION KEY
q-settlement-authorized	payment/transaction/authorized/v1/{localRegion}/>	Partitioned	txnId

Risk Management (3 subscriptions)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	PARTITION KEY
q-risk-fraud-medium	fraud/check/completed/v1/{localRegion}/medium	Non-exclusive	n/a
q-risk-fraud-high	fraud/check/completed/v1/{localRegion}/high	Non-exclusive	n/a
q-risk-alerts	fraud/alert/raised/v1/{localRegion}/>	Non-exclusive	n/a

Deliberately excludes low riskLevel to reduce processing noise.

Merchant Portal (5 subscriptions)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	PARTITION KEY
q-merchant-authorized	payment/transaction/authorized/v1/{localRegion}/>	Non-exclusive	n/a
q-merchant-captured	payment/transaction/captured/v1/{localRegion}/>	Non-exclusive	n/a
q-merchant-settled	payment/transaction/settled/v1/{localRegion}/>	Non-exclusive	n/a
q-merchant-batch	settlement/batch/completed/v1/{localRegion}	Non-exclusive	n/a
q-merchant-reports	settlement/report/generated/v1/{localRegion}/>	Non-exclusive	n/a

Internal Ops Dashboard (5 subscriptions)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	PARTITION KEY
q-ops-batch-start	settlement/batch/initiated/v1/{localRegion}	Non-exclusive	n/a
q-ops-batch-done	settlement/batch/completed/v1/{localRegion}	Non-exclusive	n/a
q-ops-fraud-high	fraud/alert/raised/v1/{localRegion}/high	Non-exclusive	n/a
q-ops-fraud-critical	fraud/alert/raised/v1/{localRegion}/critical	Non-exclusive	n/a
q-ops-risk	risk/assessment/completed/v1/{localRegion}/>	Non-exclusive	n/a

Uses selective severity subscriptions (high + critical only), not > wildcards.

Customer Notification Service (3 subscriptions, Direct messaging)

QUEUE NAME	SUBSCRIPTION PATTERN	TYPE	NOTES
q-notification-rdp	notification/customer/requested/v1/{localRegion}/>	RDP queue	Routes by channel
(same queue)	auth/twoFactor/failed/v1/{localRegion}/>	RDP queue	2FA failure alerts
(same queue)	payment/transaction/declined/v1/{localRegion}/>	RDP queue	Decline alerts

REST delivery point: rdp-notification-service . Broker pushes via HTTPS webhook.

Customer-Facing App (1 subscription)

SUBSCRIPTION PATTERN	TRANSPORT	NOTES
payment/transaction/declined/v1/{localRegion}/>	MQTT / WebSocket topic sub	Real-time decline push to end user

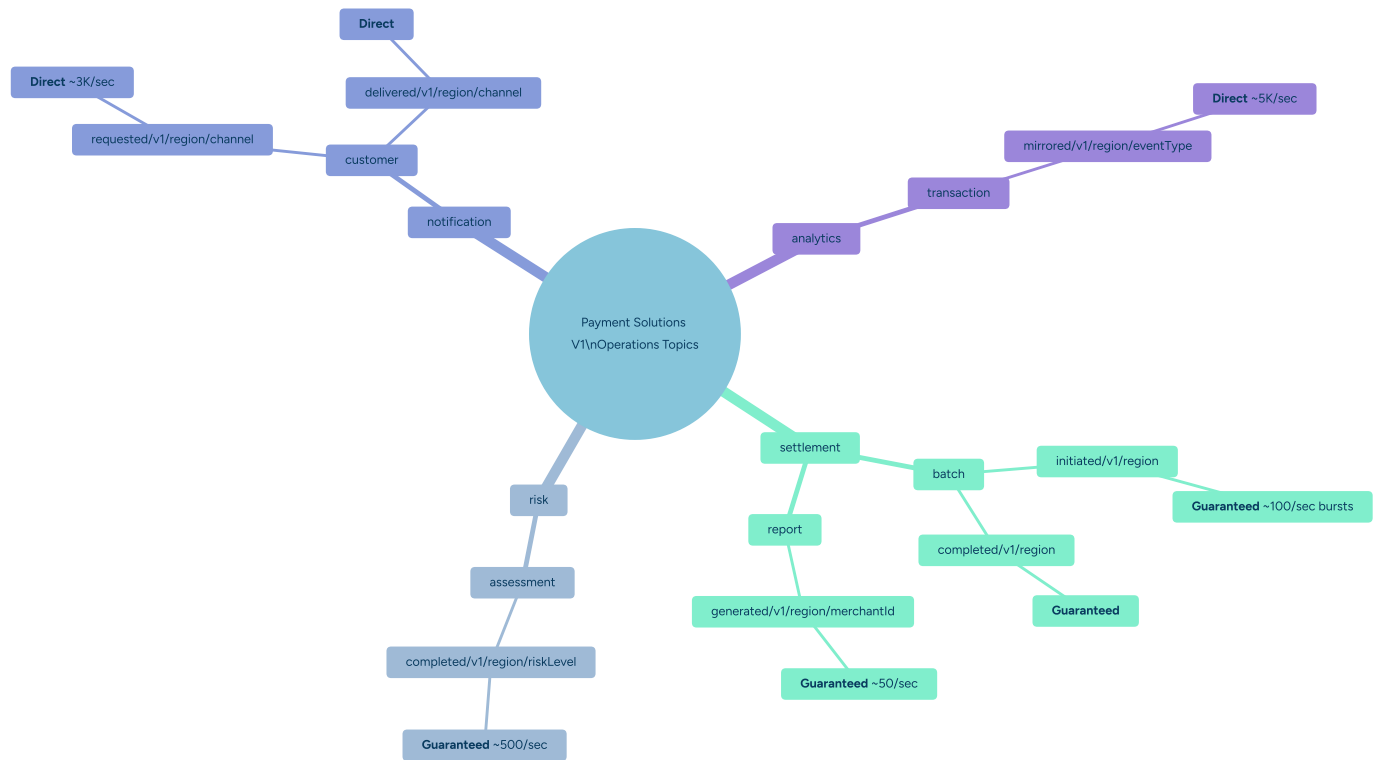
Analytics Platform (3 subscriptions, all Direct messaging)

SUBSCRIPTION PATTERN	TRANSPORT	NOTES
analytics/transaction/mirrored/v1/{localRegion}/>	SMF topic sub	~5K/sec mirror stream
notification/customer/delivered/v1/{localRegion}/>	SMF topic sub	Delivery rate tracking
payment/transaction/declined/v1/{localRegion}/>	SMF topic sub	Decline analytics

Summary

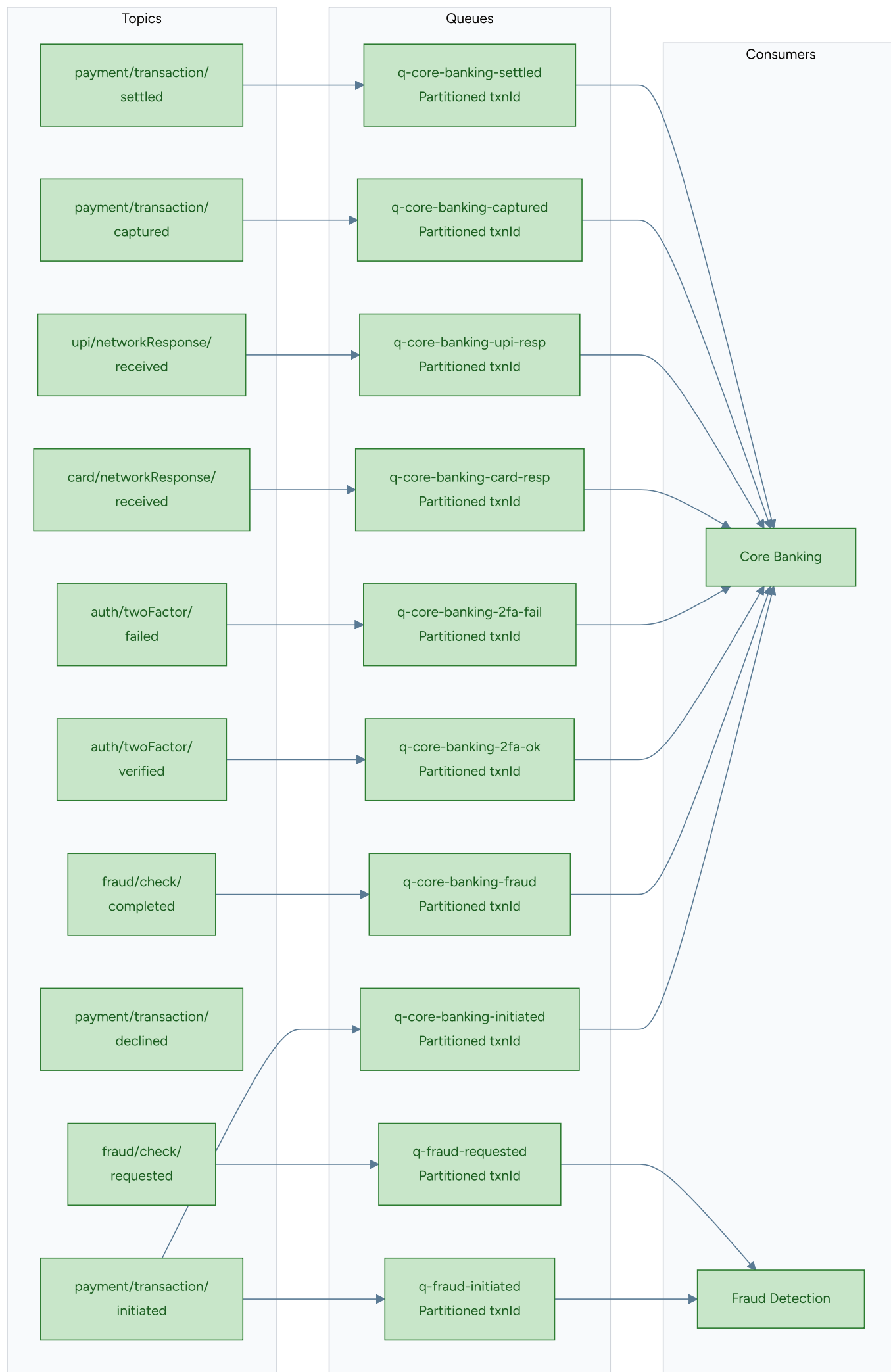
METRIC	COUNT
Total unique subscriptions	36
Partitioned queues	~10 per region
Non-exclusive queues	~13 per region
RDP queues	1 per region
Topic subscriptions (no queue)	7 (Direct messaging consumers)

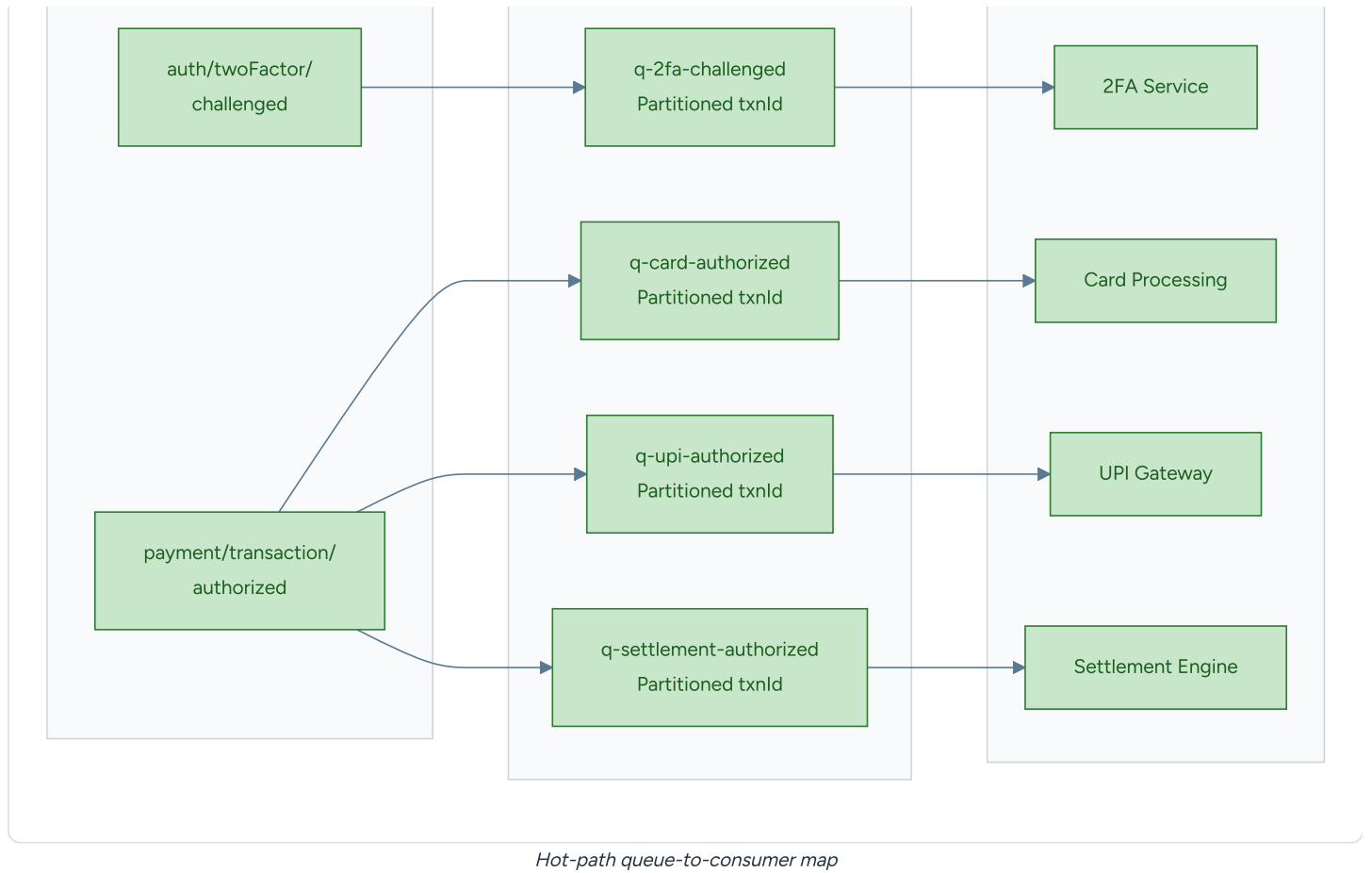
topic hierarchy O2 operations



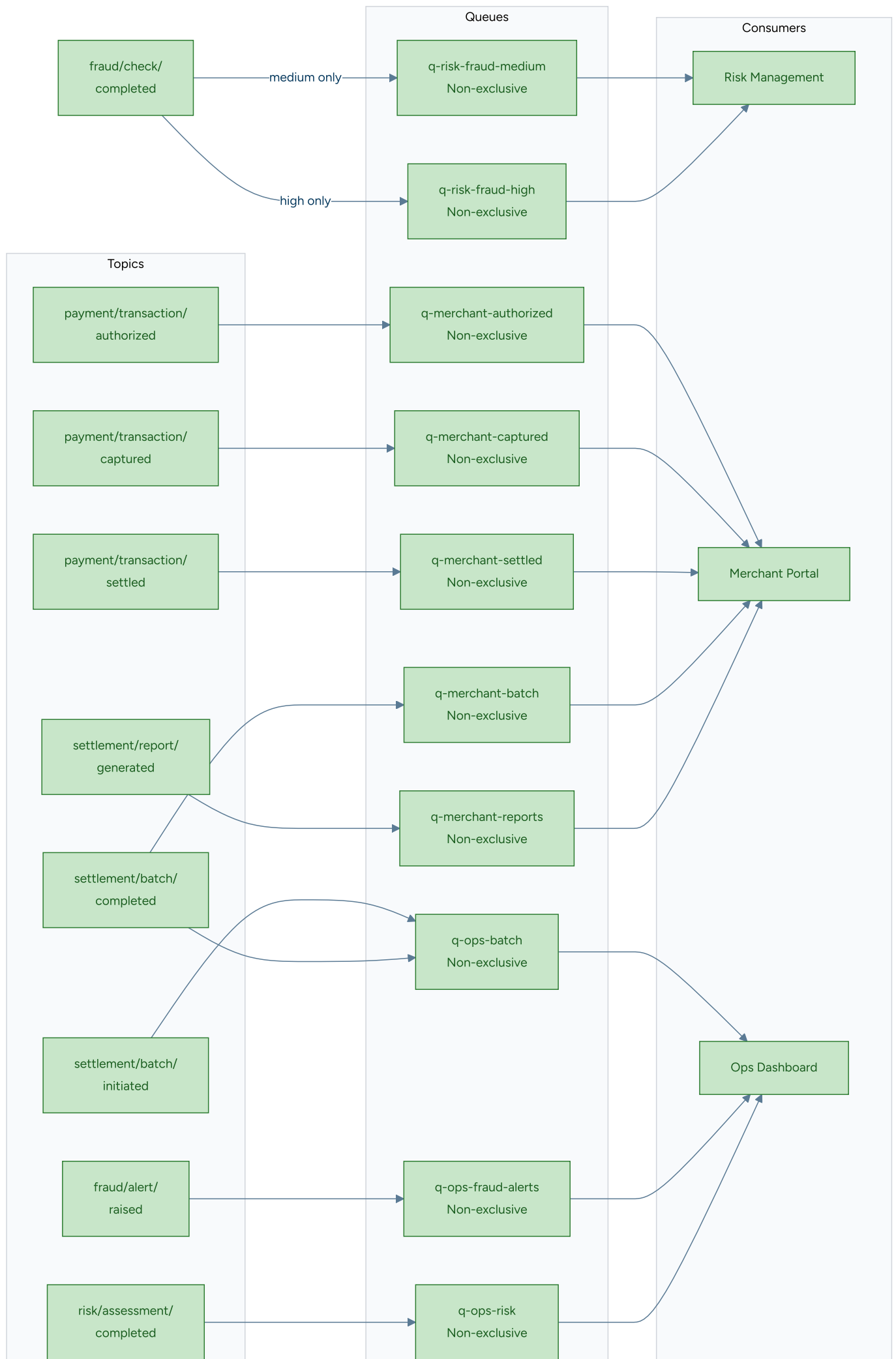
Operations mindmap: settlement, risk, notification, analytics

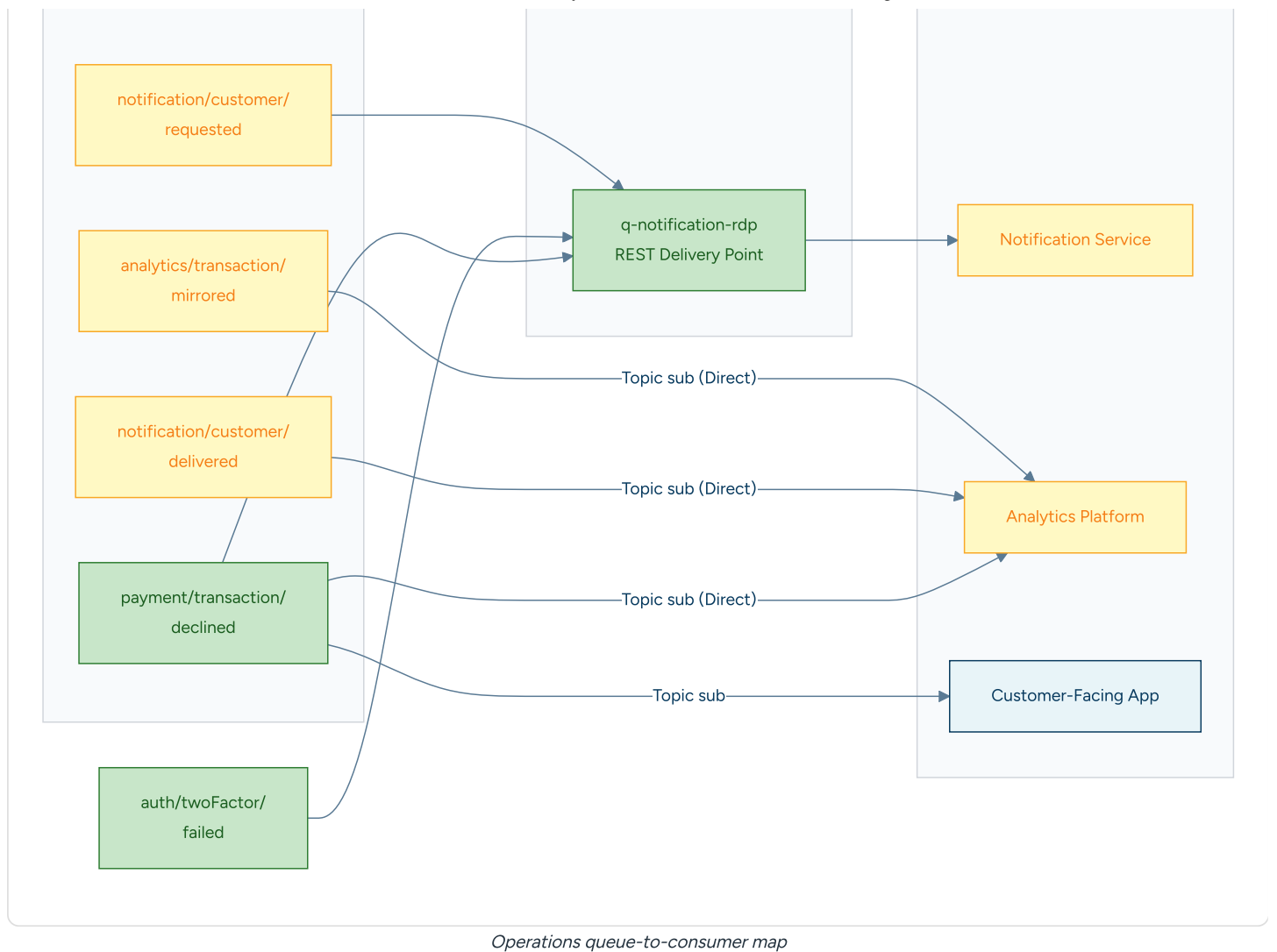
queue subscriptions





queue subscriptions 02 operations





BROKER SELECTION

Broker Selection: Payment Solutions V1

Recommendation: Event Broker Service (Cloud), Enterprise Class

Two event broker services, one per region:

- **AWS ap-south-1 (Mumbai)** — India payment data, UPI/NPCI integration, India card processing
- **AWS us-east-1 (US East)** — US payment data, US card processing

Connected via DMR external links for cross-region reference data sharing.

Why Cloud, not Software or Appliance

FACTOR	DECISION DRIVER
Team capacity	10 people, moderate Solace experience, 4-month timeline. Cannot absorb broker infrastructure ops.
Latency	Sub-second target. Cloud event broker services deliver low-millisecond latency. Appliance sub-microsecond is unnecessary.
Data residency	Solace Cloud supports AWS Mumbai and US-East regions. Each broker holds only its region's data.
Budget	Cost-optimized, OpEx preferred. Cloud subscription aligns.
HA	Built-in for Enterprise class. Three-node redundancy managed by Solace. No self-managed failover config.
DMR	Click-to-Connect wizard for external links between Mumbai and US-East.
Time to production	Provision in minutes. Software broker requires weeks for K8s + HA. Appliance requires months for hardware.

Three-way comparison

CRITERION	EVENT BROKER SERVICE (CLOUD)	SOFTWARE EVENT BROKER	APPLIANCE EVENT BROKER
Operations	Fully managed	Self-managed (K8s/Docker)	Self-managed hardware
Latency	Low millisecond	Host-dependent	Sub-microsecond
Data residency	Solace Cloud regions	Your infrastructure	Your data center
HA	Built-in (Enterprise)	Self-configured pairs	Self-configured
DMR setup	Click-to-Connect	Manual	Manual
Cost model	OpEx (subscription)	OpEx/CapEx (license + infra)	CapEx (hardware + license)
Team burden	Lowest	Medium	Highest
PCI-DSS	SOC 2 Type II (verify PCI scope with Solace)	You own infrastructure compliance	You own compliance
Time to production	Minutes	Weeks	Months

Sizing

Per-region event broker service

PARAMETER	ESTIMATE	NOTES
Service class	Enterprise	HA, DMR, replay, sufficient spool. Developer is evaluation-only.
Peak TPS	5,000 (combined across all topics)	10% YoY growth to ~8,050 in 5 years.
Avg payload	~2 KB	JSON payment events. ISO 8583/UPI XML payloads may be larger.
Peak throughput	~10 MB/sec	5K TPS x 2 KB.
Spool for 1hr replay	~36 GB	10 MB/sec x 3600 sec. Verify plan limits.
Connections	24-120 per region	12 systems x 2-10 instances.
Queues	~40 per region	36 subscriptions, some shared across queues.
Partitioned queues	~10 per region	Hot path queues with transaction ID partition key.

Growth path

YEAR	EST. TPS	ACTION
1	5,000	Enterprise class. Baseline.
2	5,500	Monitor spool and connection usage.
3	6,050	Evaluate Enterprise+ if approaching limits.
4	6,655	Upgrade to Enterprise+ if needed.
5	8,050	Enterprise+ with headroom.

DMR external link sizing

PARAMETER	ESTIMATE
Cross-region topics	Reference data only. No payment transactions.
Cross-region volume	< 100 msg/sec (aggregated analytics summaries, shared config).
Compression	Recommended. Reduces WAN bandwidth.
Bandwidth	< 1 MB/sec. Minimal.

PCI-DSS compliance note

Solace Cloud is SOC 2 Type II certified. PCI-DSS compliance for the payment system spans the entire stack: application code, broker configuration, network security, encryption, access controls, audit logging. The broker is one component. Verify Solace Cloud's specific PCI-DSS compliance scope with Solace sales before production deployment. (Architectural inference, not from Solace docs -- verify before external use.)

Operational benefits

- **Solace Insights** provides broker-level metrics exportable to Datadog (the team's existing observability stack).
- **Message replay** is available on Enterprise class for recovery and reprocessing scenarios.
- **Distributed Tracing** (OpenTelemetry) for end-to-end payment flow visibility. Requires product key.
- **Config-Sync** across HA nodes is automatic in Cloud.
- **Upgrades and patching** managed by Solace. Team focuses on application development.

PROTOCOL SELECTION

Protocol Selection Map: Payment Solutions V1

Protocol summary

PROTOCOL	SYSTEMS USING IT	COUNT
SMF (Solace native)	Core Banking, Credit Card Processing, UPI Gateway, Fraud Detection, Risk Management, 2FA Service, Settlement Engine, Analytics Platform	8
WebSocket (Solace JavaScript API)	Merchant Portal, Customer-Facing App (web), Internal Ops Dashboard	3
MQTT 5.0	Customer-Facing App (mobile)	1
REST (via REST delivery point)	Customer Notification Service	1

Selection rationale per system

SYSTEM	PROTOCOL	DELIVERY MODE	RATIONALE
Core Banking Platform	SMF (Java/Python API)	Guaranteed	Hot path orchestrator. Lowest latency. Partitioned queue access. Full Solace feature set required.
Credit Card Processing	SMF (Java API)	Guaranteed	Hot path. Sub-second. Server-side Java bridging to card networks (ISO 8583).
UPI Gateway	SMF (Java API)	Guaranteed	Hot path. Sub-second. Server-side Java bridging to NPCI. India region only.
Fraud Detection Engine	SMF (Java/Python API)	Guaranteed	Hot path inline scoring. Partitioned queue consumer for per-transaction ordering.
Risk Management	SMF (Java API)	Guaranteed	Server-side. Selective risk level subscriptions. Consistent with server-side protocol choice.
2FA Service	SMF (Java API)	Guaranteed	Hot path. Challenge/response latency within payment authorization budget.
Customer Notification Service	REST (RDP)	Direct	Makes outbound HTTP calls to providers (SMS, push, email). REST delivery point pushes events as webhooks. Simplest integration.
Merchant Portal	WebSocket (JS API)	Guaranteed	Browser-based merchant dashboard. Real-time transaction updates via WebSocket push. Queue-backed subscriptions for Guaranteed delivery.
Customer-Facing App (mobile)	MQTT 5.0	Mixed	Lightweight, battery-efficient. Session persistence across reconnects. QoS 1 for payment initiation (maps to Guaranteed). QoS 0 for status updates (maps to Direct).
Customer-Facing App (web)	WebSocket (JS API)	Mixed	Browser-based. Real-time payment status. Same topic subscriptions as mobile, WebSocket transport.
Internal Ops Dashboard	WebSocket (JS API)	Guaranteed	Browser-based monitoring. Real-time fraud alerts and settlement status.
Settlement Engine	SMF (Java API)	Guaranteed	Server-side batch and real-time settlement. Partitioned queue access.

SYSTEM	PROTOCOL	DELIVERY MODE	RATIONALE
Analytics Platform	SMF (Java/Python API)	Direct	High-volume consumer (~5K/sec). SMF maximizes throughput. Direct messaging, no queue binding. Alternative: Kafka bridge if Kafka-based analytics.

Protocol mediation paths

The Solace event broker handles protocol mediation transparently. A message published via one protocol reaches subscribers on any other supported protocol.

PRODUCER PROTOCOL	CONSUMER PROTOCOL	PATH	EXAMPLE FLOW
SMF	WebSocket	Broker mediates	Core Banking → Merchant Portal dashboard
SMF	MQTT	Broker mediates	Core Banking → Customer-Facing App (mobile)
SMF	REST (RDP)	Broker pushes via webhook	Settlement Engine → Customer Notification Service
MQTT	SMF	Broker mediates	Customer-Facing App (mobile) → Core Banking
WebSocket	SMF	Broker mediates	Customer-Facing App (web) → Core Banking

Feature asymmetry notes

FEATURE	SMF	MQTT	WEBSOCKET	REST (RDP)
Guaranteed messaging	Yes	QoS 1/2 mapped	Yes	Yes (via queue + RDP)
Direct messaging	Yes	QoS 0 mapped	Yes	Yes
Partitioned queues	Yes	No	No	No
Negative subscriptions	Yes	No	No	No
Topic wildcards	* and >	+ and # (mapped)	* and >	N/A (queue-bound)
Session persistence	Yes	Clean/persistent session	Reconnect via API	N/A

Implications:

- Partitioned queue consumers (Core Banking, Fraud Detection, Settlement Engine, Credit Card Processing, UPI Gateway, 2FA Service) must use SMF. MQTT and WebSocket clients cannot bind to partitioned queues.
- Negative subscriptions are not needed in this design. Selective property-level subscriptions achieve the same filtering.
- MQTT wildcard syntax (+ / #) is mapped automatically by the broker to Solace wildcards (* / >).

REST delivery point configuration

The Customer Notification Service receives events via REST delivery point (webhook):

PARAMETER	VALUE
RDP name	rdp-notification-service
Queue binding	Queue subscribed to notification/customer/requested/v1/{localRegion}/>
REST consumer URL	Notification service HTTP endpoint (per environment)
Authentication	TLS client certificate or OAuth (per PCI-DSS requirements)
Retry	Broker retries on non-2xx. Exponential backoff.
DMQ	Configure for failed deliveries after retry exhaustion.

PARAMETER	VALUE
Idempotency	Notification service must handle duplicate deliveries (at-least-once).

Note: REST delivery points deliver from a Guaranteed messaging queue. Even though notification topics use Direct messaging from the publisher side, the RDP queue provides buffering and retry. This hybrid is intentional: Direct at the publisher (loss-tolerant source), Guaranteed at the delivery point (reliable webhook delivery to the notification service).

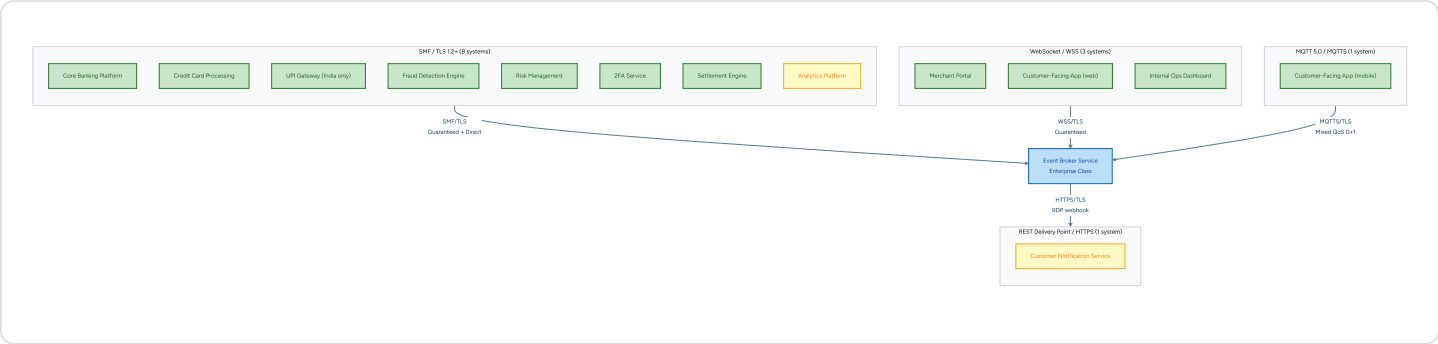
Encryption in transit

All protocols support TLS encryption. PCI-DSS requires encryption in transit for payment data.

PROTOCOL	TLS SUPPORT	CONFIGURATION
SMF	TLS 1.2+	Solace API <code>secure</code> connection properties.
MQTT	TLS 1.2+	MQTTS (port 8883).
WebSocket	WSS (TLS)	Secure WebSocket.
REST (RDP)	HTTPS	TLS for webhook delivery.

All connections to the event broker service must use TLS. Non-TLS ports should be disabled in broker configuration.

protocol stack



Protocol per component with classDef styles

DMR Topology: Payment Solutions V1

Topology pattern

Multi-site federation via DMR external links. Two event broker services connected by a single compressed external link.

- **No intra-site cluster.** Single broker per region handles peak throughput.
- **No full mesh.** Only 2 sites. Point-to-point external link.
- **Minimal cross-region traffic.** Data residency enforced at topic and subscription level.

Site inventory

SITE	CLOUD/REGION	BROKER	SYSTEMS	TRAFFIC
India	AWS ap-south-1 (Mumbai)	Event broker service, Enterprise class	All 12 systems (including UPI Gateway, India-only)	India payment transactions, UPI, India card processing
US	AWS us-east-1 (Virginia)	Event broker service, Enterprise class	11 systems (no UPI Gateway)	US payment transactions, US card processing

DMR external link

PARAMETER	VALUE
Link type	External link (between two separate event broker services)
Direction	Bidirectional
Compression	Enabled (reduces WAN bandwidth)
Guaranteed messaging bridging	Configured for reference data topics only
Direct messaging propagation	Automatic via subscription propagation for analytics summaries

Subscription propagation

What crosses the link

DIRECTION	TOPIC PATTERN	DELIVERY MODE	EST. VOLUME	PURPOSE
IN → US	reference/merchant/updated/v1/IN/>	Guaranteed	~10/sec	Merchant configs visible to US ops/analytics
IN → US	reference/rateTable/updated/v1/IN	Guaranteed	~1/min	Rate table updates
IN → US	analytics/summary/daily/v1/IN	Direct	~1/min	Aggregated India analytics for global reporting
US → IN	reference/merchant/updated/v1/US/>	Guaranteed	~10/sec	Merchant configs visible to India ops/analytics
US → IN	reference/rateTable/updated/v1/US	Guaranteed	~1/min	Rate table updates
US → IN	analytics/summary/daily/v1/US	Direct	~1/min	Aggregated US analytics for global reporting

Total cross-link: < 25 msg/sec peak. Compressed. Minimal WAN cost.

What stays local (never crosses the link)

DOMAIN	WHY
payment/*	Data residency. Payment transactions are region-specific.
fraud/*	Fraud scoring is inline with payment flow. Region-local.
auth/*	2FA is inline with payment flow. Region-local.
card/*	Card network responses are region-specific.
upi/*	India-only. No UPI in US.
settlement/*	Settlement is region-specific (per-market regulations).
risk/*	Risk assessment is region-local.
notification/*	Customer notifications are region-local.
analytics/transaction/*	High-volume transaction mirrors (~5K/sec). Stays local. Only aggregated summaries cross.

Supplementary reference topics

These topics support cross-region synchronization. They are not in the main 20-topic payment taxonomy.

#	TOPIC STRING	DELIVERY MODE	PRODUCER	CONSUMER(S)	CROSS-LINK
R1	reference/merchant/updated/v1/{region}/{merchantId}	Guaranteed	Merchant Portal	Core Banking, Settlement Engine (remote region)	Yes
R2	reference/rateTable/updated/v1/{region}	Guaranteed	Internal Ops	Core Banking, Card Processing (remote region)	Yes
R3	analytics/summary/daily/v1/{region}	Direct	Analytics Platform	Analytics Platform (remote region)	Yes

Antipattern validation

CHECK	STATUS
Single global broker	✓ PASS — two brokers, one per region
Overly broad wildcards across DMR	✓ PASS — only specific reference and analytics summary topics propagate
Symmetric high-volume bidirectional flows	✓ PASS — cross-link traffic is low-volume (~25 msg/sec), symmetric only for reference data. Not high-volume.
Payment data crossing regional boundary	✓ PASS — all payment, fraud, auth, settlement topics stay local

Mermaid diagram

See `dmr-topology.mermaid` for the visual representation.

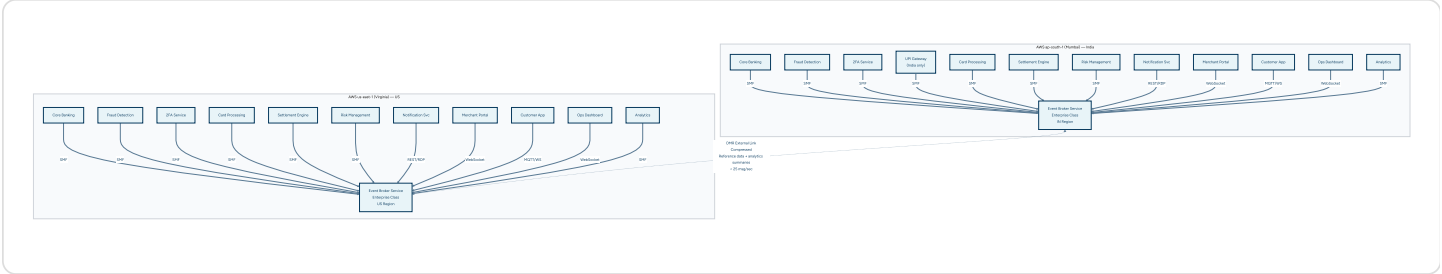
Capacity planning for external link

METRIC	CURRENT	YEAR 5 (10% YOY)
Cross-link msg/sec	~25	~40
Avg payload	~1 KB (reference data is small)	~1 KB
Bandwidth	< 50 KB/sec	< 80 KB/sec

METRIC	CURRENT	YEAR 5 (10% YOY)
With compression	< 20 KB/sec	< 35 KB/sec

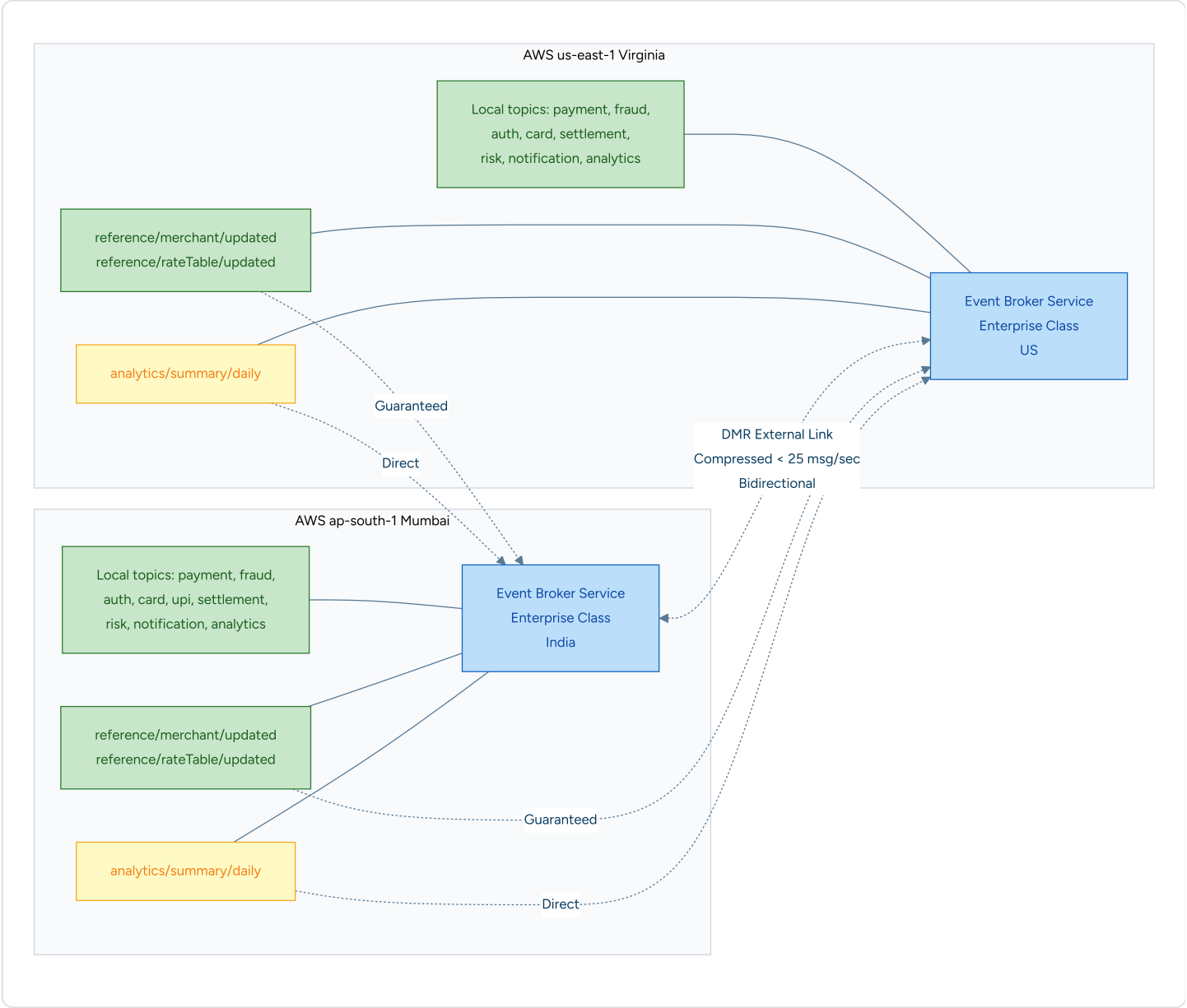
The external link is not a bottleneck. Even a minimal WAN connection handles this volume.

dmr topology



Mermaid diagram of two-site mesh topology

dmr topology



Two-site DMR federation, LR layout

HA/DR Design: Payment Solutions V1

Summary

- **HA:** Cloud Enterprise built-in. Three-node redundancy, multi-AZ, automatic failover. No team configuration needed.
- **DR:** Application-level recovery for Phase 1. The event broker is not the system of record. Core Banking ledger and card network logs are authoritative. Replication groups to a DR broker are a Phase 2 enhancement.

HA within each site

What Solace Cloud Enterprise provides

CAPABILITY	STATUS	TEAM ACTION
Three-node redundancy (primary, backup, monitoring)	Built-in	None. Solace manages.
Multi-AZ deployment	Built-in	None. Solace manages AZ placement.
Automatic failover	Built-in	Configure client reconnection settings.
Config-Sync	Built-in	None. Configuration stays consistent.
Client reconnection	Built into Solace APIs	Set reconnection parameters in client config.
Message redelivery after failover	Built-in (Guaranteed messaging)	Implement idempotent consumers.
Monitoring	Solace Insights	Export metrics to Datadog.

Team responsibilities

1. **Client reconnection configuration.** Set reconnection parameters in Solace API client config (reconnect retries, reconnect retry wait, reconnect retry delay). All 8 SMF clients, 3 WebSocket clients, and 1 MQTT client need reconnection settings.
2. **Idempotent consumers.** At-least-once delivery means messages may be redelivered after failover. Every Guaranteed messaging consumer must be idempotent. Use transaction ID as deduplication key.
3. **Queue depth monitoring.** Configure Solace Insights alerts: warning at 50% spool, critical at 80%. Export to Datadog.
4. **DMQ monitoring.** DMQ depth > 0 triggers investigation. Configure alerts in Solace Insights.
5. **Connection monitoring.** Alert on connection count drop (indicates client reconnection issues after failover).

DR strategy

Why the two regions are not DR pairs

India (Mumbai) and US (US-East) are independent payment processing deployments with data residency constraints:

- India payment data cannot leave India (RBI regulations)
- US payment data stays in US
- The regions share only reference data and aggregated analytics over DMR external links

Cross-region failover is not possible. If Mumbai fails, US cannot process India payments (no NPCI integration, no India card network connections, no India payment data).

Phase 1: Cloud HA + Application-Level Recovery

Failure scenario: single broker node failure

- HA covers this automatically. Failover to backup node in seconds.
- Clients reconnect. In-flight messages redelivered.

Failure scenario: single AZ failure

- Multi-AZ deployment covers this. Solace manages AZ redundancy.

Failure scenario: full AWS region failure

- Broker is unavailable until region recovers.
- Payment processing pauses for the affected region.
- No message loss for Guaranteed messaging (messages are in spool, recoverable when region returns).
- In-flight transactions: applications retry from last known state. Core Banking ledger is the system of record.
- Settlement Engine resumes from last committed batch checkpoint.
- Notification and analytics backfill from source after recovery.

Acceptable for Phase 1 because:

- Full AWS region failure is exceedingly rare
- The event broker is not the system of record. The ledger, card networks, and NPCI hold authoritative data.
- 4-month timeline and cost-optimized budget favor leveraging built-in capabilities
- Message replay (Enterprise class) provides recovery within the replay log window

Phase 2: Replication Groups (future enhancement)

When to add:

- If PCI-DSS auditors require documented broker-level DR
- If the business requires < 30-second RTO for broker infrastructure
- If volume growth justifies the cost of standby brokers

Design for Phase 2:

- India: standby event broker service in same region (multi-AZ provides DR within region) or in a second India-qualified region
- US: standby event broker service in AWS us-west-2
- Replication mode: asynchronous (minimizes latency impact on hot path)
- Replication groups interact with DMR: standby appears as same node to DMR topology

Data class RPO/RTO mapping

DATA CLASS	RPO	RTO	HA MECHANISM	DR MECHANISM (PHASE 1)	PHASE 2 ENHANCEMENT
Payment transactions	Near-zero	< 30 sec	Cloud HA (auto failover)	Application-level: retry from Core Banking ledger	Async replication to standby
Fraud check	Near-zero	< 30 sec	Cloud HA	Application-level: re-request from Core Banking	Async replication
2FA challenge/response	Near-zero	< 30 sec	Cloud HA	Application-level: re-challenge	Async replication
Settlement batch	< 1 min	< 5 min	Cloud HA	Batch restart from checkpoint	Async replication
Customer notifications	Minutes	< 10 min	Cloud HA	Retry from source	N/A (loss-tolerant)
Analytics	Minutes	< 30 min	Cloud HA	Backfill from source or replay log	N/A (Direct messaging)
Reference data	< 1 min	< 5 min	Cloud HA	Republish from authoritative store	Async replication

Message replay configuration

Enterprise class event broker services support message replay. Configure per region:

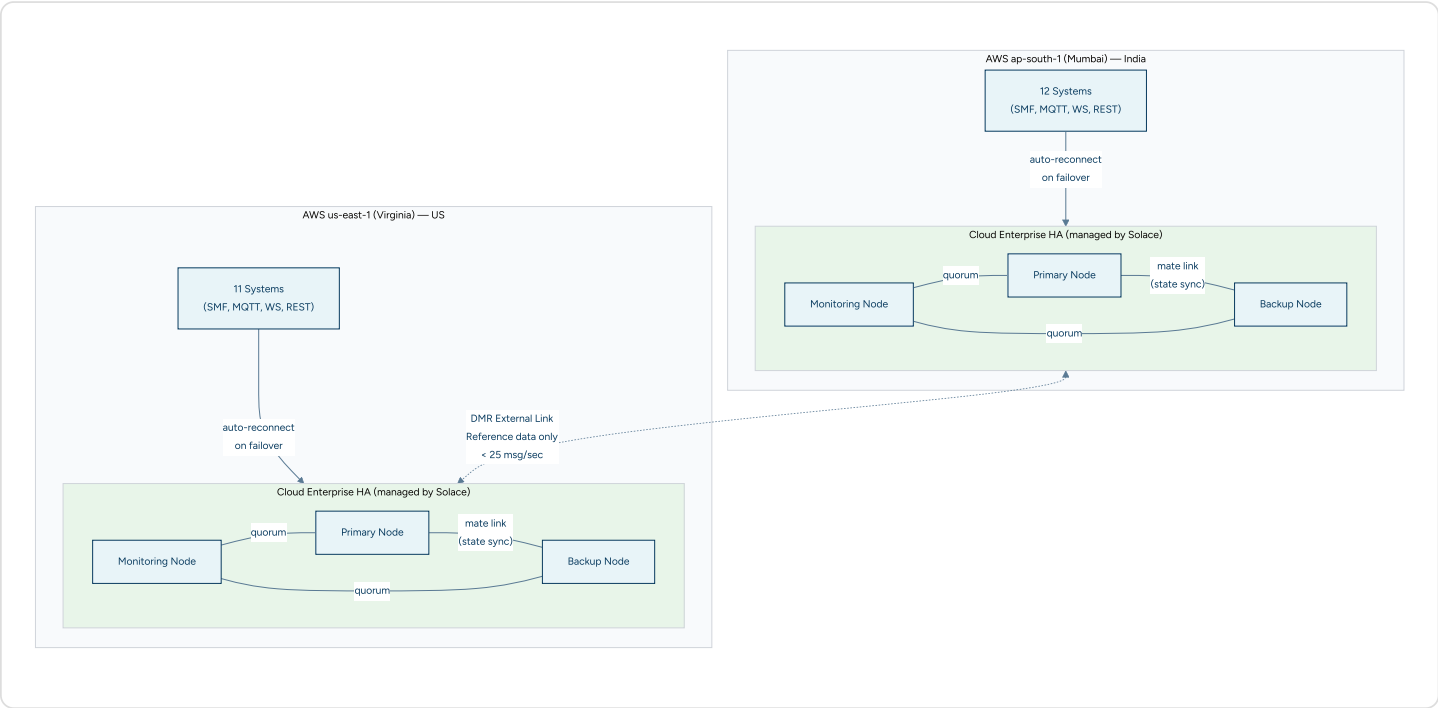
PARAMETER	VALUE	RATIONALE
Replay log enabled	Yes	Required for recovery and reprocessing.
Retention	1 hour minimum	Covers most recovery scenarios. Extend to 24 hours if spool budget permits.
Replay mode	Time-based	Most practical for recovery: "replay everything from 5 minutes ago."

Replay is not DR. It's a recovery tool for within-site failures. Messages beyond the replay window require application-level recovery from source systems.

Operational runbooks needed

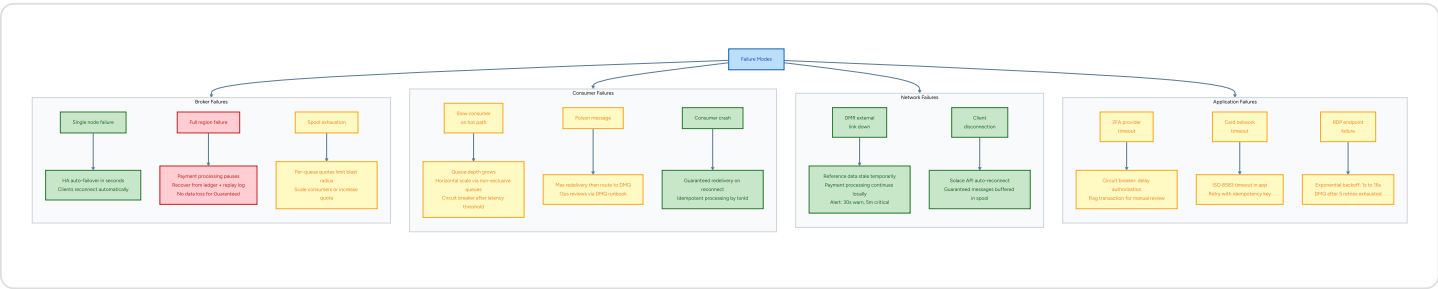
- 1. Broker failover response.** What to check when HA failover occurs. Expected: automatic, transparent. Action: verify client reconnection, check queue depths, validate no messages in DMQ.
- 2. DMQ review.** When DMQ depth > 0: inspect failed messages, identify root cause, fix consumer, republish to original topic.
- 3. Region failure response.** Communicate payment processing pause to merchants. Monitor AWS status page. Prepare for recovery when region returns. Execute replay from last checkpoint.
- 4. Spool exhaustion.** When spool reaches critical (80%): identify slow consumers, scale consumer instances, or increase max spool quota if approaching plan limits.

ha dr topology



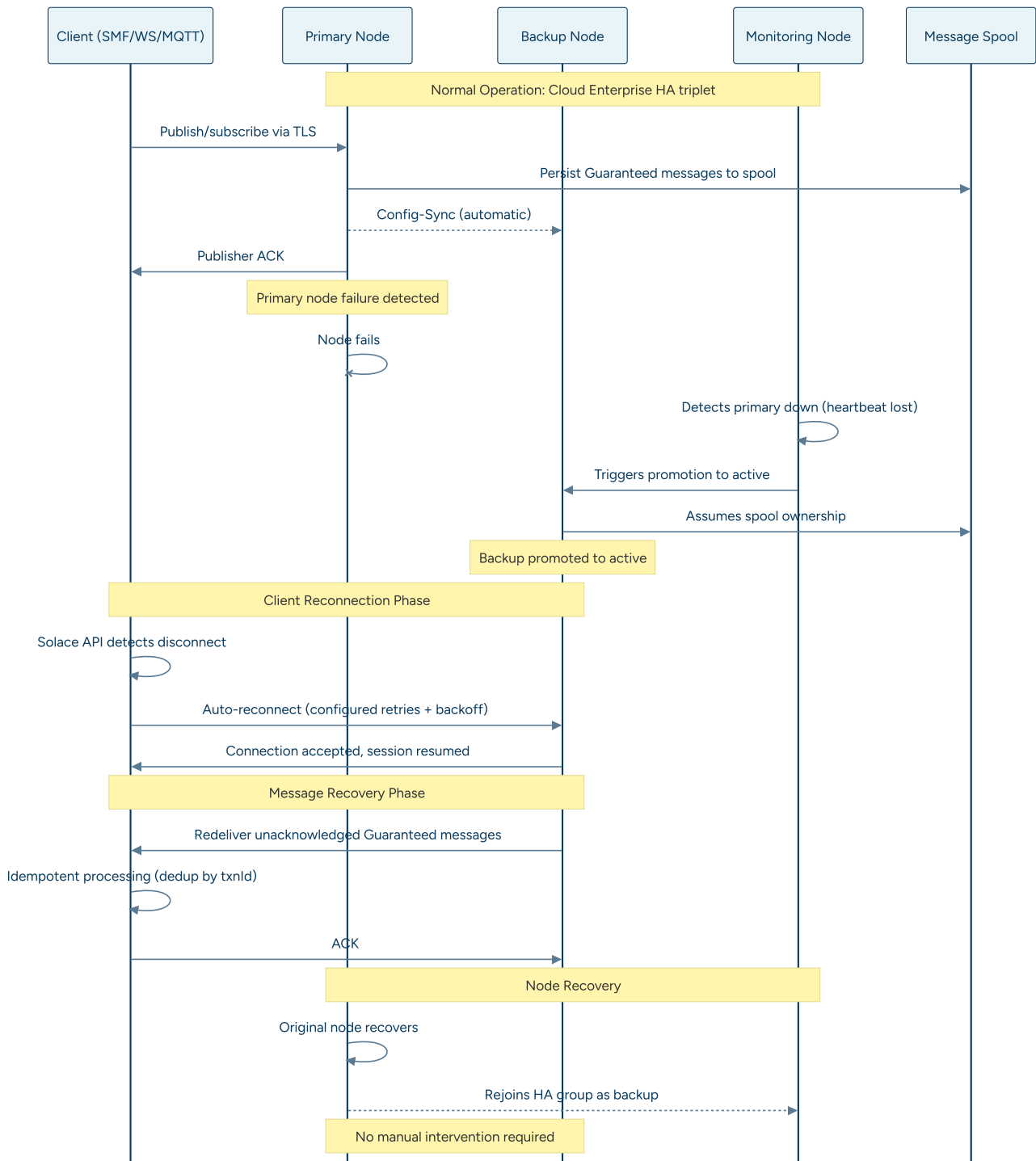
Mermaid diagram of HA topology per region

failure modes



Failure catalog as mindmap

ha failover



HA failover sequence with pool detail

INTEGRATION

Micro-Integration Map: Payment Solutions V1

Summary

Zero Micro-Integrations required. All 12 systems are custom applications connecting directly to the event broker via Solace messaging APIs. This is typical for a greenfield payment platform where every component is purpose-built.

Integration inventory

#	BACKEND SYSTEM	CONNECTION METHOD	PROTOCOL	MI REQUIRED?	RATIONALE
1	Core Banking Platform	Solace Java API (native client)	SMF	No	Custom application. Central payment orchestrator.
2	Credit Card Processing	Solace Java API	SMF	No	Custom app bridging to Visa/Mastercard/RuPay via ISO 8583.
3	UPI Gateway	Solace Java API	SMF	No	Custom app bridging to NPCI. India-only deployment.
4	Fraud Detection Engine	Solace Java/Python API	SMF	No	Custom real-time scoring engine.
5	Risk Management	Solace Java API	SMF	No	Custom risk assessment service.
6	2FA Service	Solace Java API	SMF	No	Custom app calling Twilio/MSG91 via REST for OTP delivery.
7	Customer Notification Service	REST delivery point (RDP)	REST	No	Broker pushes events to webhook endpoint. Service makes outbound calls to SMS/push/email providers.
8	Merchant Portal	Solace JavaScript API	WebSocket	No	Browser-based dashboard. Direct client connection.
9	Customer-Facing App (mobile)	Solace MQTT client	MQTT 5.0	No	Mobile app with MQTT for lightweight, battery-efficient messaging.
10	Customer-Facing App (web)	Solace JavaScript API	WebSocket	No	Browser-based client. Direct connection.
11	Internal Ops Dashboard	Solace JavaScript API	WebSocket	No	Browser-based monitoring tool. Direct connection.
12	Settlement Engine	Solace Java API	SMF	No	Custom batch and real-time settlement processor.
13	Analytics Platform	Solace Java/Python API	SMF	No*	Direct consumer via Solace API. *If Kafka-based: use Apache Kafka bridge (Broker Integrated).
14	Datadog	Solace Insights export	N/A	No	Operational monitoring. Solace Insights exports broker metrics to Datadog. Not a messaging integration.

Integration Hub catalog check

Verified against Integration Hub catalog snapshot (2026-04-29). No direct or indirect catalog matches for any payment processing backend system.

Indirect paths checked:

- None of the 12 systems publish natively to an intermediate system (Google Pub/Sub, Amazon SQS/SNS, Azure Service Bus) that has a cataloged Source Micro-Integration.
- Card networks (Visa/Mastercard/RuPay) use proprietary ISO 8583 protocols. No cataloged MI.

Conditional: Kafka bridge for analytics

If the Analytics Platform decision changes from "direct consumer" to "Kafka-based":

PARAMETER	VALUE
MI type	Apache Kafka (Broker Integrated)
Direction	Source + Target (bidirectional bridge)
Deployment	Built into the event broker. No separate runtime.
Topics bridged	<code>analytics/transaction/mirrored/v1/{region}/></code> (Direct messaging)
Fitness check	Fit. The Kafka bridge streams events from Solace topics to Kafka topics. Delivery mode: Direct messaging events bridge as non-persistent. Analytics tolerates this.

Note: If using the Kafka bridge, configure it as unidirectional (Solace → Kafka only) for analytics mirror topics. Do not bridge bidirectionally on the same topic (antipattern: infinite message loops).

REST delivery point details

The Customer Notification Service is the only system using REST delivery points (RDP):

PARAMETER	VALUE
RDP name	<code>rdp-notification-service</code> (per region)
Queue	Subscribed to <code>notification/customer/requested/v1/{localRegion}/></code>
Additional queue subscriptions	<code>auth/twoFactor/failed/v1/{localRegion}/></code> , <code>payment/transaction/declined/v1/{localRegion}/></code>
REST consumer URL	Notification service HTTP endpoint
Authentication	TLS + API key or OAuth token
Retry	Broker retries on non-2xx. Exponential backoff.
DMQ	Configured. Alert on depth > 0.
Idempotency	Notification service deduplicates by message ID.

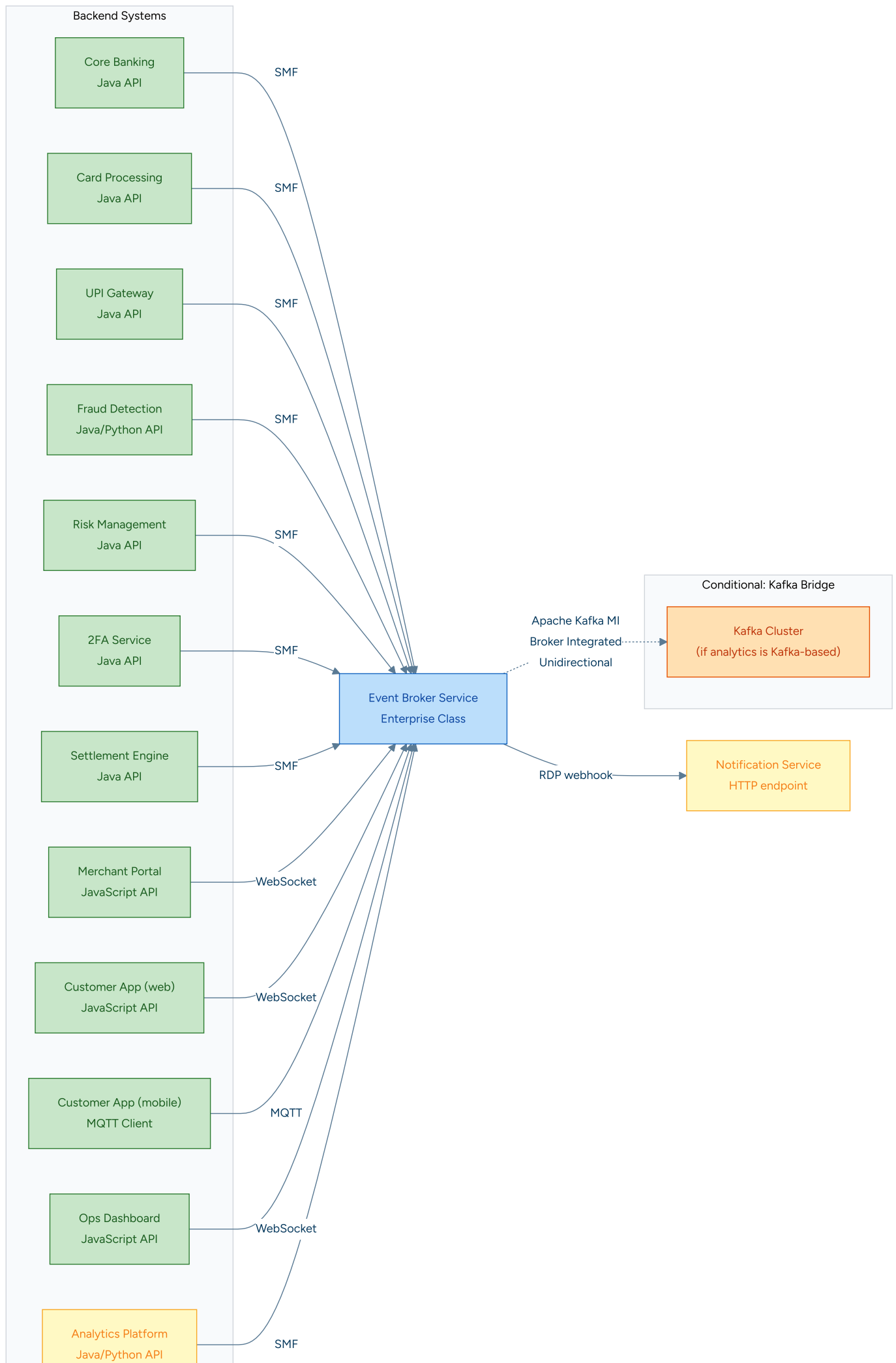
Custom integration specifications

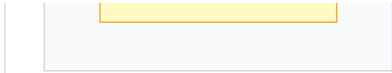
No custom Micro-Integrations are needed. All integrations are native Solace API client connections or REST delivery points.

The custom work is in the application layer:

1. **Credit Card Processing** app must implement ISO 8583 parsing/formatting for card network communication.
 2. **UPI Gateway** app must implement NPCI UPI API specification for pay/collect/mandate flows.
 3. **2FA Service** app must implement Twilio/MSG91 REST API calls for OTP delivery and verification callback handling.
 4. **Customer Notification Service** must handle the webhook (REST delivery point) contract: accept POST, return 2xx on success, handle retries idempotently.

These are application-level integrations, not Micro-Integration concerns. They connect to external systems through their own protocols and to the Solace event broker through the Solace APIs.

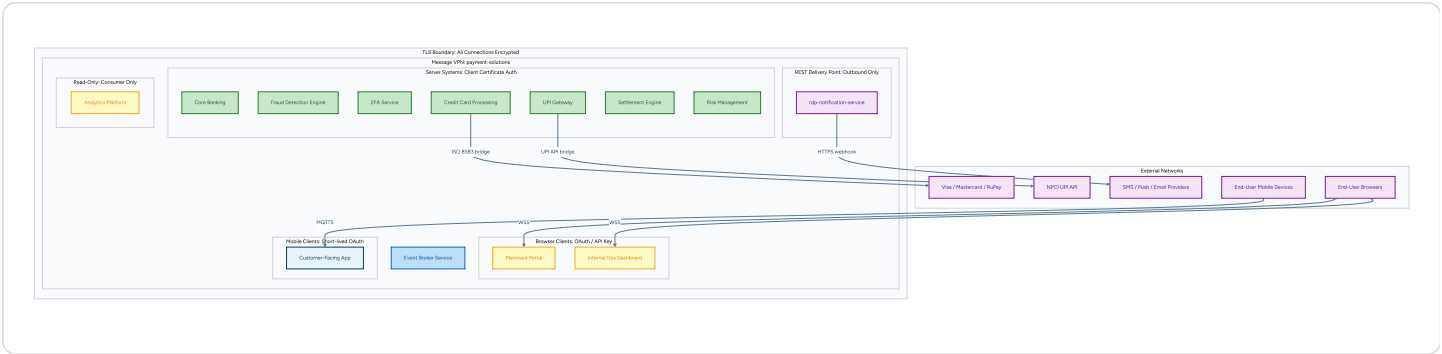




Native API connectivity, zero Micro-Integrations

REVIEWS

security boundaries



Security zones with classDef styles

Security Boundaries Detail

Companion to `security-boundaries.mermaid` . Full ACL profiles, auth methods, and credential storage per system.

ACL Profiles

SYSTEM	AUTH METHOD	PUBLISH ACL	SUBSCRIBE ACL
Core Banking	Client certificate	payment/> , fraud/check/requested/> , auth/twoFactor/challenged/> , notification/customer/requested/>	payment/transaction/initiated/> , fraud/check/completed/> , auth/twoFactor/*/v1/> , card/> , upi/> , settlement/> , risk/>
Fraud Detection Engine	Client certificate	fraud/check/completed/> , fraud/alert/>	payment/transaction/initiated/> , fraud/check/requested/>
2FA Service	Client certificate	auth/twoFactor/verified/> , auth/twoFactor/failed/>	auth/twoFactor/challenged/>
Credit Card Processing	Client certificate	card/networkResponse/>	payment/transaction/authorized/*/card
UPI Gateway	Client certificate	upi/networkResponse/>	payment/transaction/authorized/IN/upi
Settlement Engine	Client certificate	payment/transaction/captured/> , payment/transaction/settled/> , settlement/>	payment/transaction/authorized/>
Risk Management	Client certificate	(none)	fraud/check/completed/*/medium , fraud/check/completed/*/high , fraud/alert/>
Merchant Portal	OAuth / API key	(none)	payment/transaction/authorized/> , payment/transaction/captured/> , payment/transaction/settled/> , settlement/batch/completed/> , settlement/report/generated/>
Internal Ops Dashboard	OAuth / API key	(none)	settlement/batch/> , fraud/alert/*/high , fraud/alert/*/critical , risk/assessment/completed/>

SYSTEM	AUTH METHOD	PUBLISH ACL	SUBSCRIBE ACL
Customer-Facing App (mobile)	Short-lived OAuth	payment/transaction/initiated/>	payment/transaction/declined/>
Customer-Facing App (web)	OAuth	payment/transaction/initiated/>	payment/transaction/declined/>
Analytics Platform	Client certificate	(none)	analytics/> , notification/customer/delivered/> , payment/transaction/declined/>

REST Delivery Point

PARAMETER	VALUE
RDP name	rdp-notification-service (per region)
Direction	Outbound only (broker pushes to webhook)
Queue subscriptions	notification/customer/requested/> , auth/twoFactor/failed/> , payment/transaction/declined/>
Auth to endpoint	TLS + API key or OAuth token

Client Profile Resource Limits

PROFILE	MAX CONNECTIONS	MAX SUBSCRIPTIONS	MAX EGRESS RATE
cp-server-smf	20 per system	50	Unlimited
cp-browser-ws	1000 (aggregate)	10 per client	Rate-limited
cp-mobile-mqtt	10000 (aggregate)	5 per client	Rate-limited

Data Residency ACL Enforcement

BROKER REGION	BLOCKED SUBSCRIPTIONS
India (Mumbai)	*/v1/US/> on payment, fraud, auth, card, settlement topics
US (US-East)	*/v1/IN/> on payment, fraud, auth, card, settlement topics

Cross-region subscriptions are blocked via ACL deny rules. Only reference data and aggregated analytics topics are permitted to propagate via DMR external links.

Credential Storage

SYSTEM TYPE	CREDENTIAL TYPE	STORAGE
Server systems (SMF)	X.509 client certificate	Cert store, rotated via CI/CD
Browser clients (WebSocket)	OAuth access token	Browser session, short-lived
Mobile clients (MQTT)	OAuth access token	Secure keychain, short-lived
RDP webhook	API key or OAuth	Broker config, per-environment

Security Review: Payment Solutions V1

Review scope

Security assessment of the event mesh architecture for PCI-DSS compliance and payment processing security requirements.

Findings

Finding S1: TLS mandatory on all connections (confirmed)

Severity: Observation (positive) Area: Encryption in transit

All protocols (SMF, MQTT, WebSocket, REST/RDP) use TLS. Non-TLS ports to be disabled. This satisfies PCI-DSS Requirement 4 (encrypt transmission of cardholder data across open, public networks).

No action needed.

Finding S2: Encryption at rest not addressed at broker level

Severity: High Area: Encryption at rest

PCI-DSS Requirement 3 requires protection of stored cardholder data. Guaranteed messaging stores messages in broker spool (persisted to disk). The architecture does not specify encryption at rest for broker spool.

For Solace Cloud event broker services: Verify with Solace whether Cloud Enterprise class provides encryption at rest for message spool. Solace Cloud infrastructure likely provides volume-level encryption (AWS EBS encryption). Confirm this meets PCI-DSS requirements.

Resolution: Verify Solace Cloud encryption-at-rest capabilities with Solace sales. Document in the blueprint. If broker-level encryption at rest is not sufficient, consider application-level field encryption for cardholder data fields before publishing.

(Architectural inference, not from Solace docs -- verify before external use.)

Finding S3: Message VPN access control not specified

Severity: High Area: Access control

The architecture does not specify:

1. Client authentication method (basic auth, client certificate, OAuth)
2. ACL profiles per system (which client can publish/subscribe to which topics)
3. Client profile resource limits (max connections, max subscriptions per client)

PCI-DSS Requirement 7 (restrict access to cardholder data by business need-to-know) requires per-system access control.

Recommended ACL design:

SYSTEM	PUBLISH ACL	SUBSCRIBE ACL
Core Banking	payment/> , fraud/check/requested/> , auth/twoFactor/challenged/>	payment/transaction/initiated/> , fraud/check/completed/> , auth/twoFactor/*/v1/> , card/> , upi/> , settlement/> , risk/>
Fraud Detection Engine	fraud/check/completed/> , fraud/alert/>	payment/transaction/initiated/> , fraud/check/requested/>
2FA Service	auth/twoFactor/verified/> , auth/twoFactor/failed/>	auth/twoFactor/challenged/>
Credit Card Processing	card/networkResponse/>	payment/transaction/authorized/*/card
UPI Gateway	upi/networkResponse/>	payment/transaction/authorized/IN/upi
Settlement Engine	payment/transaction/captured/> , payment/transaction/settled/> , settlement/>	payment/transaction/authorized/>

SYSTEM	PUBLISH ACL	SUBSCRIBE ACL
Customer Notification Service	<code>notification/customer/delivered/></code>	(RDP-bound: <code>notification/customer/requested/></code> , <code>auth/twoFactor/failed/></code> , <code>payment/transaction/declined/></code>)
Analytics Platform	(none -- consumer only)	<code>analytics/></code> , <code>notification/customer/delivered/></code> , <code>payment/transaction/declined/></code>

Resolution: Define ACL profiles for every system. Use client certificate authentication for server-side systems (SMF). Use OAuth or API key for mobile/web clients.

Finding S4: Audit trail retention not configured at broker level

Severity: Medium **Area:** Audit logging

Discovery requires 1 year live and 6 years archive for audit trail. The broker's replay log is configured for 1 hour retention (recovery tool, not audit trail). The audit trail is the responsibility of the consuming systems, not the broker.

The event broker is not the audit trail. It's the transport. Core Banking, Settlement Engine, and the dedicated audit logging system capture and retain payment events per PCI-DSS requirements.

Recommendations:

- 1. Ensure every Guaranteed messaging consumer acknowledges messages only after persisting to its own audit store.
- 2. Add an audit logging consumer that subscribes to all payment and settlement topics and writes to long-term storage (S3, database with archival).
- 3. Configure Distributed Tracing for end-to-end message lifecycle tracking.

Finding S5: DMQ messages may contain cardholder data

Severity: Medium **Area:** Data handling

Dead message queues accumulate messages that failed processing. These messages may contain cardholder data (card numbers, CVV, account details). DMQ access must be restricted.

Recommendations:

- 1. Apply ACL restrictions to DMQ access. Only authorized ops staff and automated recovery processes should read from DMQs.
- 2. Set DMQ message TTL. Don't retain failed messages indefinitely.
- 3. DMQ review process must follow PCI-DSS data handling procedures (no logging of cardholder data in clear text).

Finding S6: Mobile client credentials management

Severity: Medium **Area:** Client security

Customer-Facing App (mobile) connects via MQTT 5.0. Mobile apps are distributed to end users. Credentials embedded in the app binary are extractable.

Recommendations:

- 1. Use short-lived OAuth tokens for mobile MQTT authentication, not embedded passwords.
- 2. Token refresh via a separate authentication service (not via the broker).
- 3. Rate-limit per-client connection attempts to prevent credential stuffing.
- 4. Use client profile resource limits to cap connections and subscriptions per mobile client.

Finding S7: Data residency enforcement is architectural, not broker-enforced

Severity: Low **Area:** Data residency

Data residency is enforced by topic design (region in topic properties) and subscription scoping (local region only). The broker does not enforce data residency natively. A misconfigured subscription could pull India data to the US broker.

Recommendations:

- 1. Use ACL profiles to prevent cross-region subscriptions. Block `*/v1/IN/>` subscriptions on the US broker and vice versa.
- 2. Audit DMR external link subscriptions periodically. Verify no payment topics propagate across the link.
- 3. Document the data residency enforcement mechanism in the compliance documentation.

Summary

SEVERITY	COUNT
Critical	0
High	2 (S2: encryption at rest, S3: access control)
Medium	3 (S4: audit trail, S5: DMQ data, S6: mobile credentials)
Low	1 (S7: data residency enforcement)
Observation	1 (S1: TLS confirmed)

Overall assessment: Two high-severity findings that must be resolved before production. Encryption at rest needs verification with Solace. ACL profiles need to be defined for every system. Both are standard PCI-DSS requirements that the architecture must address explicitly.

Operations Review: Payment Solutions V1

Review scope

Operations readiness assessment across all 6 technical design artifacts. Focus: what does the on-call engineer see at 3am?

Findings

Finding O1: Monitoring stack is defined but alert thresholds need specification

Severity: Medium **Area:** Observability

Solace Insights exports to Datadog (team's existing stack). Distributed Tracing is available with product key. The architecture documents which metrics matter but does not specify exact alert thresholds.

Recommended alert thresholds:

METRIC	WARNING	CRITICAL	ACTION
Queue depth (% of max spool)	50%	80%	Scale consumers or increase quota
DMQ depth	> 0	> 100	Investigate poison messages
Message rate deviation	+30% from baseline	+50% from baseline	Capacity check
Connection count drop	> 10% drop in 1 min	> 50% drop	Client reconnection issues
DMR external link status	Link down > 30 sec	Link down > 5 min	Reference data stale
Spool usage per region	60% of plan limit	80% of plan limit	Plan upgrade evaluation
End-to-end latency (hot path)	> 500ms p99	> 1000ms p99	Performance investigation

Resolution: Define these thresholds in the production runbook before go-live.

Finding O2: Four operational runbooks identified but not yet written

Severity: Medium **Area:** Runbooks

The HA/DR design identified 4 runbooks needed:

- 1. Broker failover response
- 2. DMQ review procedure
- 3. Region failure response
- 4. Spool exhaustion response

These are documented as needed but not yet written.

Resolution: Write runbooks before go-live. Assign to the platform team.

Finding O3: Capacity ceiling and growth path are documented

Severity: Observation (positive) **Area:** Capacity planning

Enterprise class per region. 5K TPS peak. 10% YoY growth to 8K TPS at year 5. Enterprise upgrade at year 3. DMR external link at < 25 msg/sec with room to grow. Queue count (40 per region) is well within limits.

No action needed. Growth path is clear.

Finding O4: Slow consumer risk on hot path

Severity: Medium **Area:** Failure modes

The hot path (payment initiation → fraud check → 2FA → authorization) has 6 handoffs, each via Guaranteed messaging queues. If any consumer slows down (e.g., Fraud Detection Engine under load), queue depth grows on that consumer's queue. At spool exhaustion, the broker enters reject mode for the entire message VPN.

Recommendations:

1. Set per-queue max spool quota. Don't let one slow consumer exhaust the VPN's spool.
2. Use non-exclusive queues for Fraud Detection and 2FA Service to enable horizontal scaling.
3. Monitor per-queue depth (not just aggregate spool) in Datadog.
4. Define a circuit breaker pattern: if fraud check latency exceeds a threshold, route transactions to a fallback path (e.g., delay authorization, flag for manual review).

Finding O5: MQTT session persistence for mobile clients

Severity: Low **Area:** Operations

Customer-Facing App (mobile) uses MQTT 5.0. Mobile clients disconnect frequently (network switches, app backgrounding). Persistent MQTT sessions accumulate state on the broker (subscriptions, inflight messages). At scale (thousands of mobile devices), this creates broker resource pressure.

Recommendations:

1. Set session expiry interval on MQTT clients (e.g., 1 hour). Expired sessions are cleaned up automatically.
2. Monitor MQTT session count in Solace Insights.
3. Use clean session for non-critical subscriptions (payment decline notifications are ephemeral).

Finding O6: REST delivery point retry behavior needs tuning

Severity: Low **Area:** Operations

The notification service RDP uses default retry settings. For a payment notification service that calls external providers (SMS, push, email), retry behavior should be explicit:

Recommended RDP settings:

- Max retries: 5
- Retry delay: exponential backoff (1s, 2s, 4s, 8s, 16s)
- Connection timeout: 5 seconds
- Response timeout: 10 seconds
- DMQ on retry exhaustion: configured with alerting

Summary

SEVERITY	COUNT
Critical	0
High	0
Medium	3 (O1: alert thresholds, O2: runbooks, O4: slow consumer)

SEVERITY	COUNT
Low	2 (O5: MQTT sessions, O6: RDP retry tuning)
Observation	1 (O3: capacity planning)

Overall assessment: Architecture is operable. No critical gaps. The three medium findings are standard production-readiness items that need specification before go-live. The platform team should prioritize alert threshold definition, runbook writing, and slow-consumer mitigation.

Architecture Review: Payment Solutions V1

Review scope

All 6 technical design artifacts reviewed: topic taxonomy (20 topics), broker selection (Cloud Enterprise x2), protocol map (4 protocols), mesh topology (2-site DMR), HA/DR design (Cloud HA + app-level DR), integration map (zero MIs).

Findings

Finding A1: Topic taxonomy is well-structured

Severity: Observation (positive) **Area:** Topic design

The 20-topic taxonomy follows `Domain/Noun/Verb/v1/{region}/{property}` consistently. Region as first property enforces data residency at the subscription level. Delivery modes are correctly assigned: Guaranteed for the hot path and settlement, Direct for notifications and analytics. No antipatterns detected.

No action needed.

Finding A2: Broker type selection is sound

Severity: Observation (positive) **Area:** Broker selection

Event broker service (Cloud), Enterprise class is the right call for a 10-person team with moderate Solace experience on a 4-month timeline. Sub-second latency (not sub-microsecond) is well within Cloud capability. The Enterprise+ upgrade path at year 3 accounts for 10% YoY growth.

No action needed.

Finding A3: Two open blocking questions remain from discovery

Severity: Medium **Area:** Discovery gaps

Two blocking questions from discovery are still unresolved:

- 1. **Card network connectivity model** -- direct ISO 8583 or through a payment processor/switch?
- 2. **NPCI integration model** -- PSP or TPP?

These affect protocol selection for the card and UPI subsystems. The current design assumes direct integration. If a payment processor/switch mediates, the Credit Card Processing and UPI Gateway topics and protocol choices may change.

Resolution: Resolve before finalizing the blueprint. The current architecture accommodates both models (the custom app bridges to the external system regardless). The main impact is on the application layer, not the event mesh topology.

Finding A4: Settlement report topic may cause queue proliferation

Severity: Low **Area:** Topic design

Topic #15 (`settlement/report/generated/v1/{region}/{merchantId}`) uses `merchantId` as a property. At scale (thousands of merchants), this creates high topic cardinality. The Merchant Portal subscribes to `settlement/report/generated/v1/{localRegion}/>` and filters at the application level.

This is acceptable. High cardinality in trailing properties does not affect broker routing performance (topic matching is efficient). The wildcard subscription covers all merchants without per-merchant queue configuration. Note: do not create per-merchant queues. A single queue with the wildcard subscription handles all merchants.

No action needed. Document the pattern for the dev team.

Finding A5: Direct messaging for notifications creates a silent-loss path

Severity: Medium **Area:** Delivery mode

Customer notifications use Direct messaging. If the Customer Notification Service is down or slow when a notification event is published, the message is lost. No spool, no retry.

This is an intentional design decision documented in the topic taxonomy. The discovery brief classified notifications as ephemeral. The source systems (Core Banking, Settlement Engine, Fraud Detection) can regenerate notification requests.

Recommendation: Implement a retry mechanism in the source applications. If a notification matters enough to send, the source should track whether it was delivered and re-publish if not acknowledged within a timeout. This is application-level logic, not broker-level.

Finding A6: Cross-region reference data topics not in main catalog

Severity: Low **Area:** Mesh design

The mesh design introduced 3 supplementary reference topics (R1-R3) for cross-region sync that are not in the main 20-topic taxonomy. This is correct design (they carry reference data, not payment transactions), but they should be documented in a single combined topic catalog for the blueprint.

Resolution: Include reference topics in the blueprint's consolidated topic catalog with a clear label distinguishing them from payment transaction topics.

Finding A7: No simpler topology alternative exists

Severity: Observation (positive) **Area:** Mesh design

With two regions and data residency constraints, the two-site DMR external link topology is the simplest possible. A single-broker topology cannot satisfy data residency. The external link carries < 25 msg/sec, which is minimal.

No action needed.

Summary

SEVERITY	COUNT
Critical	0
High	0
Medium	2 (A3: blocking questions, A5: notification loss path)
Low	2 (A4: merchant topic cardinality, A6: reference topics not in main catalog)
Observation	3 (A1, A2, A7: positive confirmations)

Overall assessment: Architecture is structurally sound. No critical or high-severity findings. The two medium findings are acknowledged design trade-offs with documented rationale. Resolve the two blocking discovery questions before blueprint.

Developer Experience Review: Payment Solutions V1

Review scope

Developer experience assessment: what does the development team need to build, what Solace APIs do they use, and where are the complexity hotspots?

Findings

Finding D1: Solace API selection is clear per system

Severity: Observation (positive) **Area:** Developer experience

Protocol map clearly assigns APIs per system. 8 server-side systems use Solace Java/Python API (SMF). 3 browser apps use Solace JavaScript API (WebSocket). 1 mobile app uses MQTT 5.0. 1 notification service uses REST delivery point.

Developers know which SDK to use and why. No ambiguity.

Finding D2: Idempotent consumer pattern needed for all Guaranteed messaging consumers

Severity: Medium **Area:** Application patterns

At-least-once delivery with Guaranteed messaging means every consumer must handle duplicate messages. With transaction ID as partition key, the deduplication strategy is clear: maintain a seen-transaction-ID set (with TTL) and skip duplicates.

Recommendations:

1. Build a shared idempotent consumer library that all Java/Python services use.
2. Use a bounded LRU cache or Redis set for deduplication. TTL should exceed the longest expected replay window (1 hour).
3. Document the pattern in the team's developer guide. Every new consumer must include it.

Finding D3: Hot path timeout handling is complex

Severity: Medium **Area:** Application complexity

The hot path (payment initiation → fraud check → 2FA → authorization → network response) has 6 asynchronous handoffs across 6 systems. Each handoff is a publish-subscribe pair with Guaranteed messaging. The orchestrating system (Core Banking) must:

1. Track the state of each payment transaction across handoffs
2. Handle timeouts if any downstream service doesn't respond
3. Handle out-of-order responses
4. Implement compensating actions on failure

This is a state machine. The complexity is inherent to the payment flow, not introduced by the event mesh architecture. But the team needs to be aware that event-driven payment orchestration requires careful state management.

Recommendations:

1. Implement a state machine in Core Banking for each payment transaction (states: initiated, fraud_checking, 2fa_challenged, 2fa_verified, authorizing, authorized, declined, captured, settled).
2. Set per-step timeouts (e.g., fraud check must complete within 3 seconds, 2FA within 30 seconds).
3. Use correlation via transaction ID in message properties for request-response matching.
4. Build a dead letter/manual review path for transactions stuck in intermediate states.

Finding D4: Event Portal for schema management

Severity: Low **Area:** Developer tooling

The architecture uses JSON payloads with industry-standard schemas (ISO 8583 for cards, UPI XML for NPCI). As the system evolves, schema changes will affect multiple consumers.

Recommendations:

1. Use Event Portal to catalog all 20+ topics with their schemas.
2. Use Solace Schema Registry for schema versioning (version in topic: `v1`).
3. Define schema evolution rules: additive changes are backward-compatible, breaking changes increment the version.

Finding D5: Testing strategy for multi-region architecture

Severity: Low **Area:** Testing

Multi-region deployment with DMR external links adds testing complexity. Developers need:

1. **Local development:** Single broker (Developer class) for each developer. Simulate single-region flow.
2. **Integration testing:** Two brokers with DMR external link in a test environment. Verify cross-region reference data sync.
3. **Performance testing:** SDKPerf for throughput benchmarking. Simulate 5K TPS load against Enterprise class broker.
4. **Chaos testing:** Simulate broker failover, DMR link failure, slow consumer scenarios.

Recommendations:

1. Provision a shared Developer-class event broker service for the team.
2. Provision an integration test environment with two Enterprise-class brokers and DMR external link.

- 3. Include SDKPerf in CI pipeline for regression testing.

Finding D6: WebSocket client considerations for merchant and ops dashboards

Severity: Low Area: Client development

Merchant Portal and Ops Dashboard use Solace JavaScript API over WebSocket. Browser-based clients have specific considerations:

- 1. Browser tab suspension (background tabs throttle WebSocket connections)
- 2. Connection management on page reload
- 3. Subscription recovery after reconnect

Recommendations:

- 1. Use the Solace JavaScript API's built-in reconnection with session properties.
- 2. Implement subscription recovery on reconnect (re-subscribe to all topics).
- 3. Handle browser visibility API to detect tab suspension and refresh state on tab activation.

Summary

SEVERITY	COUNT
Critical	0
High	0
Medium	2 (D2: idempotent consumers, D3: hot path state machine)
Low	3 (D4: Event Portal, D5: testing strategy, D6: WebSocket clients)
Observation	1 (D1: API selection clarity)

Overall assessment: No critical developer experience gaps. The two medium findings are standard event-driven payment processing patterns that the team will need to implement. The shared idempotent consumer library and Core Banking state machine are the two highest-effort development items.

Validation

Validation Report: Payment Solutions V1

Cross-artifact consistency checks

1. Topic taxonomy vs subscription map

Check	Status	Detail
Every topic in taxonomy has at least one subscriber	✓ PASS	All 20 topics have mapped subscriptions.
Every subscription maps to a real topic	✓ PASS	All 36 subscriptions resolve to topics in the catalog.
Delivery modes consistent between topic and subscriber	✓ PASS	Guaranteed topics have queue-bound subscribers. Direct topics have topic-subscription subscribers.
Region scoping consistent	✓ PASS	Every subscription filters by local region.

2. Broker selection vs scale requirements

Check	Status	Detail
Broker can handle peak TPS	✓ PASS	Enterprise class handles 5K TPS with headroom.
Connection count within limits	✓ PASS	24-120 connections per region. Well within Enterprise.
Spool capacity for Guaranteed messaging	✓ PASS	~36 GB for 1hr replay. Enterprise class accommodates.
DMR external link supported	✓ PASS	Enterprise class includes DMR.

3. Protocol map vs topic taxonomy

Check	Status	Detail
Every producer has a protocol assigned	✓ PASS	All 12 systems have protocol selection.
Protocol supports required delivery mode	✓ PASS	SMF supports Guaranteed + Direct. MQTT QoS 1 maps to Guaranteed. WebSocket supports both. REST/RDP via queue.
Partitioned queue consumers use SMF	✓ PASS	All hot path consumers (Core Banking, Fraud Detection, Settlement, etc.) use SMF.

4. Mesh topology vs data residency

Check	Status	Detail
No payment topics cross DMR link	✓ PASS	Only reference and analytics summary topics propagate.
India data stays in Mumbai broker	✓ PASS	All IN-region subscriptions are local.
US data stays in US-East broker	✓ PASS	All US-region subscriptions are local.
UPI topics restricted to India	✓ PASS	UPI Gateway hardcoded to IN region.

5. HA/DR vs broker selection

CHECK	STATUS	DETAIL
HA mechanism matches broker type	✓ PASS	Cloud Enterprise includes built-in HA.
DR strategy documented	✓ PASS	Phase 1: app-level recovery. Phase 2: replication groups.
Replay log configured	✓ PASS	1hr minimum retention.

6. Integration map vs system inventory

CHECK	STATUS	DETAIL
Every system from discovery has an integration path	✓ PASS	All 12 systems mapped to connection method and protocol.
MI availability checked against catalog	✓ PASS	Zero MIs needed. All native API connections.
REST delivery point configured where needed	✓ PASS	Notification service has RDP specification.

7. Review findings vs architecture

CHECK	STATUS	DETAIL
All critical/high findings have resolution paths	⚠ PARTIAL	2 high findings from security review need resolution before blueprint: encryption at rest (S2) and ACL profiles (S3).
Medium findings acknowledged	✓ PASS	10 medium findings across 4 reviews. All have documented recommendations.

8. Naming conventions

CHECK	STATUS	DETAIL
"Micro-Integration" (capital M, hyphenated)	✓ PASS	Used correctly throughout.
"Direct messaging" / "Guaranteed messaging"	✓ PASS	No QoS references.
"Event broker service" for Cloud	✓ PASS	Used correctly.
"DMR" / "external links"	✓ PASS	Used correctly.
Topic taxonomy follows Domain/Noun/Verb/Version/Properties	✓ PASS	Consistent across all 20 topics.

9. Antipattern check (final)

ANTIPATTERN	STATUS
Environment names in topics	✓ PASS
Tracing IDs in topics	✓ PASS
Overly broad wildcards across DMR	✓ PASS
Properties before version	✓ PASS
Selector-based filtering	✓ PASS
Single global broker	✓ PASS
Mixed delivery modes on same topic	✓ PASS
Custom MI when cataloged path exists	✓ PASS (no MIs in use)
Direct messaging chosen where Guaranteed required	✓ PASS
Missing DMQ on production queues	⚠ FLAG — DMQ specified in RDP config but not explicitly specified for all production queues. Add DMQ to all Guaranteed messaging queues.

Validation summary

CATEGORY	CHECKS	PASSED	FLAGGED
Cross-artifact consistency	24	23	1 (high findings need resolution)
Naming conventions	5	5	0
Antipattern check	10	9	1 (DMQ on all queues)
Total	39	37	2

Items to resolve before blueprint

1. **[High] Encryption at rest** (S2) — Verify Solace Cloud encryption-at-rest for message spool. Document in blueprint.
2. **[High] ACL profiles** (S3) — Define publish/subscribe ACLs per system. Include in blueprint.
3. **[Medium] DMQ on all production queues** — Specify DMQ configuration for every Guaranteed messaging queue, not just the RDP queue.

Architecture completeness score

8.5 / 10 — All technical design artifacts are complete and internally consistent. Two high-severity security items and one operational item need resolution for a complete production-ready blueprint. No structural or topological gaps.

Architecture Blueprint: Payment Solutions V1

Executive Summary

Payment Solutions V1 is a greenfield event-driven payment processing platform supporting credit/debit cards (Visa, Mastercard, RuPay) and UPI. It processes transactions across two regions: India (AWS Mumbai) and US (AWS US-East), with strict data residency enforcement at the topic level. The platform targets 5,000 TPS peak with sub-second latency on the hot path (payment initiation through fraud check, 2FA, and authorization).

The event mesh uses two Solace Cloud event broker services (Enterprise class), one per region, connected by a compressed DMR external link that carries only reference data and aggregated analytics (< 25 msg/sec). All 12 systems connect via native Solace APIs. Zero Micro-Integrations are needed. The topic taxonomy spans 20 topics across 9 domains, with 17 using Guaranteed messaging for PCI-DSS audit trail and 3 using Direct messaging for ephemeral notifications and analytics.

Key architectural decisions: per-key partitioned queues with transaction ID as partition key for per-transaction ordering; Cloud Enterprise built-in HA with application-level DR for Phase 1; region as the first property after version in every topic string, enforcing data residency at the subscription layer.

System Context

Systems

#	SYSTEM	ROLE	PROTOCOL	REGION
1	Core Banking Platform	Central payment orchestrator	SMF	IN + US
2	Credit Card Processing	Visa/Mastercard/RuPay ISO 8583 bridge	SMF	IN + US
3	UPI Gateway	NPCI UPI integration	SMF	IN only
4	Fraud Detection Engine	Real-time transaction scoring	SMF	IN + US
5	Risk Management	Risk scoring and regulatory reporting	SMF	IN + US
6	2FA Service	OTP and app-based two-factor auth	SMF	IN + US
7	Customer Notification Service	SMS, push, email delivery	REST (RDP)	IN + US
8	Merchant Portal	Merchant dashboard	WebSocket	IN + US
9	Customer-Facing App (mobile)	End-customer mobile app	MQTT 5.0	IN + US
10	Customer-Facing App (web)	End-customer browser app	WebSocket	IN + US
11	Internal Ops Dashboard	Ops monitoring and reconciliation	WebSocket	IN + US
12	Settlement Engine	Batch and real-time settlement	SMF	IN + US
13	Analytics Platform	Transaction analytics and BI	SMF	IN + US

Boundaries

- Data residency boundary:** India payment data stays in Mumbai broker. US payment data stays in US-East broker. No payment transaction events cross the DMR external link.
- Network boundary:** All broker connections use TLS. Non-TLS ports disabled. Card networks and NPCI accessed through custom application bridges, not through the event broker.
- PCI-DSS scope:** The event broker carries cardholder data in transit. Encryption in transit (TLS) and encryption at rest (verify with Solace) are required. ACL profiles restrict which system can publish/subscribe to which topics.

Hot Path

The payment authorization hot path traverses 6 asynchronous handoffs:

1. Customer-Facing App publishes `payment/transaction/initiated`

- 2. Core Banking receives, publishes `fraud/check/requested`
- 3. Fraud Detection scores, publishes `fraud/check/completed`
- 4. Core Banking receives, publishes `auth/twoFactor/challenged`
- 5. 2FA Service verifies, publishes `auth/twoFactor/verified` (or `failed`)
- 6. Core Banking authorizes, publishes `payment/transaction/authorized`

Target: all 6 handoffs complete in under 1 second. Each handoff uses Guaranteed messaging with partitioned queues keyed by transaction ID.

Event Mesh Design

Broker Type and Topology

Two event broker services (Cloud), Enterprise class:

SITE	AWS REGION	SYSTEMS	TRAFFIC
India	ap-south-1 (Mumbai)	All 12 + UPI Gateway	India payments, UPI, India card processing
US	us-east-1 (Virginia)	11 (no UPI Gateway)	US payments, US card processing

Why Cloud: 10-person team, moderate Solace experience, 4-month timeline. Cannot absorb broker infrastructure ops. Cloud provisions in minutes, includes built-in HA, and exports metrics to Datadog via Solace Insights.

DMR external link: Single compressed link between Mumbai and US-East. Bidirectional. Carries reference data (merchant configs, rate tables) and aggregated analytics summaries. Volume: < 25 msg/sec. Payment, fraud, auth, settlement, and risk topics never cross the link.

Growth path: Enterprise class through year 2. Evaluate Enterprise+ at year 3 (~6,050 TPS). Enterprise+ accommodates growth to ~8,050 TPS at year 5.

Topic Taxonomy

20 topics across 9 domains. Naming convention: `Domain/Noun/Verb/v1/{region}/{property}` .

#	TOPIC	DELIVERY MODE	PRODUCER	CONSUMER(S)
1	<code>payment/transaction/initiated/v1/{region}/{paymentMethod}</code>	Guaranteed	Customer-Facing App	Core Banking, Fraud Detection
2	<code>fraud/check/requested/v1/{region}/{paymentMethod}</code>	Guaranteed	Core Banking	Fraud Detection
3	<code>fraud/check/completed/v1/{region}/{riskLevel}</code>	Guaranteed	Fraud Detection	Core Banking, Risk Management
4	<code>auth/twoFactor/challenged/v1/{region}/{method}</code>	Guaranteed	Core Banking	2FA Service
5	<code>auth/twoFactor/verified/v1/{region}/{method}</code>	Guaranteed	2FA Service	Core Banking
6	<code>auth/twoFactor/failed/v1/{region}/{method}</code>	Guaranteed	2FA Service	Core Banking, Notification Service
7	<code>payment/transaction/authorized/v1/{region}/{paymentMethod}</code>	Guaranteed	Core Banking	Card/UPI Processing, Settlement, Merchant Portal, Notification
8	<code>payment/transaction/declined/v1/{region}/{paymentMethod}</code>	Guaranteed	Core Banking	Customer App, Notification, Analytics
9	<code>card/networkResponse/received/v1/{region}/{network}</code>	Guaranteed	Card Processing	Core Banking
10	<code>upi/networkResponse/received/v1/{region}/{flowType}</code>	Guaranteed	UPI Gateway	Core Banking
11	<code>payment/transaction/captured/v1/{region}/{paymentMethod}</code>	Guaranteed	Settlement Engine	Core Banking, Merchant Portal, Analytics

#	TOPIC	DELIVERY MODE	PRODUCER	CONSUMER(S)
12	payment/transaction/settled/v1/{region}/{paymentMethod}	Guaranteed	Settlement Engine	Core Banking, Merchant Portal
13	settlement/batch/initiated/v1/{region}	Guaranteed	Settlement Engine	Core Banking, Ops Dashboard
14	settlement/batch/completed/v1/{region}	Guaranteed	Settlement Engine	Merchant Portal, Ops Dashboard
15	settlement/report/generated/v1/{region}/{merchantId}	Guaranteed	Settlement Engine	Merchant Portal
16	fraud/alert/raised/v1/{region}/{severity}	Guaranteed	Fraud Detection	Risk Management, Ops Dashboard
17	risk/assessment/completed/v1/{region}/{riskLevel}	Guaranteed	Risk Management	Core Banking, Ops Dashboard
18	notification/customer/requested/v1/{region}/{channel}	Direct	Core Banking, Settlement, Fraud Detection	Notification Service
19	notification/customer/delivered/v1/{region}/{channel}	Direct	Notification Service	Analytics
20	analytics/transaction/mirrored/v1/{region}/{eventType}	Direct	All producers	Analytics Platform

Supplementary reference topics (cross-region sync, not in the payment taxonomy):

#	TOPIC	DELIVERY MODE	CROSS-LINK
R1	reference/merchant/updated/v1/{region}/{merchantId}	Guaranteed	Yes
R2	reference/rateTable/updated/v1/{region}	Guaranteed	Yes
R3	analytics/summary/daily/v1/{region}	Direct	Yes

Delivery Mode Strategy

- **Guaranteed messaging (17 topics):** Payment transactions, fraud checks, 2FA, settlement, risk events. At-least-once delivery with application-level idempotency using transaction ID as deduplication key. PCI-DSS requires lossless delivery for audit trail.
- **Direct messaging (3 topics):** Customer notifications (#18, #19) and analytics mirror (#20). Ephemeral data. Source systems can regenerate on loss.

Subscription Strategy

36 subscriptions across 12 consumers. Every subscription scoped to local region.

Key patterns:

- **Partitioned queues (~10 per region):** Hot path consumers (Core Banking, Fraud Detection, 2FA Service, Card Processing, UPI Gateway, Settlement Engine) use partitioned queues with transaction ID as partition key. All events for one transaction processed in order by one consumer instance.
- **Non-exclusive queues:** Risk Management, Merchant Portal, Ops Dashboard. Multiple consumer instances for scale, no ordering requirement.
- **Topic subscriptions (no queue):** Analytics Platform and Customer Notification Service on Direct messaging topics. No queue binding, no spool.

Selective filtering: Risk Management subscribes only to `fraud/check/completed/v1/{localRegion}/medium` and `.../high` , excluding low-risk results. Ops Dashboard subscribes to `fraud/alert/raised/v1/{localRegion}/high` and `.../critical` only, not all severities.

Integration Design

Connection Summary

Zero Micro-Integrations. All systems connect via native Solace APIs.

PROTOCOL	COUNT	SYSTEMS
SMF (Solace Java/Python API)	8	Core Banking, Card Processing, UPI Gateway, Fraud Detection, Risk Management, 2FA Service, Settlement Engine, Analytics Platform
WebSocket (Solace JavaScript API)	3	Merchant Portal, Customer-Facing App (web), Ops Dashboard
MQTT 5.0	1	Customer-Facing App (mobile)
REST delivery point	1	Customer Notification Service

Protocol Selection Rationale

- **SMF** for all server-side systems: lowest latency, full feature access including partitioned queues. Required for hot path consumers.
- **WebSocket** for browser apps: native browser support, real-time push, supports both Guaranteed and Direct messaging.
- **MQTT 5.0** for mobile: lightweight, battery-efficient, persistent sessions across reconnects. MQTT QoS 1 maps to Guaranteed messaging for payment initiation.
- **REST delivery point** for notifications: the Notification Service makes outbound HTTP calls to SMS/push/email providers. REST delivery point pushes events as webhooks. Simplest integration pattern for a service that already speaks HTTP.

Protocol Mediation

The broker handles all protocol mediation transparently. Example flows:

- Core Banking (SMF) publishes `payment/transaction/authorized` -> Merchant Portal (WebSocket) receives dashboard update
- Customer-Facing App (MQTT) publishes `payment/transaction/initiated` -> Core Banking (SMF) begins authorization
- Settlement Engine (SMF) publishes `notification/customer/requested` -> Notification Service receives via REST delivery point webhook

REST Delivery Point Configuration

PARAMETER	VALUE
RDP name	<code>rdp-notification-service</code> (per region)
Queue subscriptions	<code>notification/customer/requested/v1/{localRegion}/></code> , <code>auth/twoFactor/failed/v1/{localRegion}/></code> , <code>payment/transaction/declined/v1/{localRegion}/></code>
Authentication	TLS + API key or OAuth token
Retry	Exponential backoff (1s, 2s, 4s, 8s, 16s), max 5 retries
DMQ	Configured with alerting on depth > 0

Conditional: Kafka Bridge for Analytics

If the Analytics Platform decision changes to Kafka-based: use the Apache Kafka (Broker Integrated) Micro-Integration. Unidirectional (Solace -> Kafka only) for `analytics/transaction/mirrored/v1/{region}/>` . Do not bridge bidirectionally on the same topic.

Reliability Design

High Availability

Cloud Enterprise HA is built-in per region:

CAPABILITY	STATUS	TEAM ACTION
Three-node redundancy (primary, backup, monitoring)	Built-in	None
Multi-AZ deployment	Built-in	None
Automatic failover	Built-in	Configure client reconnection settings
Config-Sync across HA nodes	Built-in	None
Message redelivery after failover	Built-in (Guaranteed)	Implement idempotent consumers

Team responsibilities:

- 1. Configure reconnection parameters in all Solace API clients (SMF, WebSocket, MQTT)
- 2. Implement idempotent consumers using transaction ID as deduplication key
- 3. Monitor queue depth, DMQ depth, and connection count via Solace Insights -> Datadog

Disaster Recovery

Phase 1: Cloud HA + Application-Level Recovery

The two regions are NOT DR pairs. Data residency prevents cross-region failover: if Mumbai goes down, US-East cannot process India payments (no NPCI integration, no India card network connections, no India payment data).

FAILURE	RECOVERY
Single broker node	HA auto-failover in seconds. Clients reconnect.
Single AZ	Multi-AZ deployment handles this.
Full AWS region	Payment processing pauses for affected region. Broker spool is durable; messages recover when region returns. Applications retry from Core Banking ledger (system of record).

Phase 2: Replication Groups (future)

- Standby event broker service within the same data residency boundary
- Asynchronous replication to minimize hot-path latency impact
- Triggers: PCI-DSS auditor requirement for broker-level DR, or business requires < 30-second RTO

Data Class RPO/RTO

DATA CLASS	RPO	RTO	PHASE 1 MECHANISM
Payment transactions	Near-zero	< 30 sec	Cloud HA auto-failover
Fraud check	Near-zero	< 30 sec	Cloud HA + re-request from Core Banking
2FA challenge/response	Near-zero	< 30 sec	Cloud HA + re-challenge
Settlement batch	< 1 min	< 5 min	Batch restart from checkpoint
Customer notifications	Minutes	< 10 min	Retry from source
Analytics	Minutes	< 30 min	Backfill from source or replay log
Reference data	< 1 min	< 5 min	Republish from authoritative store

Message Replay

Enterprise class supports message replay, configured per region:

- Replay log: enabled
- Retention: 1 hour minimum (extend to 24 hours if spool budget permits)
- Mode: time-based ("replay from 5 minutes ago")
- Replay is a recovery tool for within-site failures, not DR

Security Model

Encryption

LAYER	MECHANISM	STATUS
In transit	TLS 1.2+ on all connections (SMF, MQTTS, WSS, HTTPS)	Configured. Non-TLS ports disabled.

LAYER	MECHANISM	STATUS
At rest	Solace Cloud infrastructure encryption (AWS EBS)	Verify with Solace. PCI-DSS Requirement 3 requires protection of stored cardholder data. Confirm that Cloud Enterprise message spool encryption meets PCI-DSS. If insufficient, implement application-level field encryption for cardholder data before publishing.

Authentication

CLIENT TYPE	RECOMMENDED METHOD
Server-side SMF clients (8 systems)	Client certificate authentication
WebSocket browser clients (3 apps)	OAuth or API key
MQTT mobile client	Short-lived OAuth tokens (not embedded passwords). Token refresh via separate auth service.
REST delivery point	TLS + API key or OAuth

Access Control (ACL Profiles)

Per-system publish/subscribe ACL profiles are required (PCI-DSS Requirement 7: restrict access by business need-to-know).

SYSTEM	PUBLISH ACL	SUBSCRIBE ACL
Core Banking	payment/> , fraud/check/requested/> , auth/twoFactor/challenged/>	payment/transaction/initiated/> , fraud/check/completed/> , auth/twoFactor/*/v1/> , card/> , upi/> , settlement/> , risk/>
Fraud Detection	fraud/check/completed/> , fraud/alert/>	payment/transaction/initiated/> , fraud/check/requested/>
2FA Service	auth/twoFactor/verified/> , auth/twoFactor/failed/>	auth/twoFactor/challenged/>
Credit Card Processing	card/networkResponse/>	payment/transaction/authorized/*/card
UPI Gateway	upi/networkResponse/>	payment/transaction/authorized/IN/upi
Settlement Engine	payment/transaction/captured/> , payment/transaction/settled/> , settlement/>	payment/transaction/authorized/>
Notification Service	notification/customer/delivered/>	(RDP-bound)
Analytics Platform	(none)	analytics/> , notification/customer/delivered/> , payment/transaction/declined/>

DMQ Security

Dead message queues may contain cardholder data. Restrict DMQ access to authorized ops staff and automated recovery processes. Set DMQ message TTL. DMQ review must follow PCI-DSS data handling procedures.

Data Residency Enforcement

Enforced at three layers:

- 1. Topic level:** Region is first property after version (v1/{region}/...)
- 2. Subscription level:** Every consumer subscribes only to local region
- 3. ACL level:** Block cross-region subscriptions (e.g., */v1/IN/> blocked on US broker)

Periodic audit: verify no payment topics propagate across the DMR external link.

Operational Model

Monitoring

Solace Insights exports broker metrics to Datadog. Distributed Tracing (OpenTelemetry) available with product key for end-to-end payment flow visibility.

Alert thresholds:

Metric	Warning	Critical	Action
Queue depth (% of max spool)	50%	80%	Scale consumers or increase quota
DMQ depth	> 0	> 100	Investigate poison messages
Message rate deviation	+30% from baseline	+50% from baseline	Capacity check
Connection count drop	> 10% drop in 1 min	> 50% drop	Client reconnection issues
DMR external link status	Down > 30 sec	Down > 5 min	Reference data stale
Spool usage per region	60% of plan limit	80% of plan limit	Plan upgrade evaluation
End-to-end latency (hot path)	> 500ms p99	> 1000ms p99	Performance investigation

Capacity Planning

Parameter	Current	Year 3	Year 5
Peak TPS	5,000	6,050	8,050
Broker class	Enterprise	Enterprise (evaluate Enterprise+)	Enterprise+
Spool for 1hr replay	~36 GB	~44 GB	~58 GB
Connections per region	24-120	30-150	40-200
Queues per region	~40	~50	~60
DMR cross-link	< 25 msg/sec	< 33 msg/sec	< 40 msg/sec

Upgrade Path

Enterprise class through year 2. At year 3 (~6,050 TPS), evaluate Enterprise+ based on actual spool usage, connection count, and queue count. Enterprise+ provides higher limits across all dimensions. Upgrade is a plan change in Solace Cloud Console, not a migration.

Open Questions and Risks

Blocking (resolve before production)

- Card network connectivity model:** Direct ISO 8583 or through a payment processor/switch? Current design assumes direct. Impact: application layer, not event mesh topology.
- NPCI integration model:** PSP or TPP? Current design assumes PSP model. Impact: UPI topic flows and settlement logic.
- [High] Encryption at rest (S2):** Verify Solace Cloud encryption-at-rest for message spool meets PCI-DSS Requirement 3. Document in compliance artifacts.
- [High] ACL profiles (S3):** Define and provision publish/subscribe ACLs per system before production. The recommended profiles are documented in the Security Model section above.

Medium Priority (resolve before go-live)

- DMQ on all production queues:** Specify DMQ configuration for every Guaranteed messaging queue, not just the RDP queue.
- Alert thresholds:** Finalize and configure the thresholds from the Operational Model section in Datadog.
- Runbooks:** Write the 4 operational runbooks identified in HA/DR design (broker failover, DMQ review, region failure, spool exhaustion).
- Slow consumer mitigation:** Set per-queue max spool quota. Use non-exclusive queues for Fraud Detection and 2FA Service to enable horizontal scaling.
- Notification loss path:** Implement application-level retry in source systems for lost Direct messaging notifications.

Advisory (Phase 2 or later)

- Merchant onboarding: self-service (currently ops-managed)

- 11. Dispute/chargeback handling: event-driven (currently batch)
- 12. Analytics platform: Kafka bridge if stack changes to Kafka
- 13. Replication groups for broker-level DR
- 14. Event Portal for schema management and versioning

Appendices

A. Decision Log

#	DECISION	VALUE	RATIONALE
1	Project type	New build	Greenfield payment platform
2	Delivery mode	Mixed (17 Guaranteed, 3 Direct)	PCI-DSS audit trail for financial events; notifications/analytics ephemeral
3	Ordering	Per-key partitioned queues (txnId)	Per-transaction ordering at 5K TPS without global bottleneck
4	Latency tier	Sub-second hot path	UPI/card network SLAs, user waiting on payment
5	Topology	Multi-region (Mumbai + US-East)	India/US data residency hard requirement
6	Topic taxonomy	20 topics, 9 domains	Region-first properties for data residency enforcement
7	Broker type	Cloud Enterprise x2	Team capacity, timeline, cost model
8	Broker sizing	Enterprise, upgrade to Enterprise+ year 3	5K TPS with 10% YoY growth
9	Protocol strategy	SMF/WebSocket/MQTT/REST	Match protocol to client type and feature needs
10	Encryption in transit	TLS mandatory, all protocols	PCI-DSS Requirement 4
11	Mesh topology	2-site DMR external link	Data residency; < 25 msg/sec cross-link
12	Cross-region policy	No payment data crosses link	RBI (India), US data residency
13	HA strategy	Cloud Enterprise built-in	Three-node redundancy, multi-AZ, auto-failover
14	DR strategy	Phase 1: app-level recovery	Regions not DR pairs; broker not system of record
15	Replay log	1hr minimum retention	Recovery and reprocessing tool
16	Integration strategy	Zero Micro-Integrations	All custom apps via native APIs
17	Analytics integration	Direct consumer (Kafka bridge conditional)	Advisory decision, default to Solace API
18	Subscription strategy	36 subs, region-scoped	No cross-region payment data over DMR
19	Analytics delivery mode	Direct messaging	Fan-out data, loss-tolerant

B. Review Findings Summary

REVIEW	FINDINGS	CRITICAL	HIGH	MEDIUM	LOW	POSITIVE
Architecture	7	0	0	2	2	3
Operations	6	0	0	3	2	1
Security	7	0	2	3	1	1
Developer	6	0	0	2	3	1
Total	26	0	2	10	8	6

C. Validation Score

Broker Provisioning Parameters: Payment Solutions V1

Event Broker Service Instances

India (Mumbai)

PARAMETER	VALUE
Cloud provider	AWS
Region	ap-south-1 (Mumbai)
Service class	Enterprise
Message VPN name	payment-solutions-in
HA	Built-in (three-node, multi-AZ)
DMR	External link to US-East broker
Replay log	Enabled, 1hr retention
Non-TLS ports	Disabled

US (US-East)

PARAMETER	VALUE
Cloud provider	AWS
Region	us-east-1 (Virginia)
Service class	Enterprise
Message VPN name	payment-solutions-us
HA	Built-in (three-node, multi-AZ)
DMR	External link to Mumbai broker
Replay log	Enabled, 1hr retention
Non-TLS ports	Disabled

DMR External Link

PARAMETER	VALUE
Link name	in-us-reference-link
Direction	Bidirectional
Compression	Enabled
Subscriptions propagated	reference/> , analytics/summary/>
Subscriptions blocked	All payment, fraud, auth, settlement, risk, notification topics

Queue Configuration

Partitioned Queues (per region)

QUEUE NAME	SUBSCRIPTION	PARTITION KEY	CONSUMERS
q-core-banking-initiated	payment/transaction/initiated/v1/{region}/>	txnId	Core Banking
q-core-banking-fraud-result	fraud/check/completed/v1/{region}/>	txnId	Core Banking
q-core-banking-2fa-verified	auth/twoFactor/verified/v1/{region}/>	txnId	Core Banking
q-core-banking-2fa-failed	auth/twoFactor/failed/v1/{region}/>	txnId	Core Banking
q-core-banking-card-response	card/networkResponse/received/v1/{region}/>	txnId	Core Banking
q-core-banking-upi-response	upi/networkResponse/received/v1/{region}/>	txnId	Core Banking (IN only)
q-core-banking-captured	payment/transaction/captured/v1/{region}/>	txnId	Core Banking
q-core-banking-settled	payment/transaction/settled/v1/{region}/>	txnId	Core Banking
q-core-banking-risk	risk/assessment/completed/v1/{region}/>	txnId	Core Banking
q-fraud-detection-initiated	payment/transaction/initiated/v1/{region}/>	txnId	Fraud Detection
q-fraud-detection-requested	fraud/check/requested/v1/{region}/>	txnId	Fraud Detection
q-2fa-challenged	auth/twoFactor/challenged/v1/{region}/>	txnId	2FA Service
q-card-processing-authorized	payment/transaction/authorized/v1/{region}/card	txnId	Card Processing
q-upi-gateway-authorized	payment/transaction/authorized/v1/IN/upi	txnId	UPI Gateway (IN only)
q-settlement-authorized	payment/transaction/authorized/v1/{region}/>	txnId	Settlement Engine

Non-Exclusive Queues (per region)

QUEUE NAME	SUBSCRIPTION	CONSUMERS
q-risk-fraud-medium	fraud/check/completed/v1/{region}/medium	Risk Management
q-risk-fraud-high	fraud/check/completed/v1/{region}/high	Risk Management
q-risk-alerts	fraud/alert/raised/v1/{region}/>	Risk Management
q-merchant-authorized	payment/transaction/authorized/v1/{region}/>	Merchant Portal
q-merchant-captured	payment/transaction/captured/v1/{region}/>	Merchant Portal
q-merchant-settled	payment/transaction/settled/v1/{region}/>	Merchant Portal
q-merchant-batch	settlement/batch/completed/v1/{region}	Merchant Portal
q-merchant-reports	settlement/report/generated/v1/{region}/>	Merchant Portal
q-ops-batch-init	settlement/batch/initiated/v1/{region}	Ops Dashboard
q-ops-batch-complete	settlement/batch/completed/v1/{region}	Ops Dashboard
q-ops-fraud-high	fraud/alert/raised/v1/{region}/high	Ops Dashboard
q-ops-fraud-critical	fraud/alert/raised/v1/{region}/critical	Ops Dashboard
q-ops-risk	risk/assessment/completed/v1/{region}/>	Ops Dashboard
q-core-banking-batch	settlement/batch/initiated/v1/{region}	Core Banking

All Queues: Common Settings

PARAMETER	VALUE
Max spool (per queue)	Set per queue based on expected volume. Do not leave at VPN default.
Max message size	10 MB

PARAMETER	VALUE
Max redelivery count	5
DMQ	Enabled on every Guaranteed messaging queue
DMQ name	#DEAD_MSG_QUEUE (default) or named per queue
Respect TTL	Enabled
Access type	Partitioned or Non-exclusive as specified above

REST Delivery Point

PARAMETER	INDIA	US
RDP name	rdp-notification-service-in	rdp-notification-service-us
Queue	q-notification-rdp-in	q-notification-rdp-us
Subscriptions	notification/customer/requested/v1/IN/> , auth/twoFactor/failed/v1/IN/> , payment/transaction/declined/v1/IN/>	Same with US
REST consumer URL	(per environment)	(per environment)
Max retries	5	
Retry delay	Exponential backoff (1s base)	
Connection timeout	5 seconds	
Response timeout	10 seconds	
DMQ	Enabled with alerting	

Client Profiles

PROFILE	MAX CONNECTIONS	MAX SUBSCRIPTIONS	MAX SPOOL (AGGREGATE)
cp-server-smf	50 per client username	100	Per-queue limits
cp-browser-ws	1000 per client username	10	Per-queue limits
cp-mobile-mqtt	10000	5	Per-session limits
cp-rdp	10	20	Per-queue limits

Solace Insights Configuration

EXPORT	TARGET	FREQUENCY
Broker metrics	Datadog	Continuous (Solace Insights native export)
Distributed Tracing	OpenTelemetry collector	Per-message (requires product key)

Topic Taxonomy: Payment Solutions V1

Topic naming convention

Domain/Noun/Verb/v1/Properties where properties are ordered least-specific to most-specific.

- **Region** (IN , US) is always the first property. It is the primary routing dimension for data residency enforcement.
- **Secondary properties** vary by domain: `paymentMethod` (card, upi), `riskLevel` (low, medium, high), `method` (otp, app), `channel` (sms, push, email), `network` (visa, mastercard, rupay), `flowType` (pay, collect, mandate), `severity` (low, medium, high, critical), `merchantId` , `eventType` .
- Hard limits: 250 characters max, 128 levels max. camelCase used throughout.

Topic catalog

#	TOPIC STRING	DELIVERY MODE	PRODUCER	CONSUMER(S)	RATIONALE
1	payment/transaction/initiated/v1/{region}/{paymentMethod}	Guaranteed	Customer-Facing App	Core Banking, Fraud Detection Engine	Hot path entry point. Region + method enables per-broker, per-method filtering.
2	fraud/check/requested/v1/{region}/{paymentMethod}	Guaranteed	Core Banking	Fraud Detection Engine	Inline fraud scoring on every transaction before authorization.
3	fraud/check/completed/v1/{region}/{riskLevel}	Guaranteed	Fraud Detection Engine	Core Banking, Risk Management	riskLevel property lets Risk Management subscribe selectively to medium+high only.
4	auth/twoFactor/challenged/v1/{region}/{method}	Guaranteed	Core Banking	2FA Service	method = otp or app. 2FA Service routes to correct verification channel.
5	auth/twoFactor/verified/v1/{region}/{method}	Guaranteed	2FA Service	Core Banking	Success path. Core Banking resumes payment authorization flow.
6	auth/twoFactor/failed/v1/{region}/{method}	Guaranteed	2FA Service	Core Banking, Customer Notification Service	Failure triggers decline flow and customer notification.

#	TOPIC STRING	DELIVERY MODE	PRODUCER	CONSUMER(S)	RATIONALE
7	payment/transaction/authorized/v1/{region}/{paymentMethod}	Guaranteed	Core Banking	Credit Card Processing / UPI Gateway, Settlement Engine, Merchant Portal, Customer Notification	Fan-out to downstream processors. paymentMethod routes to correct network handler.
8	payment/transaction/declined/v1/{region}/{paymentMethod}	Guaranteed	Core Banking	Customer-Facing App, Customer Notification, Analytics	Decline events carry reason codes in payload for dispute tracking.
9	card/networkResponse/received/v1/{region}/{network}	Guaranteed	Credit Card Processing	Core Banking	network = visa, mastercard, rupay. ISO 8583 response from card network.
10	upi/networkResponse/received/v1/{region}/{flowType}	Guaranteed	UPI Gateway	Core Banking	flowType = pay, collect, mandate per NPCI specification. India region only.
11	payment/transaction/captured/v1/{region}/{paymentMethod}	Guaranteed	Settlement Engine	Core Banking, Merchant Portal, Analytics	Capture confirmation after network-level settlement.
12	payment/transaction/settled/v1/{region}/{paymentMethod}	Guaranteed	Settlement Engine	Core Banking, Merchant Portal	Final settlement. Audit trail anchor for PCI-DSS compliance.
13	settlement/batch/initiated/v1/{region}	Guaranteed	Settlement Engine	Core Banking, Internal Ops Dashboard	End-of-day or scheduled batch processing start.
14	settlement/batch/completed/v1/{region}	Guaranteed	Settlement Engine	Merchant Portal, Internal Ops Dashboard	Batch completion with summary metrics.
15	settlement/report/generated/v1/{region}/{merchantId}	Guaranteed	Settlement Engine	Merchant Portal	Per-merchant settlement report. merchantId is most-specific property.

#	TOPIC STRING	DELIVERY MODE	PRODUCER	CONSUMER(S)	RATIONALE
16	fraud/alert/raised/v1/{region}/{severity}	Guaranteed	Fraud Detection Engine	Risk Management, Internal Ops Dashboard	severity = low, medium, high, critical. Ops subscribes to high+critical.
17	risk/assessment/completed/v1/{region}/{riskLevel}	Guaranteed	Risk Management	Core Banking, Internal Ops Dashboard	Post-fraud-check risk scoring for authorization decisions.
18	notification/customer/requested/v1/{region}/{channel}	Direct	Core Banking, Settlement Engine, Fraud Detection	Customer Notification Service	channel = sms, push, email. Direct: loss triggers retry from source app.
19	notification/customer/delivered/v1/{region}/{channel}	Direct	Customer Notification Service	Analytics	Delivery confirmation for tracking delivery rates.
20	analytics/transaction/mirrored/v1/{region}/{eventType}	Direct	All producing systems	Analytics Platform	eventType mirrors source event name. Direct: analytics tolerates gaps.

Domain breakdown

DOMAIN	TOPICS	DELIVERY MODE	VOLUME (PEAK)
payment	5 topics (#1, 7, 8, 11, 12)	Guaranteed	~7,000/sec combined
fraud	3 topics (#2, 3, 16)	Guaranteed	~4,050/sec combined
auth	3 topics (#4, 5, 6)	Guaranteed	~2,000/sec combined
card	1 topic (#9)	Guaranteed	~1,200/sec
upi	1 topic (#10)	Guaranteed	~800/sec
settlement	3 topics (#13, 14, 15)	Guaranteed	~250/sec
risk	1 topic (#17)	Guaranteed	~500/sec
notification	2 topics (#18, 19)	Direct	~6,000/sec combined
analytics	1 topic (#20)	Direct	~5,000/sec

Delivery mode summary

- **Guaranteed messaging (17 topics):** All payment processing, fraud, 2FA, settlement, risk events. PCI-DSS audit trail requires lossless delivery. At-least-once with application-level idempotency using transaction ID.
- **Direct messaging (3 topics):** Customer notifications, notification delivery confirmations, analytics mirror. These are ephemeral or recoverable from source.

Data residency enforcement

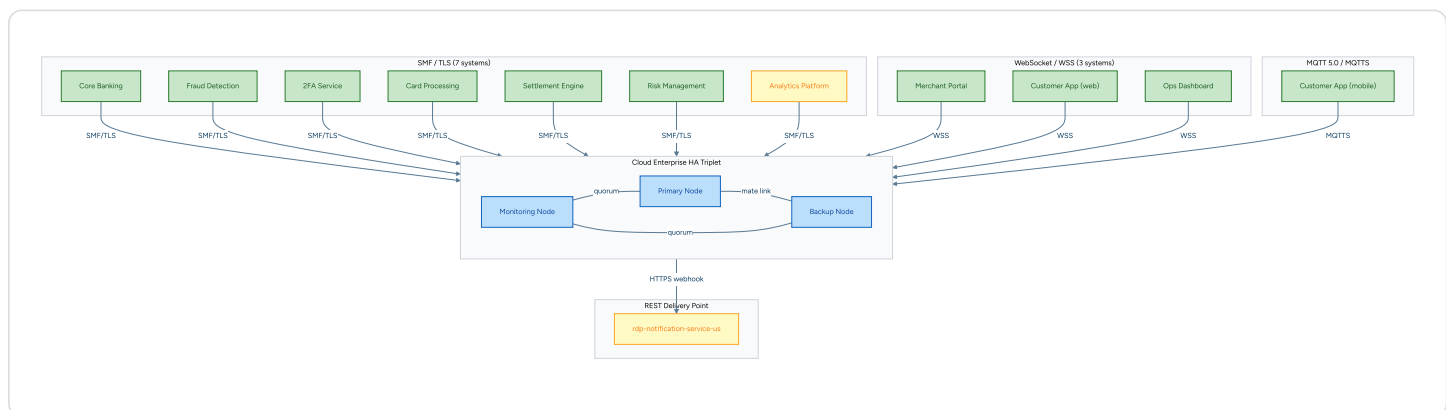
Region is the first property after version in every topic. This enables:

- Mumbai broker: subscribes only to topics with **IN** region
- US-East broker: subscribes only to topics with **US** region
- No payment transaction data crosses DMR external links
- Cross-region flows limited to shared reference data (not in this topic catalog; handled at the application layer)

Ordering strategy

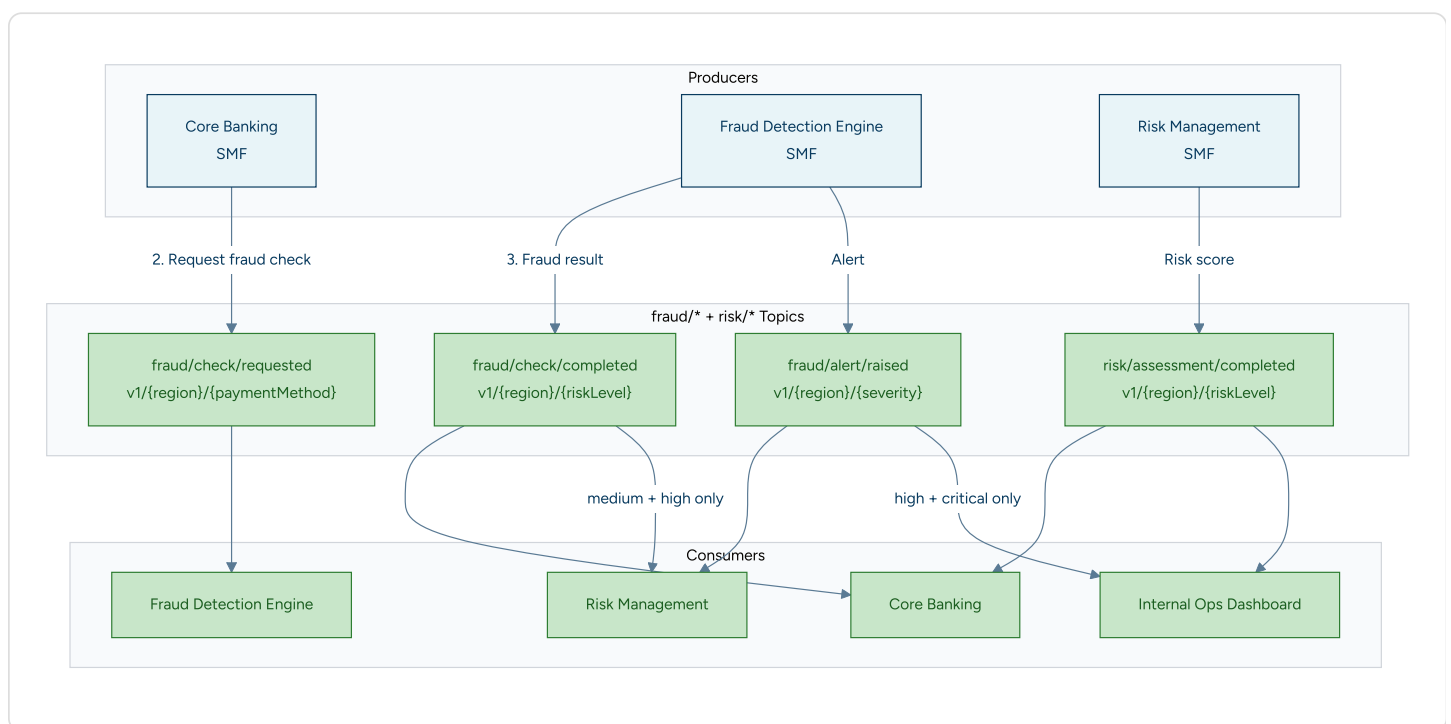
Per-key partitioned queues with transaction ID as partition key. The partition key is carried in message properties (not in the topic string). All events for a single payment transaction are processed in order by the same consumer instance within a consumer group.

broker topology us



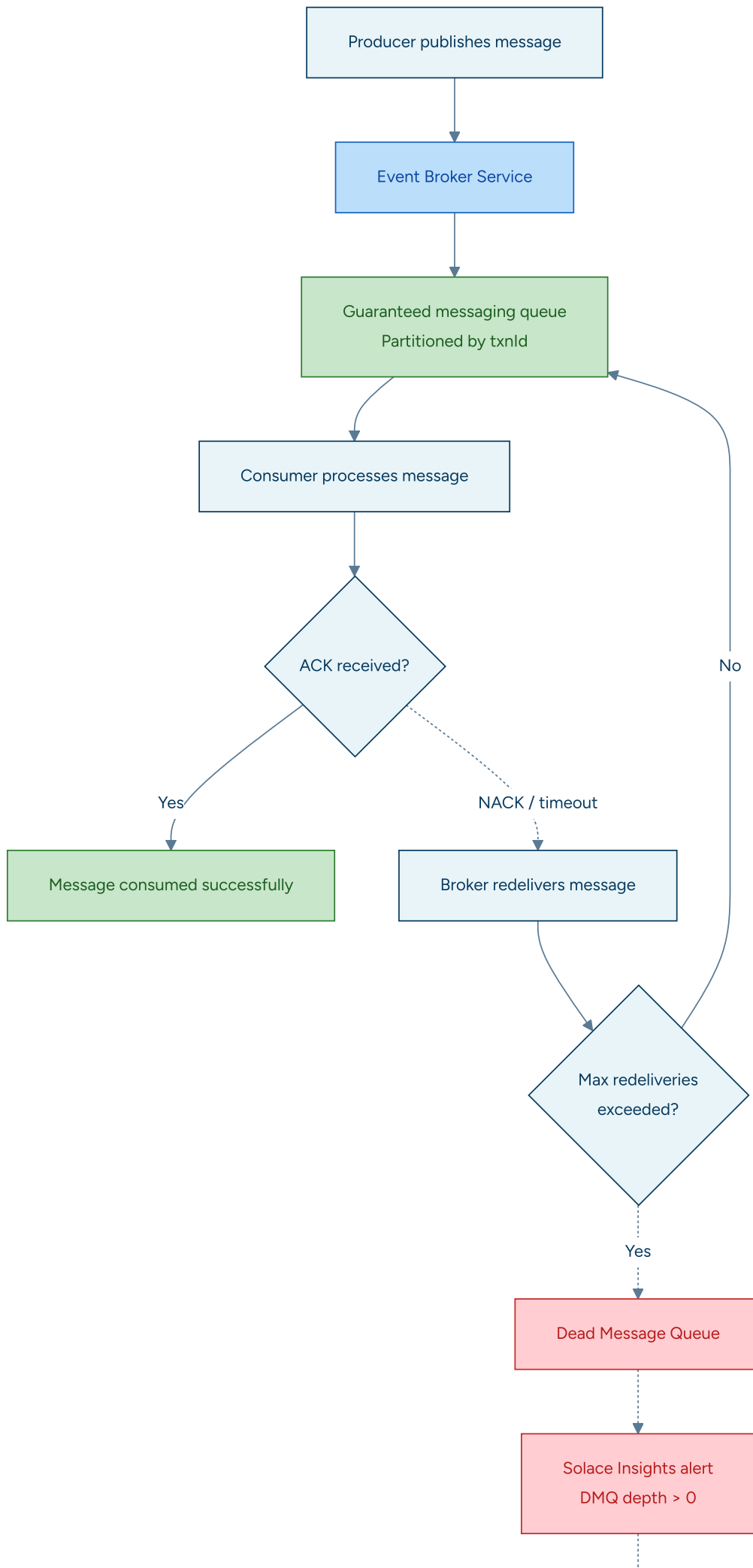
US region: 11 systems, HA triplet

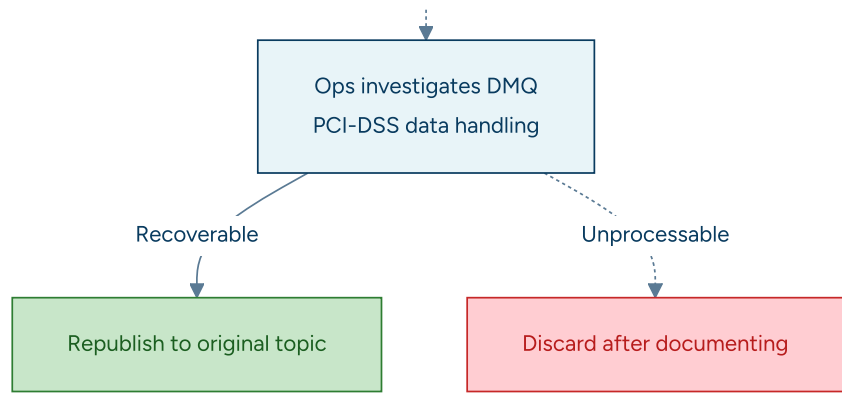
data flow 02 fraud risk



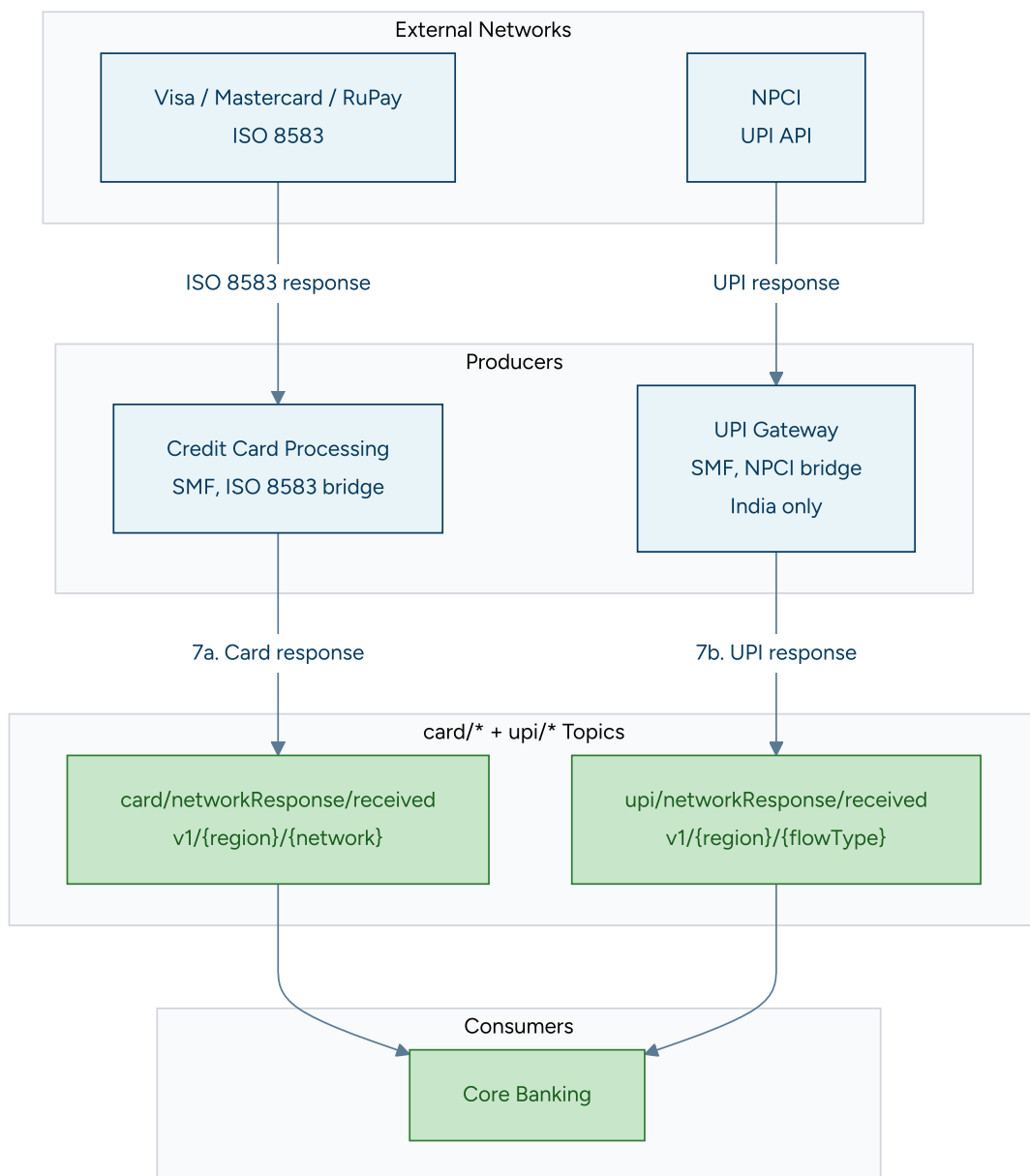
Fraud + risk domains: 4 topics

dlq flow

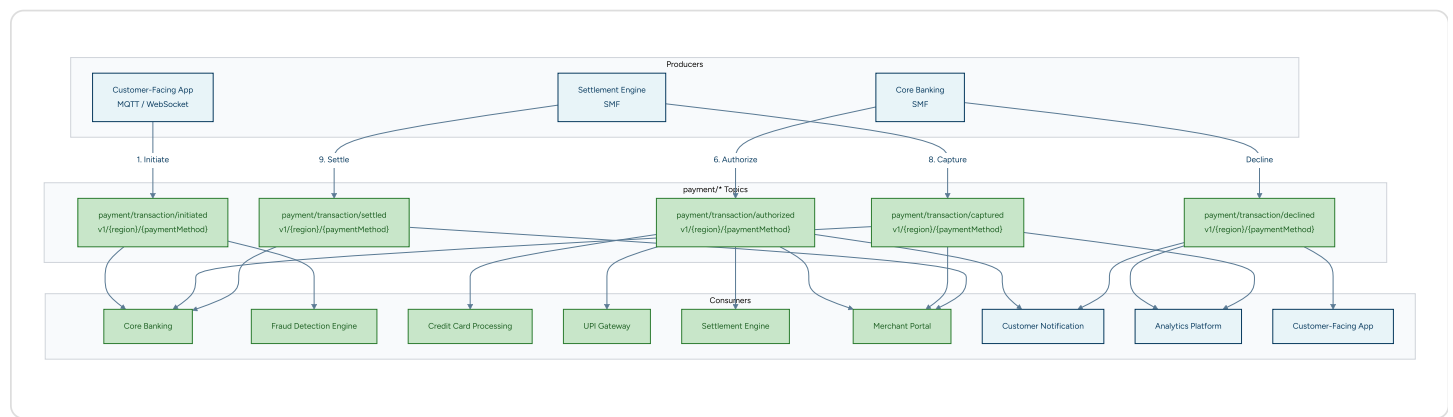


*Dead letter queue flow with classDef styles*

data flow 04 network

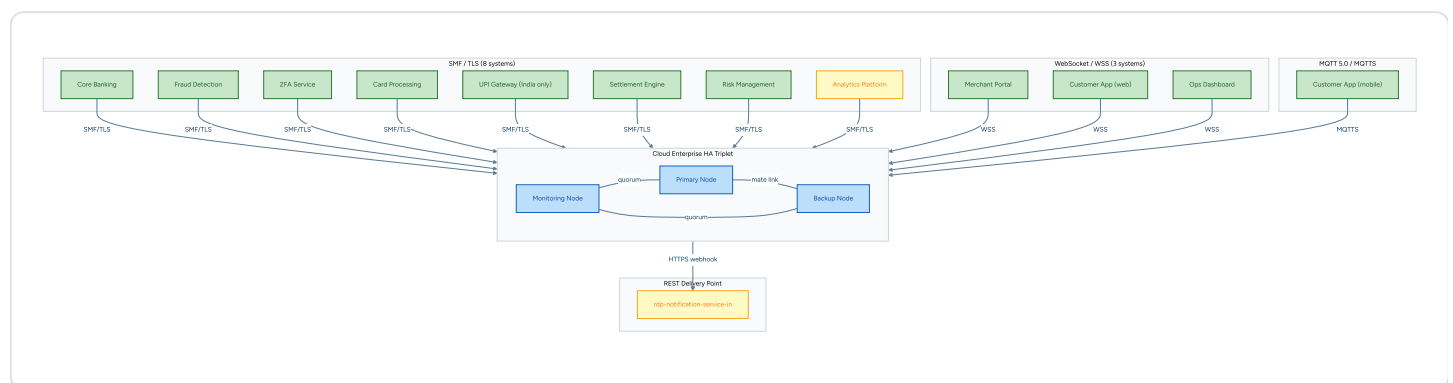
*Card + UPI network response flow*

data flow 01 payment



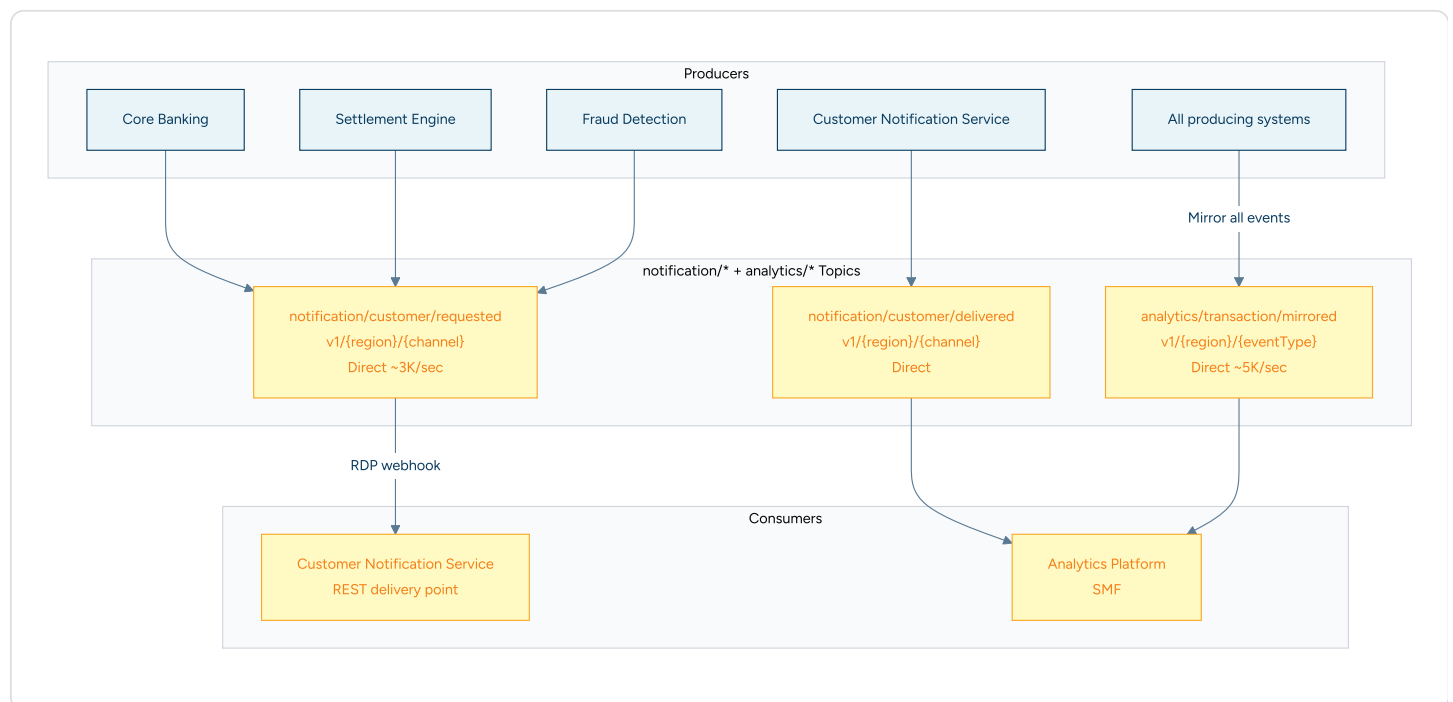
Payment domain: 5 topics, ~7K/sec

broker topology india



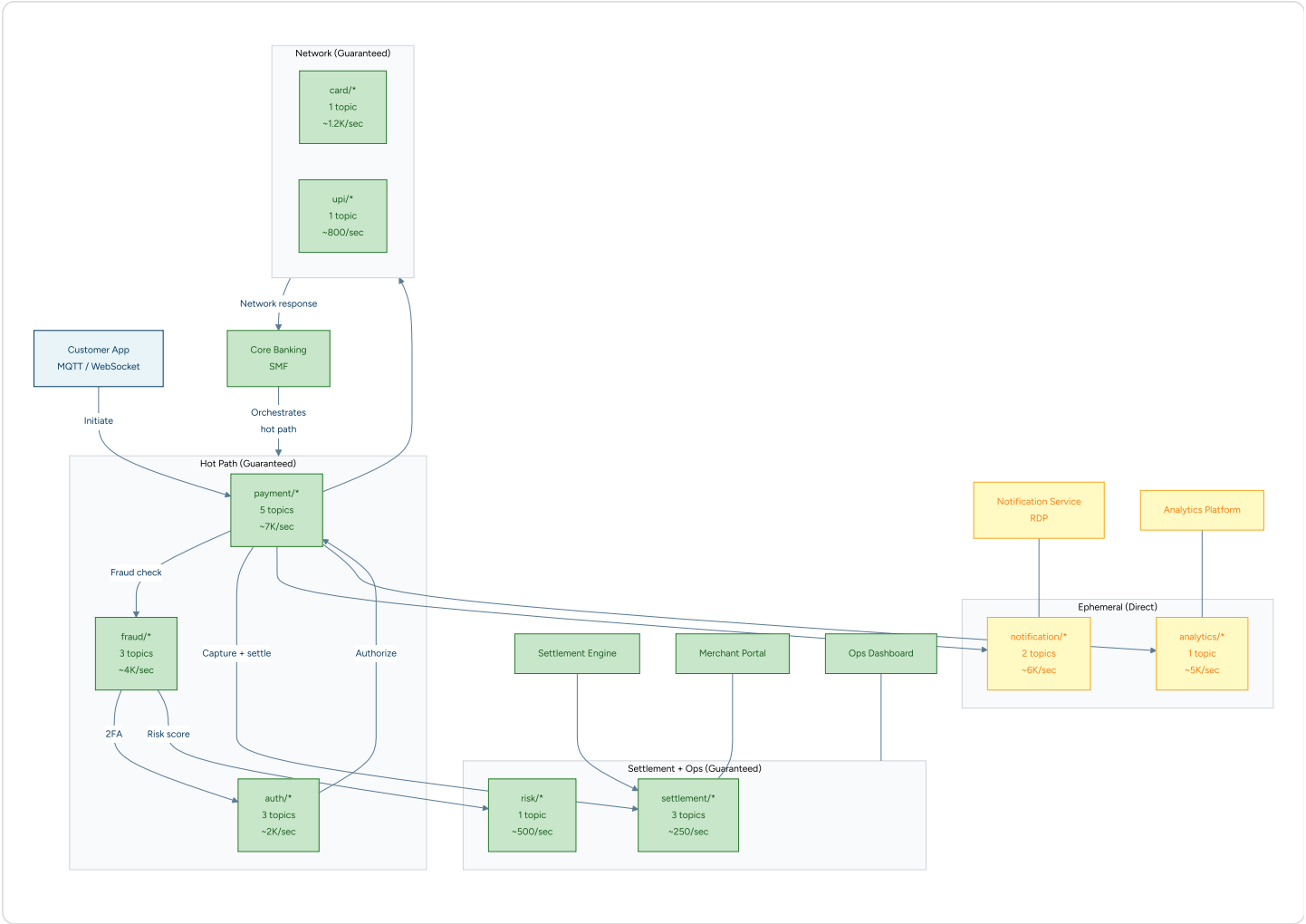
India region: 12 systems, HA triplet

data flow 06 notification analytics



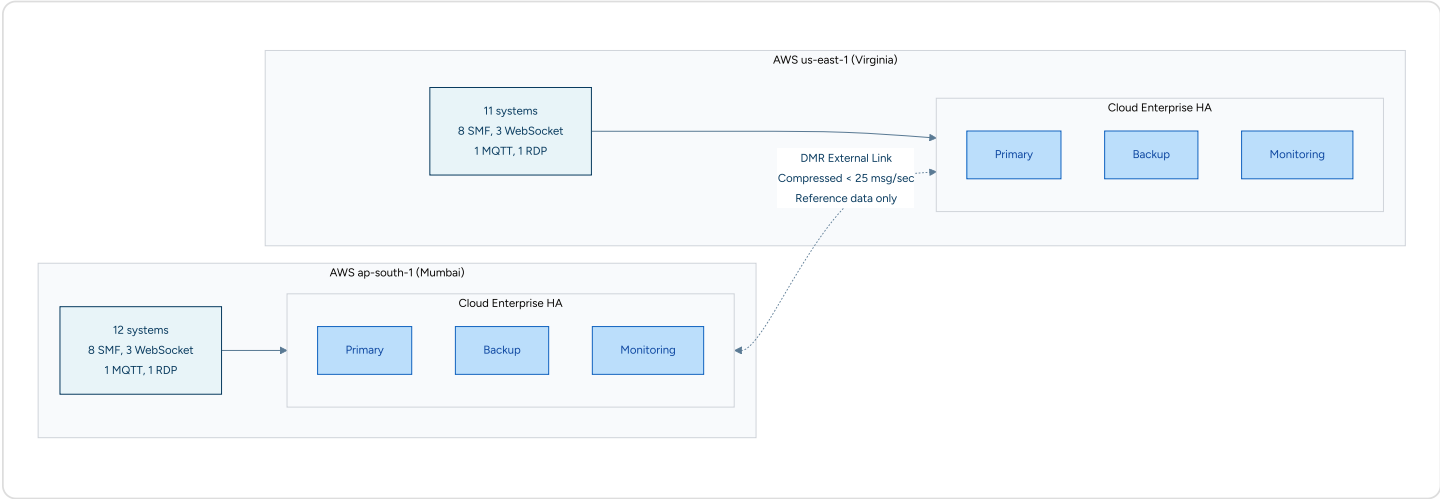
Notification + analytics: Direct messaging

data flow



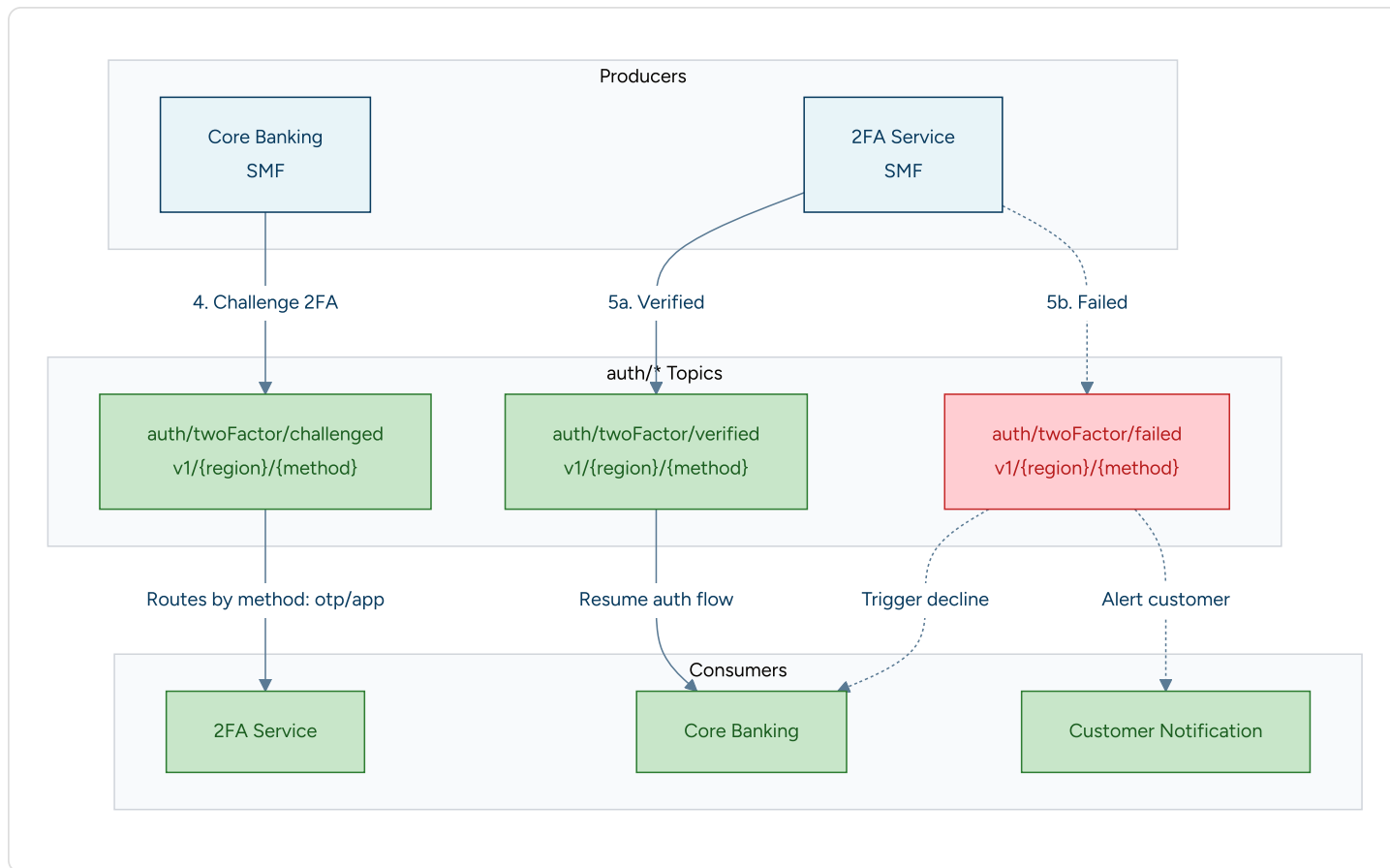
Domain-level overview (9 domains)

broker topology



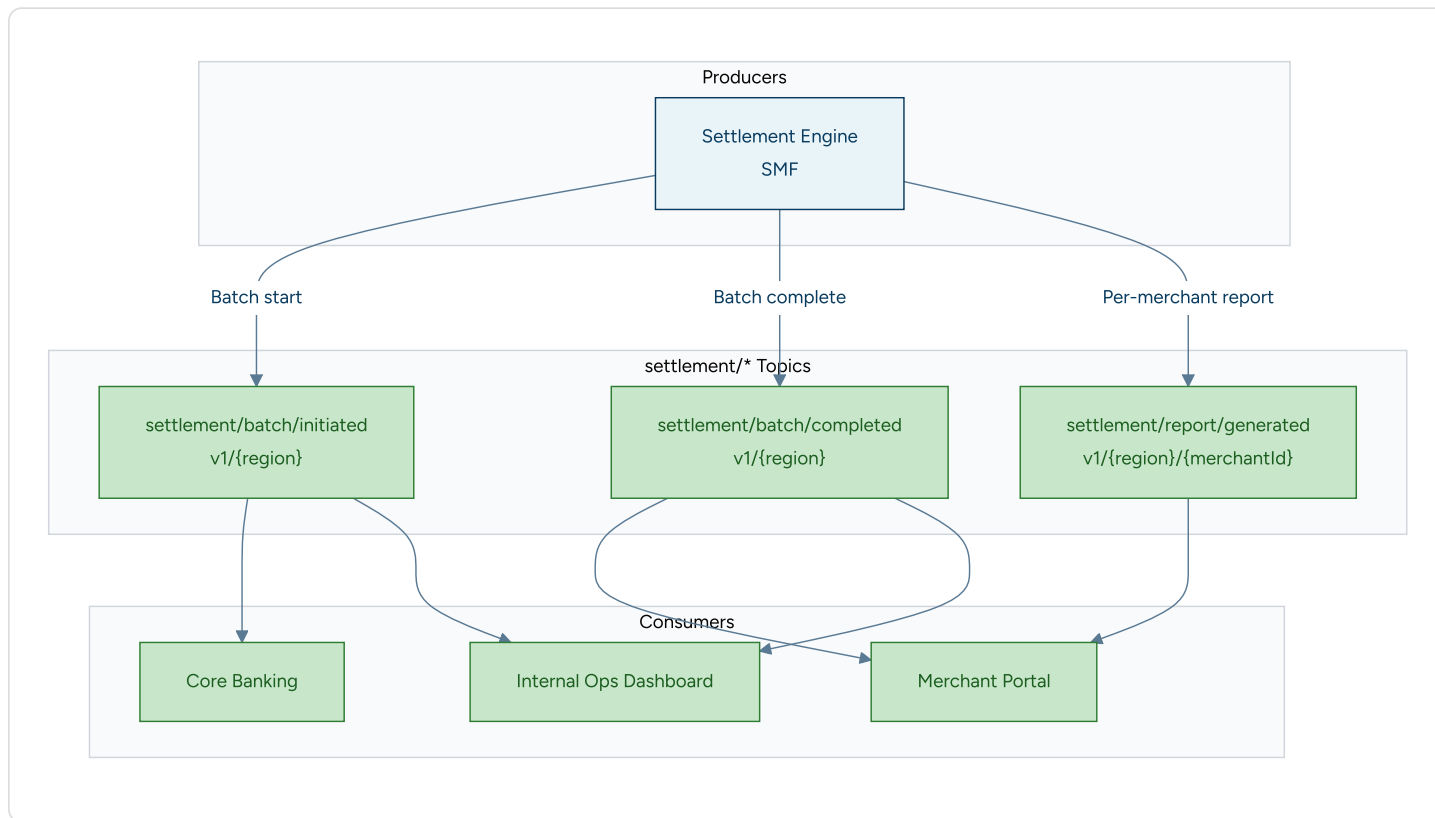
Two-region overview with DMR link

data flow 03 auth



Auth domain: 3 topics, 2FA flow

data flow 05 settlement



Settlement domain: 3 topics

Operational Runbook: Payment Solutions V1

1. Monitoring Setup

Metrics Pipeline

Solace Insights -> Datadog. All broker metrics exported automatically. Configure Datadog dashboards and alerts.

Dashboard: Payment Mesh Health

PANEL	METRIC	SOURCE
Hot path latency (p50/p95/p99)	End-to-end time from <code>payment/transaction/initiated</code> to <code>payment/transaction/authorized</code>	Application tracing (OpenTelemetry)
Queue depth by consumer	Per-queue spool usage	Solace Insights
DMQ depth	Messages in dead message queues	Solace Insights
Connection count	Active client connections per region	Solace Insights
DMR link status	External link up/down, message rate	Solace Insights
Spool utilization	Total spool used vs plan limit	Solace Insights
TPS by topic domain	Message rate per domain	Solace Insights

Alert Thresholds

METRIC	WARNING	CRITICAL	ACTION
Queue depth (% of max spool)	50%	80%	Scale consumers or increase per-queue quota
DMQ depth	> 0	> 100	Investigate poison messages (see Section 3)
Message rate deviation	+30% from baseline	+50%	Capacity review
Connection count drop	> 10% drop in 1 min	> 50% drop	Client reconnection check
DMR external link	Down > 30 sec	Down > 5 min	Reference data going stale
Spool usage per region	60% of plan limit	80%	Plan upgrade evaluation
Hot path latency	> 500ms p99	> 1000ms p99	Performance investigation
MQTT session count	> 10,000	> 50,000	Set session expiry intervals

2. Failure Modes

2.1 Broker Node Failure

Detection: Solace Insights connection count drop. Solace Cloud status notification. **Impact:** Auto-failover to backup node. Clients reconnect. In-flight Guaranteed messages redelivered. **Recovery:**

1. Verify failover completed (Solace Cloud Console)
2. Check client reconnection status (connection count returning to normal)
3. Monitor queue depths for message backlog from failover window
4. Check DMQ for messages that failed during transition
5. Verify DMR external link re-established

Expected duration: Seconds. Transparent to applications with reconnection configured.

2.2 Single AZ Failure

Detection: AWS status page. Solace Cloud status notification. **Impact:** Cloud Enterprise multi-AZ deployment handles this. Same as node failure from application perspective. **Recovery:** Same as 2.1. Solace manages AZ placement.

2.3 Full AWS Region Failure

Detection: AWS status page. All connections from affected region drop. DMR link goes down. **Impact:** Payment processing pauses for affected region. Other region unaffected. **Recovery:**

1. Confirm region failure via AWS status page
2. Communicate payment processing pause to merchants and internal stakeholders
3. Monitor AWS status for recovery ETA
4. When region returns: a. Verify broker comes back online (Solace Cloud Console) b. Verify client auto-reconnection c. Check queue depths for accumulated messages d. Execute message replay from last known good timestamp if needed e. Settlement Engine resumes from last committed batch checkpoint f. Verify DMR external link re-established g. Backfill notifications and analytics from source systems

Expected duration: Hours (depends on AWS). No data loss for Guaranteed messages (spool is durable).

2.4 Slow Consumer

Detection: Queue depth rising on specific consumer's queue. Hot path latency increasing. **Impact:** If unchecked, queue fills to max spool. At spool exhaustion, broker enters reject mode for the message VPN. **Recovery:**

1. Identify which consumer queue is growing (Solace Insights per-queue depth)
2. Check consumer application health (CPU, memory, downstream dependency latency)
3. Scale consumer horizontally if using non-exclusive queue
4. If partitioned queue: add consumer instances up to partition count
5. If consumer is permanently impaired: activate circuit breaker, route transactions to fallback path
6. If spool approaching limit: increase per-queue quota temporarily

Prevention: Set per-queue max spool quota so one slow consumer cannot exhaust the VPN.

2.5 DMR External Link Failure

Detection: Solace Insights link status alert. Reference data not updating in remote region. **Impact:** Cross-region reference data (merchant configs, rate tables) becomes stale. Payment processing within each region continues normally. **Recovery:**

1. Check link status in Solace Cloud Console
2. Verify both brokers are healthy
3. If one-sided: check network connectivity between regions
4. Link auto-reconnects when connectivity restores
5. After reconnection: verify subscription propagation resumed
6. Check if reference data needs manual republish

Expected duration: Seconds to minutes for transient network issues. Hours if underlying infrastructure issue.

2.6 Poison Message (DMQ Accumulation)

Detection: DMQ depth > 0 alert. **Recovery:**

1. Browse DMQ messages (do NOT consume yet)
2. Identify pattern: same topic? same producer? same payload structure?
3. Fix root cause in consumer application
4. Republish recoverable messages to original topic
5. Discard truly unprocessable messages after documenting
6. **PCI-DSS:** Do not log cardholder data from DMQ messages in clear text

2.7 REST Delivery Point Failure

Detection: Notification service queue depth rising. Delivery failures in RDP stats. **Impact:** Customer notifications delayed. Payment processing unaffected. **Recovery:**

1. Check Notification Service health (is the HTTP endpoint responding?)
2. Check RDP connection status in Solace Cloud Console
3. If endpoint down: fix the service. RDP auto-retries with exponential backoff.
4. If retries exhausted: messages move to DMQ. Follow poison message procedure.

5. After recovery: check DMQ for notifications that need manual republish.

3. Escalation Paths

SEVERITY	CONDITION	RESPONSE	CONTACT
P1 (Critical)	Payment processing halted, full region failure, spool exhaustion in reject mode	Page on-call immediately. Engage Solace support.	On-call engineer + Solace support ticket (severity 1)
P2 (High)	Hot path latency > 1s, connection count > 50% drop, DMQ > 100	Page on-call within 15 min.	On-call engineer
P3 (Medium)	Queue depth warning, DMR link down > 5 min, single DMQ message	Alert on-call. Investigate during business hours.	Platform team
P4 (Low)	MQTT session count high, RDP retry warning	Track in ticket. No page.	Platform team (next sprint)

4. Capacity Thresholds

METRIC	CURRENT HEADROOM	ACTION TRIGGER
TPS per region	5K of Enterprise limit	At 70% of plan limit
Spool per region	~36 GB of Enterprise allocation	At 60% of plan limit
Connections per region	24-120 of Enterprise limit	At 50% of plan limit
Queues per region	~40 of Enterprise limit	At 50% of plan limit
DMR cross-link bandwidth	< 1 MB/sec	At 50% of link capacity

When to upgrade: Monitor quarterly. If any metric consistently above 60% of plan limit across a billing cycle, evaluate Enterprise+ upgrade. Enterprise+ is a plan change in Solace Cloud Console, not a migration.

5. Upgrade Procedures

Broker Plan Upgrade (Enterprise -> Enterprise+)

1. Coordinate with Solace account team
2. Schedule maintenance window (Solace handles the upgrade)
3. Verify new plan limits in Solace Cloud Console
4. Update monitoring thresholds to new limits
5. No application changes needed

Client SDK Upgrade

1. Test new SDK version in Developer-class event broker service
2. Run integration tests with two Enterprise-class brokers (test DMR)
3. Deploy to one consumer instance, monitor for 1 hour
4. Roll out to remaining instances
5. Verify reconnection behavior after upgrade

6. Certificate Rotation

TLS Certificates

CERTIFICATE	OWNER	ROTATION
Broker TLS (server cert)	Solace Cloud (managed)	Automatic

CERTIFICATE	OWNER	ROTATION
Client certificates (SMF auth)	Platform team	Annual. Rotate 30 days before expiry.
OAuth tokens (mobile, browser)	Auth service	Short-lived (minutes to hours). Automatic refresh.
RDP webhook TLS	Notification service team	Annual. Coordinate with platform team.

Client certificate rotation procedure:

1. Generate new certificate signed by same CA
2. Add new certificate to broker's allowed client certificate list
3. Deploy new certificate to application
4. Remove old certificate from broker's allowed list after all instances updated
5. Verify connections using new certificate

7. Disaster Recovery Procedures

Phase 1: Application-Level Recovery

Trigger: Full AWS region failure affecting one broker.

Steps:

1. Confirm region failure (AWS status page, Solace Cloud Console)
2. Communicate to stakeholders: payment processing paused for affected region
3. Unaffected region continues independently (data residency: regions are isolated)
4. When region recovers: a. Verify broker comes online b. Check client auto-reconnection c. Execute replay: `solcache replay --from <last-known-good-time> --to <failure-time>` d. Settlement Engine: resume from last batch checkpoint e. Verify DMR external link re-established f. Republish stale reference data if needed g. Monitor queue depths for 30 minutes after recovery
5. Post-incident review: document timeline, message loss (if any), recovery actions

Phase 2: Replication Groups (future)

Not yet implemented. When provisioned:

- Standby broker in same data residency boundary
- Failover: automatic or manual (per configuration)
- After failover: verify clients connected to standby, verify spool state
- After primary recovery: decide to fail back or continue on standby

Validation Report: Payment Solutions V1

Cross-artifact consistency checks

1. Topic taxonomy vs subscription map

CHECK	STATUS	DETAIL
Every topic in taxonomy has at least one subscriber	✓ PASS	All 20 topics have mapped subscriptions.
Every subscription maps to a real topic	✓ PASS	All 36 subscriptions resolve to topics in the catalog.
Delivery modes consistent between topic and subscriber	✓ PASS	Guaranteed topics have queue-bound subscribers. Direct topics have topic-subscription subscribers.

CHECK	STATUS	DETAIL
Region scoping consistent	✓ PASS	Every subscription filters by local region.

2. Broker selection vs scale requirements

CHECK	STATUS	DETAIL
Broker can handle peak TPS	✓ PASS	Enterprise class handles 5K TPS with headroom.
Connection count within limits	✓ PASS	24-120 connections per region. Well within Enterprise.
Spool capacity for Guaranteed messaging	✓ PASS	~36 GB for 1hr replay. Enterprise class accommodates.
DMR external link supported	✓ PASS	Enterprise class includes DMR.

3. Protocol map vs topic taxonomy

CHECK	STATUS	DETAIL
Every producer has a protocol assigned	✓ PASS	All 12 systems have protocol selection.
Protocol supports required delivery mode	✓ PASS	SMF supports Guaranteed + Direct. MQTT QoS 1 maps to Guaranteed. WebSocket supports both. REST/RDP via queue.
Partitioned queue consumers use SMF	✓ PASS	All hot path consumers (Core Banking, Fraud Detection, Settlement, etc.) use SMF.

4. Mesh topology vs data residency

CHECK	STATUS	DETAIL
No payment topics cross DMR link	✓ PASS	Only reference and analytics summary topics propagate.
India data stays in Mumbai broker	✓ PASS	All IN-region subscriptions are local.
US data stays in US-East broker	✓ PASS	All US-region subscriptions are local.
UPI topics restricted to India	✓ PASS	UPI Gateway hardcoded to IN region.

5. HA/DR vs broker selection

CHECK	STATUS	DETAIL
HA mechanism matches broker type	✓ PASS	Cloud Enterprise includes built-in HA.
DR strategy documented	✓ PASS	Phase 1: app-level recovery. Phase 2: replication groups.
Replay log configured	✓ PASS	1hr minimum retention.

6. Integration map vs system inventory

CHECK	STATUS	DETAIL
Every system from discovery has an integration path	✓ PASS	All 12 systems mapped to connection method and protocol.
MI availability checked against catalog	✓ PASS	Zero MIs needed. All native API connections.
REST delivery point configured where needed	✓ PASS	Notification service has RDP specification.

7. Review findings vs architecture

CHECK	STATUS	DETAIL
All critical/high findings have resolution paths	⚠ PARTIAL	2 high findings from security review need resolution before blueprint: encryption at rest (S2) and ACL profiles (S3).
Medium findings acknowledged	✓ PASS	10 medium findings across 4 reviews. All have documented recommendations.

8. Naming conventions

CHECK	STATUS	DETAIL
"Micro-Integration" (capital M, hyphenated)	✓ PASS	Used correctly throughout.
"Direct messaging" / "Guaranteed messaging"	✓ PASS	No QoS references.
"Event broker service" for Cloud	✓ PASS	Used correctly.
"DMR" / "external links"	✓ PASS	Used correctly.
Topic taxonomy follows Domain/Noun/Verb/Version/Properties	✓ PASS	Consistent across all 20 topics.

9. Antipattern check (final)

ANTIPATTERN	STATUS
Environment names in topics	✓ PASS
Tracing IDs in topics	✓ PASS
Overly broad wildcards across DMR	✓ PASS
Properties before version	✓ PASS
Selector-based filtering	✓ PASS
Single global broker	✓ PASS
Mixed delivery modes on same topic	✓ PASS
Custom MI when cataloged path exists	✓ PASS (no MIs in use)
Direct messaging chosen where Guaranteed required	✓ PASS
Missing DMQ on production queues	⚠ FLAG — DMQ specified in RDP config but not explicitly specified for all production queues. Add DMQ to all Guaranteed messaging queues.

Validation summary

CATEGORY	CHECKS	PASSED	FLAGGED
Cross-artifact consistency	24	23	1 (high findings need resolution)
Naming conventions	5	5	0
Antipattern check	10	9	1 (DMQ on all queues)
Total	39	37	2

Items to resolve before blueprint

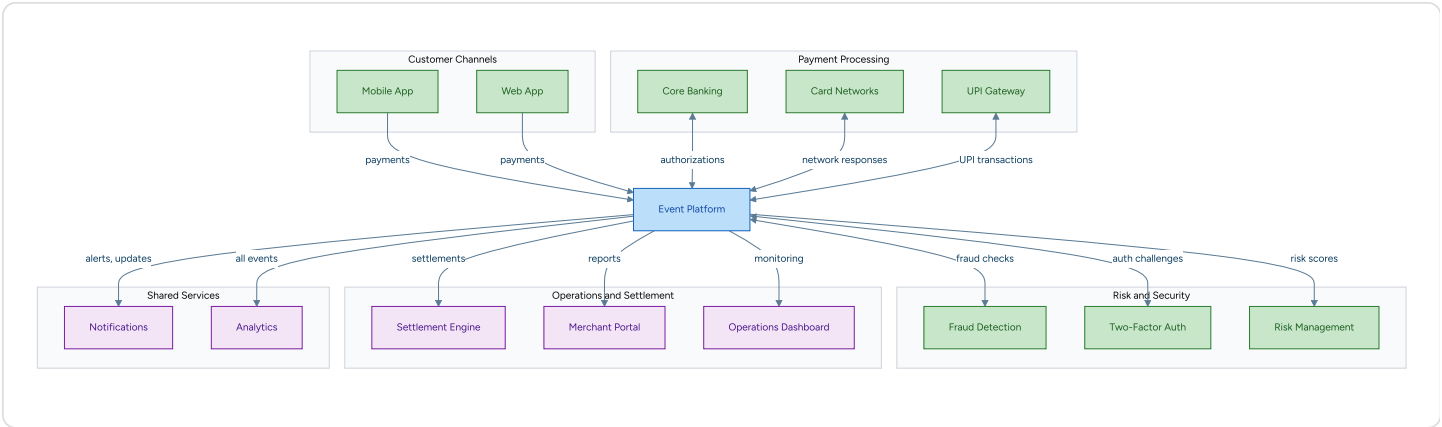
1. [High] Encryption at rest (S2) — Verify Solace Cloud encryption-at-rest for message spool. Document in blueprint.
2. [High] ACL profiles (S3) — Define publish/subscribe ACLs per system. Include in blueprint.
3. [Medium] DMQ on all production queues — Specify DMQ configuration for every Guaranteed messaging queue, not just the RDP queue.

Architecture completeness score

8.5 / 10 — All technical design artifacts are complete and internally consistent. Two high-severity security items and one operational item need resolution for a complete production-ready blueprint. No structural or topological gaps.

BUSINESS CASE

business architecture



Simplified one-page architecture visual (14 nodes, business labels)

Executive Summary: Payment Solutions V1

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The Opportunity

Payment Solutions V1 is a new payment processing platform that will enable the organization to offer credit card payments (Visa, Mastercard, RuPay) and UPI transactions to customers in India and the United States. The platform is designed to process thousands of transactions per second with sub-second authorization times, matching the speed expectations of modern digital payment experiences.

This initiative replaces a pattern of direct API-based connections between systems with a centralized event-driven platform. Instead of each system talking directly to every other system it needs, all 12 business systems publish and receive events through a single managed platform. This eliminates the fragile web of point-to-point connections that becomes harder to maintain, monitor, and secure as the business grows.

The architecture has been designed, reviewed across four dimensions (architecture, operations, security, and developer experience), validated for internal consistency, and assembled into a complete technical blueprint. The design is production-ready, pending resolution of two security verification items documented in this report.

Current State Challenges

- Point-to-point fragility.** Every new system or capability requires building, testing, and maintaining direct connections to every other system it depends on. Adding one system to a landscape of 12 means up to 11 new integration points.
- No message delivery guarantees.** REST-based connections have no built-in mechanism to ensure a payment event reaches all systems that need it. A network hiccup or a receiving system being briefly unavailable means lost data and manual reconciliation.
- No built-in failover.** If a system goes down, other systems discover this only when their API calls fail. Recovery is manual, reactive, and slow.
- Compliance exposure.** Payment data flows across direct connections without centralized encryption, access control, or audit trail. Meeting PCI-DSS requirements means retrofitting security into every connection individually.
- Limited real-time visibility.** Without centralized event routing, there is no single place to monitor all payment flows, detect anomalies, or measure end-to-end processing times.

Recommended Solution

The recommended platform uses Solace, a cloud-managed event broker service, deployed in two regions: India (AWS Mumbai) and US (AWS US-East). All 12 business systems connect to the platform using standard protocols suited to their deployment context: server-side applications, browser dashboards, and mobile apps each use the protocol best suited to their needs. The platform handles all translation between protocols transparently.

The design delivers zero-message-loss guarantees for all financial transactions. Every payment initiation, fraud check, authorization, and settlement event is delivered exactly as intended, with automatic recovery if a receiving system is temporarily unavailable. Notifications and analytics use a high-speed delivery mode appropriate for data that can be regenerated if lost.

Data residency is enforced at the platform level. India payment data stays in the India region. US payment data stays in the US region. Only shared reference data (merchant configurations, rate tables) and aggregated analytics cross the regional boundary. This is enforced by the platform's routing rules and access controls, not by application code that could be misconfigured.

Business Value

Operational Resilience

The platform includes automatic failover within each region. If a component fails, the system recovers in seconds without manual intervention. There is no single point of failure. If an entire cloud region experiences an outage, payment data is preserved on durable storage and processing resumes automatically when the region recovers. The system of record remains the core banking ledger, so no financial data is lost even in extreme scenarios.

A phased disaster recovery plan covers immediate needs (Phase 1: built-in failover plus application-level recovery) and future requirements (Phase 2: dedicated standby environments if regulators or business needs require faster recovery targets).

Integration Efficiency

All 12 systems connect through the central platform. Adding a new system requires connecting it to the platform once, not building connections to every other system. The platform's structured event routing ensures that each system receives exactly the data it needs, automatically.

The design uses zero pre-built integration modules because this is a greenfield platform where all systems are custom-built. This is a deliberate choice: the team controls the entire stack. If the analytics platform later adopts a different technology, a pre-built bridge is available to connect it without re-engineering.

Time to Market

Standardized connection patterns mean every new system follows the same integration approach. The developer experience review identified clear patterns for building new consumers: per-transaction ordering guarantees, a defined hot path for payment authorization (6 steps, sub-second total), and a testing strategy that validates each integration independently.

New payment methods, merchant categories, or regional expansions follow the same pattern. Adding a new payment rail means adding topics to the existing taxonomy and connecting the new gateway, not rebuilding the integration fabric.

Regulatory and Compliance

The platform delivers PCI-DSS alignment across several requirements:

- **Data protection in transit.** All connections are encrypted. Unencrypted ports are disabled.
- **Access control.** Each system has defined permissions specifying exactly which events it can send and receive. The fraud detection engine cannot access settlement data. The merchant portal cannot publish payment authorizations. These boundaries are enforced by the platform, not by trusting application code.
- **Audit trail.** Every financial event is delivered with zero-message-loss guarantees, providing a complete record of all payment processing activity.
- **Data residency.** India and US payment data are isolated at the platform level, meeting RBI and US data handling requirements.

Two items require verification before production: confirming that stored message encryption meets PCI-DSS requirements, and provisioning the access control profiles documented in the security design.

Risk Assessment

Deployment Risk

This is a greenfield platform with no legacy migration. The primary deployment risk is the 4-month timeline with a 10-person team that has moderate experience with event-driven platforms. Mitigations: cloud-managed infrastructure eliminates broker operations, standardized patterns reduce per-system implementation effort, and the complete technical blueprint provides a defined build plan.

Operational Risk

The architecture handles most failure scenarios automatically. Three operational items need attention before production:

1. **Monitoring thresholds.** Alert levels for queue depth, connection drops, and processing latency need to be configured and tested in Datadog.

- 2. **Operational runbooks.** Four runbooks are needed: broker failover procedures, failed message review, regional failure response, and storage capacity management.
- 3. **Consumer scaling.** Under sustained load spikes, the fraud detection and two-factor authentication services need a defined scaling policy to prevent processing delays on the payment authorization path.

Risk Mitigation

- **Automatic failover** handles infrastructure failures without human intervention.
- **Failed message recovery** captures any messages that cannot be processed, routes them to a review queue, and alerts the operations team. Nothing is silently lost.
- **Message replay** allows reprocessing of recent events from any point in the last hour, supporting recovery from application errors without data loss.
- **Structured monitoring** exports all platform metrics to the existing Datadog installation, providing a single pane of glass for the operations team.

Investment Overview

Scope

- 12 business systems connected across 2 regions
- 20 event types covering payments, fraud detection, authorization, settlement, notifications, and analytics
- Multi-protocol support: server applications, browser dashboards, and mobile apps
- Full encryption, access control, and compliance infrastructure

Cost Drivers

- **Platform licensing.** Two cloud-managed event broker instances (Enterprise class), one per region. This covers high-availability infrastructure, monitoring integration, message replay, and platform management. Upgrade to the next tier is anticipated around year 3 as transaction volumes grow.
- **Implementation effort.** 12 system integrations, each following a standardized pattern. The hot path (payment authorization flow) involves 6 event handoffs across 5 systems and represents the most complex integration work.
- **Operational overhead.** Cloud-managed deployment eliminates broker infrastructure operations. The team maintains application-level connections, monitoring dashboards, and the 4 operational runbooks. No dedicated broker administration staff required.

ROI Indicators

- **Reduced integration maintenance.** Centralized platform replaces point-to-point connections. Each new system requires one connection instead of up to 11.
- **Faster incident recovery.** Automatic failover reduces mean-time-to-recovery from manual intervention (minutes to hours) to seconds.
- **Accelerated delivery.** Standardized integration patterns reduce per-system development time. New payment methods follow the existing pattern.
- **Compliance cost avoidance.** Platform-level encryption, access control, and audit trail eliminate the need to retrofit security into each individual connection.

Note: Specific dollar figures require organizational cost data not captured during architecture design. A fillable ROI worksheet is provided as a companion document for your finance team to populate with actuals.

Implementation Timeline

Phase Overview

- **Phase 1: Foundation (months 1-2).** Deploy cloud platform in both regions. Build the payment authorization hot path: payment initiation, fraud detection, two-factor authentication, and authorization. Connect Core Banking, Fraud Detection, 2FA Service, and Card Processing. Validate sub-second end-to-end latency.
- **Phase 2: Full connectivity (months 2-3).** Connect remaining systems: UPI Gateway, Settlement Engine, Merchant Portal, Customer Apps (mobile and web), Notification Service, Risk Management, Analytics Platform, and Operations Dashboard. Configure cross-region reference data synchronization.
- **Phase 3: Hardening (months 3-4).** Finalize access control profiles, configure monitoring and alerting in Datadog, write and test operational runbooks, conduct PCI-DSS verification for stored message encryption, and run load tests at projected peak volumes.

Key Milestones

- First payment authorization processed end-to-end through the platform

Open Items

- The following business decisions are needed before or during implementation:
- 1. Card network connectivity model.** How will the platform connect to Visa, Mastercard, and RuPay: directly via the card network protocol, or through a third-party payment processor? This affects the integration approach for card processing but does not change the platform architecture.
 - 2. UPI integration model.** Will the organization operate as a Payment Service Provider (PSP) or Third Party Provider (TPP) under NPCI? This affects the UPI transaction flows and settlement logic.
 - 3. Stored data encryption verification.** Confirm that the cloud platform's storage encryption meets PCI-DSS Requirement 3 for protection of stored cardholder data. If it does not, application-level encryption of sensitive fields will be added before data enters the platform.
 - 4. Access control provisioning.** The security design specifies exactly which system can send and receive which events. These profiles need to be provisioned on the platform before production.

Items 1 and 2 are business decisions. Items 3 and 4 are implementation tasks with clear resolution paths documented in the technical blueprint.

Recommendation

Proceed with implementation. The architecture design is complete, internally consistent (37 of 39 cross-artifact validation checks passed), and has been reviewed across architecture, operations, security, and developer experience dimensions. No critical issues were found. The two high-severity security items have clear resolution paths and should be addressed in Phase 3.

The platform delivers the foundation for a modern payment processing capability: real-time transaction flow across 12 systems, automatic failover, regulatory-grade security, and a design that scales with the business. The single most important outcome is this: every payment transaction is tracked, delivered, and recoverable from end to end, with no silent data loss and no manual intervention required for routine operations.

This executive summary was generated from the complete architecture design produced by Solace Architect. The full technical blueprint is available for engineering teams.

ROI Discussion Guide

This guide walks you through building a business case in five steps. Architecture-derived values are pre-filled from your design. Fill in your organization's cost data to calculate ROI automatically.

FOUNDATION Architecture Indicators			
These values are derived from the architecture design. They anchor all cost and value estimates below.			
12 SYSTEMS CONNECTED Each avoids up to 11 point-to-point integrations	20 (17 Guaranteed + 3 Direct) EVENT TYPES Structured routing replaces custom dispatch	5,000 PEAK TRANSACTIONS/SECOND Platform handles scale; apps focus on business logic	2 (India + US) REGIONS Data residency enforced by platform, not app code
Seconds (Cloud Enterprise HA) AUTOMATIC FAILOVER TIME vs manual recovery (minutes to hours)	1 hour MESSAGE REPLAY WINDOW Recovery from application errors without data loss	0 (greenfield) PRE-BUILT INTEGRATIONS Full team control; Kafka bridge available if needed	12 (one per system) SECURITY PROFILES Centralized ACL with explicit permissions

PCI - DSS

COMPLIANCE FRAMEWORKS

TLS, access control, audit trail built in

Sub-second

HOT-PATH LATENCY

6-step payment authorization across 5 systems

Payment data

stays in-region

CROSS-REGION DATA POLICY

RBI + US requirements met at platform level

Phase 1: Cloud

HA + app

recovery

DR STRATEGY

Phased investment matching risk tolerance

STEP 1

Cost of Current State (Annual)

Estimate what your organization spends today due to the limitations of your current integration approach. Each row tells you **what to measure**, **who to ask**, and gives a **worked example**.

#	CATEGORY	ESTIMATE (\$)	ARCHITECTURE BASIS
C1	Unplanned downtime Count payment processing outages over the past 12 months where the cause was integration failure, missed messages, or manual recovery delays. Multiply by average duration and your cost-per-hour (lost revenue + recovery labor + SLA penalties). Example: 4 incidents/yr x 3 hours avg x \$50,000/hr = \$600,000	600000	No built-in failover; recovery is manual
C2	Integration maintenance Count FTEs who spend significant time keeping point-to-point connections alive between your 12 systems. Include connection monitoring, retry logic, format translation, break-fix, and on-call rotation for integration issues. Your current architecture has up to 66 possible point-to-point connections (12 x 11 / 2). Example: 2.5 FTE x \$180,000/yr loaded cost = \$450,000	450000	12 systems, up to 66 point-to-point connections
C3	Failed/lost transactions Estimate how many payment transactions fail silently each month because a REST call didn't reach its destination (network blip, receiver down, timeout). Multiply by the business cost per failed transaction (reversal cost, customer service call, potential customer loss). Example: 200 failed/month x \$150/transaction = \$360,000/yr	360000	REST calls with no delivery guarantee
C4	Compliance remediation How much did the last PCI-DSS audit cost in remediation effort? With no centralized encryption, access control, or audit trail, each connection must be secured individually. Example: 6 person-months x \$25,000/month = \$150,000	150000	PCI-DSS retrofit across individual connections
C5	Developer integration time When a new system is added, how many days does it take to build, test, and deploy each new point-to-point connection? Multiply by daily developer cost and expected new integrations per year. Currently each new system needs up to 11 new connections. Example: 15 days/integration x \$1,200/day x 3 new integrations/yr = \$54,000	54000	Each new system needs up to 11 new connections
Total current state cost		\$1,614,000	

STEP 2

Cost of New Platform (Annual)

Estimate what the new platform will cost to license, implement, and operate. Contact Solace sales for licensing quotes; use your project manager for implementation scope.

#	CATEGORY	ESTIMATE (\$)	ARCHITECTURE BASIS
P1	Platform licensing (annual) <i>Request a quote from Solace sales for: Cloud Enterprise class, 2 regional instances (AWS Mumbai + US-East), with the throughput and connection requirements from the technical blueprint (5K TPS, 12 systems, 24-120 connections).</i> Example: \$120,000/yr for 2 Enterprise instances (varies by contract)	120000	Cloud Enterprise class, 2 instances
P2	Implementation (one-time, amortize /3yr) <i>The architecture defines the scope: 12 system integrations, 4-month build, 10-person team. Multiply average team member cost by duration. Enter the one-time total — the calculator amortizes it over 3 years automatically.</i> Example: 10 people x \$15,000/month x 4 months = \$600,000 one-time (\$200,000/yr amortized)	600000	12 integrations, 4 months, 10-person team
P3	Incremental operational staff <i>Cloud-managed deployment means no dedicated broker admins. Estimate incremental ops effort: dashboard monitoring, runbook execution, incident response for the event platform. This is typically a fraction of an existing role, not a new hire.</i> Example: 0.25 FTE x \$180,000/yr = \$45,000	45000	Cloud-managed; no dedicated broker admin
P4	Training and enablement <i>The team has moderate event-driven experience but needs training on Solace protocols (SMF, MQTT), topic design patterns, and the operational model. Estimate training vendor cost + team downtime.</i> Example: \$30,000 vendor training + 2 weeks x 10 people x \$1,500/wk opportunity cost = \$60,000	60000	Moderate event-driven experience
P5	Platform upgrade (year 3+) <i>The architecture plans an upgrade from Enterprise to Enterprise+ as transaction volumes grow past 6K TPS. This is a plan upgrade (no migration, no downtime). Enter the estimated annual licensing delta.</i> Example: \$40,000/yr incremental licensing starting year 3	40000	Enterprise to Enterprise+ at ~year 3
Total new platform cost		\$465,000	

STEP 3 Value Delivered (Annual)

Values marked **AUTO-FILLED** are computed from your Step 1 inputs using architecture-derived ratios. Edit any field to override with your own estimate. V3 and V5 require manual input.

#	CATEGORY	ESTIMATE (\$)	ARCHITECTURE BASIS
V1	Reduced downtime AUTO-FILLED Auto: 90% of C1 (downtime eliminated). Edit to override; double-click to restore. <i>With automatic failover (seconds, not hours), how many downtime hours per year would be eliminated? Multiply by the same cost-per-hour used in C1. The value is C1 minus the residual downtime cost on the new platform.</i> Example: If C1 was \$600K and new platform reduces downtime by 90% → \$540,000 saved	540000	Automatic failover in seconds vs manual hours
V2	Integration maintenance savings AUTO-FILLED Auto: 80% of C2 (FTEs redirected). Edit to override; double-click to restore. <i>With 1 connection per system (vs up to 11), how many FTEs can be redirected from integration maintenance to product work?</i> Example: 2 of the 2.5 FTE from C2 redirected → 2 x \$180,000 = \$360,000 saved	360000	1 connection per system vs up to 11

#	CATEGORY	ESTIMATE (\$)	ARCHITECTURE BASIS
V3	Faster time-to-market With standardized integration patterns, how many weeks faster can new payment features ship? Multiply by the number of new features per year and the weekly cost of delay (lost revenue opportunity, competitive impact). Example: 3 features/yr x 4 weeks saved x \$25,000/week opportunity cost = \$300,000	300000	Standardized patterns for new capabilities
V4	Compliance cost avoidance AUTO-FILLED Auto: 100% of C4 (compliance replaced). Edit to override; double-click to restore. Platform provides TLS encryption, 12 per-system ACL profiles, and zero-message-loss audit trail. Estimate the annual compliance spend eliminated (audit remediation, security patching across connections, penetration testing of individual integration points). Example: \$150K remediation (from C4) + \$50K annual security testing savings = \$200,000	150000	Platform-level TLS, ACL, audit trail
V5	Developer productivity With defined patterns (6-step hot path, protocol mediation, topic-based routing), how many developer hours per week are saved across the team? Count time previously spent on: retry logic, message format translation, connection debugging, and protocol adaptation. Example: 8 developers x 4 hrs/wk saved x 50 weeks x \$85/hr = \$136,000	136000	Defined hot-path pattern, protocol mediation
V6	Transaction recovery AUTO-FILLED Auto: 95% of C3 (transactions recovered). Edit to override; double-click to restore. With Guaranteed messaging and 1-hour replay, how many transactions that currently fail silently would now be recovered? Use the same per-transaction cost from C3. Example: If C3 was 200 failed/month and 95% are now recovered → 190 x 12 x \$150 = \$342,000	342000	Guaranteed messaging + 1-hour replay
Total annual value		\$1,828,000	

STEP 4Results

These metrics are calculated automatically from your inputs above. They update live as you change any value.

NET ANNUAL BENEFIT \$1,363,000 Value delivered minus platform cost	IMPLEMENTATION COST \$600,000 P2 one-time cost (amortized over 3 years in platform total)	PAYBACK PERIOD 6 months Months until implementation cost recovered
3-YEAR NET VALUE \$3,489,000 (Net benefit x 3) minus implementation	5-YEAR NET VALUE \$6,175,000 (Net benefit x 5) minus impl minus upgrade	ROI PERCENTAGE 293% Net benefit / platform cost x 100

STEP 5What-If Scenarios

Each card shows the isolated impact of a single variable change. The **Combined Scenario** card at the bottom compounds all active adjustments together for a realistic view.

STEP 6Download

Export your completed analysis as an Excel workbook with formulas for all calculated fields.

Includes formulas, sensitivity scenarios, and architecture indicators

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